BARBM312 · BACHELOR · FHEQ LEVEL 6 · 5 ECTS · EU BUSINESS SCHOOL

BARBM312

AI in Investment and Business Growth

Framework Sheet Library

Master Index · 28 Slibraries · 140 Reference Sheets

7 Course Units
strategy, data, costs, IP, agility, finance, scaling

28 Slibraries
thematic clusters, 4–5 per unit

140 Framework Reference Sheets
print-ready A4, nine-section layout

280+ Harvard-referenced sources
peer-reviewed and practitioner

Design philosophy. Each sheet follows a consistent nine-section layout: professional definition; who developed it; why it was developed; a bespoke illustrative figure; how, when and by whom it is used; the innovation relative to the predecessor tool; related frameworks in this course; and Harvard-referenced sources. The library is built in the JD Meier "Thumbnail Thinking" tradition (Meier, 2025) — treat the 140 sheets as a mental Slibrary that scales from single-framework lookup to full strategic reasoning across connected tools. Core messages are clarified after Bostelaar.

Dr. Hildegard Haas · EU Business School · BARBM312

MASTER INDEX

The 28 Slibraries

MASTER INDEX · CONTINUED

The 28 Slibraries

DIKW Pyramid

Data → Information → Knowledge → Wisdom

PROFESSIONAL DEFINITION

The DIKW Pyramid is a hierarchical model that arranges Data, Information, Knowledge and Wisdom in ascending order of context and value, showing how raw signals must be progressively interpreted before they can inform a decision.

WHO DEVELOPED IT

Popularised by Russell Ackoff in his 1989 address 'From Data to Wisdom', with roots traced to earlier information-science writing and an oft-cited stanza by T.S. Eliot.

WHY IT WAS DEVELOPED

Ackoff argued that organisations were drowning in data while starving for understanding. The model was developed to make explicit the work required to convert data into the judgment on which action depends.

FIGURE 1 · DIKW PYRAMID

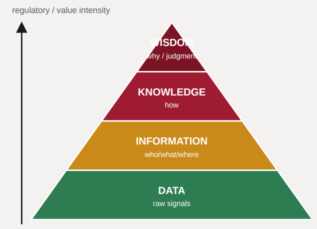

Figure 1. A schematic of the DIKW Pyramid as applied in BARBM312 (illustrative).

HOW IT IS USED

Each customer signal is located on the pyramid and traced upward: is it raw data, structured information, actionable knowledge, or decision-ready wisdom? Gaps reveal where value is being lost.

WHEN IT IS USED

At the start of any data or analytics initiative, when scoping what an AI system must actually deliver, and when auditing why dashboards fail to change behaviour.

BY WHOM IT IS USED

Data leaders, analytics teams, product managers and executives deciding what to instrument and what to ignore.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with treating 'big data' as inherently valuable, DIKW reframes value as the climb toward judgment — clarifying that AI accelerates the lower rungs but rarely supplies wisdom by itself.

RELATED FRAMEWORKS IN THIS COURSE

- **OODA Loop** · Slibrary 1
- **Decision-Intelligence Stack** · Slibrary 1
- **Data Value Chain** · Slibrary 1
- **Insight-to-Action Gap** · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Ackoff, R.L. (1989) 'From data to wisdom', Journal of Applied Systems Analysis, 16, pp. 3–9.

Rowley, J. (2007) 'The wisdom hierarchy', Journal of Information Science, 33(2), pp. 163–180.

OODA Loop

Observe → Orient → Decide → Act, at AI speed

PROFESSIONAL DEFINITION

The OODA Loop is a four-stage decision cycle — Observe, Orient, Decide, Act — that models how an actor adapts to a changing environment by continuously re-cycling through the four stages faster than rivals.

WHO DEVELOPED IT

Developed by United States Air Force colonel and military strategist John Boyd in the 1970s–80s from his study of fighter-pilot decision-making and the dynamics of conflict.

WHY IT WAS DEVELOPED

Boyd sought to explain why outnumbered pilots could prevail: not by superior force, but by completing the decision cycle faster, repeatedly disorienting the opponent.

FIGURE 1 · OODA LOOP

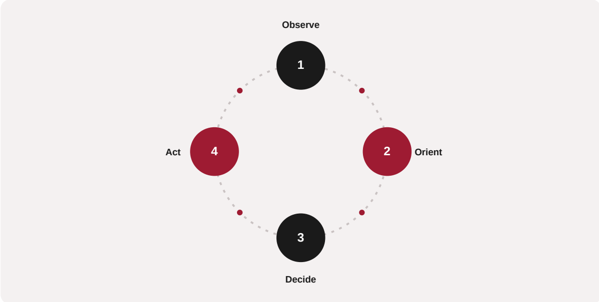

Figure 1. A schematic of the OODA Loop as applied in BARBM312 (illustrative).

HOW IT IS USED

Map the four stages to a real decision process, then look for delays. AI typically compresses Observe and Orient; the discipline is to keep Decide and Act human and fast.

WHEN IT IS USED

In fast-moving competitive or operational contexts — pricing, fraud, content, incident response — where the speed of re-decision matters more than the size of any single decision.

BY WHOM IT IS USED

Operations leaders, strategists, and anyone designing real-time or near-real-time AI decision systems.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Unlike linear plan-then-execute models, OODA is explicitly cyclical and adversarial, treating tempo itself as a source of advantage — a natural fit for AI-accelerated decision loops.

RELATED FRAMEWORKS IN THIS COURSE

- **DIKW Pyramid** · Slibrary 1
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- **Data Value Chain** · Slibrary 1
- **Insight-to-Action Gap** · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Boyd, J. (1986) 'Patterns of Conflict' (unpublished briefing).

Osinga, F. (2007) Science, Strategy and War: The Strategic Theory of John Boyd. London: Routledge.

Decision-Intelligence Stack

Signals, models, judgment, action layered for growth

PROFESSIONAL DEFINITION

The Decision-Intelligence Stack is a layered model that arranges raw signals, models, human judgment and action into a pipeline, making explicit each step required to turn data into a decision that creates business value.

WHO DEVELOPED IT

A practitioner synthesis associated with the emerging discipline of decision intelligence, drawing on Cassie Kozyrkov's framing of decision-making as an engineering problem.

WHY IT WAS DEVELOPED

It was developed to counter the tendency to invest in models while neglecting the surrounding pipeline — the data feeding them and the human action that follows.

FIGURE 1 · DECISION-INTELLIGENCE STACK

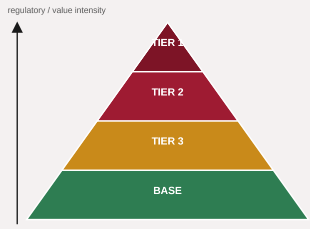

Figure 1. A schematic of the Decision-Intelligence Stack as applied in BARBM312 (illustrative).

HOW IT IS USED

Each layer is audited for fitness: are the signals trustworthy, the models appropriate, the judgment well-framed, and the action actually taken? The weakest layer caps the value of the whole.

WHEN IT IS USED

When designing or diagnosing an end-to-end AI decision system, especially where good models fail to produce business outcomes.

BY WHOM IT IS USED

Product owners, data-science leads and executives responsible for AI return on investment.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It shifts attention from the model in isolation to the full decision pipeline, treating judgment and action as first-class components rather than afterthoughts.

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- **Data Value Chain** · Slibrary 1
- **Insight-to-Action Gap** · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Kozyrkov, C. (2019) 'Introduction to Decision Intelligence', Google Cloud.

Pratt, L. and Malcolm, N. (2022) The Decision Intelligence Handbook. Sebastopol: O'Reilly.

Data Value Chain

Capture → store → process → analyse → act

PROFESSIONAL DEFINITION

The Data Value Chain describes the sequence of activities — capture, store, process, analyse and act — through which raw data is progressively transformed into business value, mirroring Porter's value-chain logic applied to data.

WHO DEVELOPED IT

Adapted from Michael Porter's value chain (1985) by data-management scholars and bodies such as the OECD, which formalised a data value cycle.

WHY IT WAS DEVELOPED

It was developed to help organisations see data handling as a chain of value-adding steps rather than a cost centre, and to locate where value leaks.

FIGURE 1 · DATA VALUE CHAIN

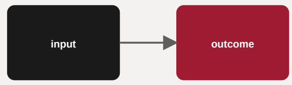

Figure 1. A schematic of the Data Value Chain as applied in BARBM312 (illustrative).

HOW IT IS USED

Each link is examined for cost, quality and value contribution; investment is directed to the link that most constrains the eventual decision.

WHEN IT IS USED

During data-strategy design, platform investment decisions, and when justifying data spend to finance.

BY WHOM IT IS USED

Chief data officers, data engineers and finance partners assessing the return on data infrastructure.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

By borrowing Porter's margin logic, it reframes data from a static asset to a flow whose value depends on the whole chain, not any single tool.

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- **Insight-to-Action Gap** · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Porter, M.E. (1985) Competitive Advantage. New York: Free Press.

OECD (2015) Data-Driven Innovation: Big Data for Growth and Well-Being. Paris: OECD.

Insight-to-Action Gap

Why analysis rarely changes behaviour

PROFESSIONAL DEFINITION

The Insight-to-Action Gap is the recurring shortfall between the insights an organisation generates and the decisions or behaviours it actually changes — the distance between knowing and doing.

WHO DEVELOPED IT

Articulated by Jeffrey Pfeffer and Robert Sutton in The Knowing-Doing Gap (2000) and echoed across analytics-adoption research.

WHY IT WAS DEVELOPED

Pfeffer and Sutton observed that firms accumulate knowledge yet fail to act on it; the concept names this failure so it can be managed rather than assumed away.

FIGURE 1 · INSIGHT-TO-ACTION GAP

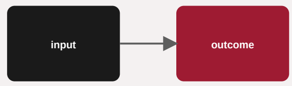

Figure 1. A schematic of the Insight-to-Action Gap as applied in BARBM312 (illustrative).

HOW IT IS USED

Insights are tracked through to the decisions they were meant to change; where no behaviour shifts, the analysis is treated as waste and the cause diagnosed.

WHEN IT IS USED

When reviewing analytics or AI programmes that produce reports nobody acts on, and when designing for adoption rather than mere production of insight.

BY WHOM IT IS USED

Analytics leaders, change managers and executives accountable for realising value from data.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It relocates the bottleneck from analysis to action, challenging the assumption that better insight automatically yields better outcomes.

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- **Decision-Intelligence Stack** · Slibrary 1
- **Data Value Chain** · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Pfeffer, J. and Sutton, R.I. (2000) The Knowing-Doing Gap. Boston: HBS Press.

Davenport, T.H. (2013) 'Keep up with your quants', Harvard Business Review, 91(7).

Customer Journey Map

End-to-end touchpoints from awareness to advocacy

PROFESSIONAL DEFINITION

A Customer Journey Map is a visual representation of the end-to-end experience a customer has with an organisation, charting stages, touchpoints, emotions and pain points from first awareness through to advocacy.

WHO DEVELOPED IT

Rooted in service-design and design-thinking practice, popularised through the 2000s by consultancies and the work of design firms such as IDEO.

WHY IT WAS DEVELOPED

It was developed to replace inside-out, org-chart thinking with an outside-in view that follows the customer across silos, surfacing breakdowns that no single department owns.

FIGURE 1 · CUSTOMER JOURNEY MAP

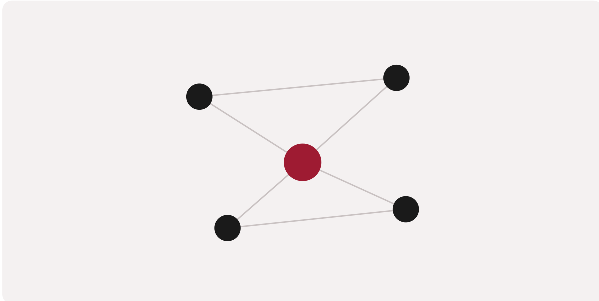

Figure 1. A schematic of the Customer Journey Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Stages are laid along a timeline; at each, touchpoints, customer goals, emotions and friction are recorded. AI opportunities are mapped where data or personalisation could remove friction.

WHEN IT IS USED

During experience redesign, before personalisation initiatives, and when diagnosing churn or low conversion.

BY WHOM IT IS USED

CX and marketing teams, service designers and product managers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Unlike funnel metrics that aggregate behaviour, the journey map preserves the lived sequence and emotion, revealing where AI should intervene and where it should not.

RELATED FRAMEWORKS IN THIS COURSE

- **Moments of Truth** · Slibrary 2
- **Engagement Flywheel** · Slibrary 2
- **Personalisation Ladder** · Slibrary 2
- **Jobs To Be Done** · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

Stickdorn, M. and Schneider, J. (2011) This Is Service Design Thinking. Amsterdam: BIS.

Richardson, A. (2010) 'Using customer journey maps to improve customer experience', Harvard Business Review.

Moments of Truth

Carlzon's high-stakes service encounters

PROFESSIONAL DEFINITION

Moments of Truth are the brief, high-stakes interactions where a customer forms a decisive impression of a brand — points at which the relationship is won or lost in seconds.

WHO DEVELOPED IT

Coined by Jan Carlzon, former CEO of Scandinavian Airlines, in his 1987 book of the same name; later extended by Procter & Gamble and Google's 'Zero Moment of Truth'.

WHY IT WAS DEVELOPED

Carlzon used the idea to refocus a struggling airline on the frontline encounters that customers actually remember, rather than internal process.

FIGURE 1 · MOMENTS OF TRUTH

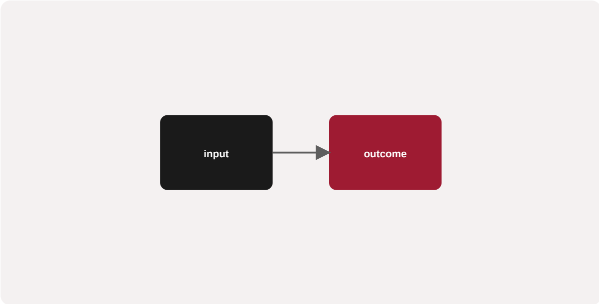

Figure 1. A schematic of the Moments of Truth as applied in BARBM312 (illustrative).

HOW IT IS USED

The journey is scanned for the handful of interactions that disproportionately shape perception; resources and AI assistance are concentrated there.

WHEN IT IS USED

When prioritising limited CX investment, designing service recovery, and deciding where automation helps versus harms.

BY WHOM IT IS USED

Service leaders, CX designers and frontline managers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It directs attention away from uniform investment across all touchpoints toward the decisive few — a sharp prioritisation lens for AI-enabled service.

RELATED FRAMEWORKS IN THIS COURSE

- **Customer Journey Map** · Slibrary 2
- **Engagement Flywheel** · Slibrary 2
- **Personalisation Ladder** · Slibrary 2
- **Jobs To Be Done** · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

Carlzon, J. (1987) Moments of Truth. Cambridge: Ballinger.

Lecinski, J. (2011) Winning the Zero Moment of Truth. Google.

Engagement Flywheel

Attract → engage → delight → repeat

PROFESSIONAL DEFINITION

The Engagement Flywheel is a cyclical model in which attracting, engaging and delighting customers feed one another, so that satisfied customers generate the energy that attracts the next cohort.

WHO DEVELOPED IT

Popularised by HubSpot (notably Brian Halligan) as a successor to the linear marketing funnel, drawing conceptually on Jim Collins's flywheel metaphor.

WHY IT WAS DEVELOPED

It was developed to capture the compounding, self-reinforcing nature of modern customer growth, where word of mouth and retention drive acquisition.

FIGURE 1 · ENGAGEMENT FLYWHEEL

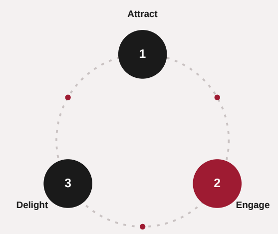

Figure 1. A schematic of the Engagement Flywheel as applied in BARBM312 (illustrative).

HOW IT IS USED

Each phase is measured for the force it adds and the friction it removes; investment targets whichever reduces friction most, accelerating the whole wheel.

WHEN IT IS USED

When designing growth strategy for products with strong referral or retention dynamics, and when moving beyond funnel thinking.

BY WHOM IT IS USED

Growth, marketing and product leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Unlike the funnel, which discards the customer at purchase, the flywheel treats existing customers as the primary engine of future growth.

RELATED FRAMEWORKS IN THIS COURSE

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- **Moments of Truth** · Slibrary 2
- **Personalisation Ladder** · Slibrary 2
- **Jobs To Be Done** · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

Halligan, B. and Shah, D. (2018) 'The flywheel', HubSpot.
Collins, J. (2001) Good to Great. New York: HarperBusiness.

Personalisation Ladder

Segment → recommend → predict → 1-to-1

PROFESSIONAL DEFINITION

The Personalisation Ladder is a maturity model describing increasing degrees of tailoring — from broad segmentation, through recommendation and prediction, to true one-to-one personalisation.

WHO DEVELOPED IT

A synthesis from CRM and marketing-technology practice, building on Peppers and Rogers's one-to-one marketing.

WHY IT WAS DEVELOPED

It was developed to help firms locate their current personalisation maturity and avoid over-reaching into intrusive tailoring before the data and trust foundations exist.

FIGURE 1 · PERSONALISATION LADDER

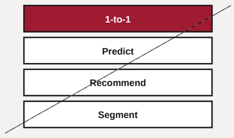

Figure 1. A schematic of the Personalisation Ladder as applied in BARBM312 (illustrative).

HOW IT IS USED

The organisation places its current capability on a rung and identifies the data, consent and modelling required to climb without eroding trust.

WHEN IT IS USED

When planning personalisation investment and when balancing relevance against the risk of 'creepy' targeting.

BY WHOM IT IS USED

CX, CRM and data teams, and privacy or compliance partners.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames personalisation as a graduated capability with a trust ceiling, rather than a switch to be flipped on.

RELATED FRAMEWORKS IN THIS COURSE

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- **Moments of Truth** · Slibrary 2
- **Engagement Flywheel** · Slibrary 2
- **Jobs To Be Done** · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

Peppers, D. and Rogers, M. (1993) The One to One Future. New York: Doubleday.

Arora, N. et al. (2008) 'Putting one-to-one marketing to work', Marketing Letters, 19.

Jobs To Be Done

What the customer hires the product to do

PROFESSIONAL DEFINITION

Jobs To Be Done (JTBD) is a theory holding that customers 'hire' products and services to make progress on a specific job in a particular circumstance, so innovation should target the job rather than the demographic.

WHO DEVELOPED IT

Developed by Clayton Christensen and colleagues (notably Bob Moesta and Tony Ulwick's related outcome-driven variant), articulated in Competing Against Luck (2016).

WHY IT WAS DEVELOPED

Christensen argued that correlation-based segmentation misled innovators; framing demand as a job clarifies why customers switch and what would make them switch to you.

FIGURE 1 · JOBS TO BE DONE

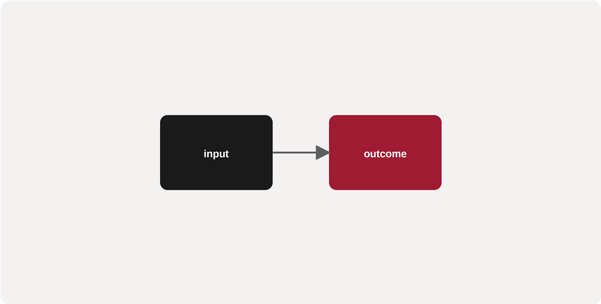

Figure 1. A schematic of the Jobs To Be Done as applied in BARBM312 (illustrative).

HOW IT IS USED

The team identifies the functional, emotional and social job, the circumstances that trigger it, and the obstacles the current solution leaves unresolved.

WHEN IT IS USED

During product strategy, opportunity sensing and when AI features risk being built for activity rather than genuine customer progress.

BY WHOM IT IS USED

Product, innovation and marketing leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It replaces demographic and feature-led thinking with a causal account of demand, sharpening where AI should add value.

RELATED FRAMEWORKS IN THIS COURSE

- **Customer Journey Map** · Slibrary 2
- **Moments of Truth** · Slibrary 2
- **Engagement Flywheel** · Slibrary 2
- **Personalisation Ladder** · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

Christensen, C.M. et al. (2016) Competing Against Luck. New York: HarperBusiness.

Ulwick, A. (2005) What Customers Want. New York: McGraw-Hill.

Embedded vs. AI-Native 2×2

Bolt-on intelligence vs. built-in intelligence

PROFESSIONAL DEFINITION

The Embedded vs. AI-Native matrix contrasts products that bolt AI onto an existing architecture with those designed around AI from the ground up, mapping the depth of intelligence against the legacy of the underlying system.

WHO DEVELOPED IT

A contemporary strategy framing arising from the 2020s debate over how incumbents should respond to generative AI, discussed widely by venture and product strategists.

WHY IT WAS DEVELOPED

It was developed to help leaders decide whether to retrofit AI into legacy products or rebuild, clarifying that the two paths change what decisions are even possible.

FIGURE 1 · EMBEDDED VS. AI-NATIVE 2×2

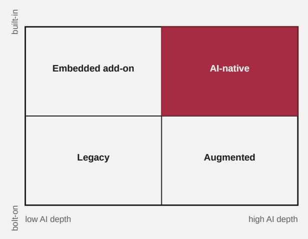

Figure 1. A schematic of the Embedded vs. AI-Native 2×2 as applied in BARBM312 (illustrative).

HOW IT IS USED

A product is placed on the grid by AI depth and architectural legacy; the position implies a different investment, talent and risk profile.

WHEN IT IS USED

When setting product architecture strategy and prioritising modernisation versus rebuild.

BY WHOM IT IS USED

Product and engineering leaders, CTOs and strategists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It names a distinction obscured by generic 'AI adoption' talk: bolt-on intelligence lifts information quality, while AI-native design changes the product's possibility space.

RELATED FRAMEWORKS IN THIS COURSE

- **Build · Buy · Partner** · Slibrary 3
- **Legacy Modernisation Curve** · Slibrary 3
- **Value Proposition Canvas** · Slibrary 3
- **Data Network Effects** · Slibrary 3

SOURCES (HARVARD REFERENCING STYLE)

Bornstein, M., Appenzeller, G. and Casado, M. (2023) 'Emerging architectures for LLM applications', a16z.

Furr, N. and Shipilov, A. (2019) 'Digital doesn't have to be disruptive', Harvard Business Review.

Build · Buy · Partner

Sourcing the intelligence layer

PROFESSIONAL DEFINITION

The Build–Buy–Partner framework structures the sourcing decision for a capability: develop it internally, purchase a finished solution, or collaborate with a partner, according to strategic value and internal capability.

WHO DEVELOPED IT

A long-standing corporate-strategy heuristic formalised in make-or-buy and transaction-cost economics, extended to alliances by scholars such as Williamson and later strategy texts.

WHY IT WAS DEVELOPED

It was developed to discipline sourcing decisions that are often made by default, ensuring scarce build capacity is reserved for what is genuinely strategic.

FIGURE 1 · BUILD · BUY · PARTNER

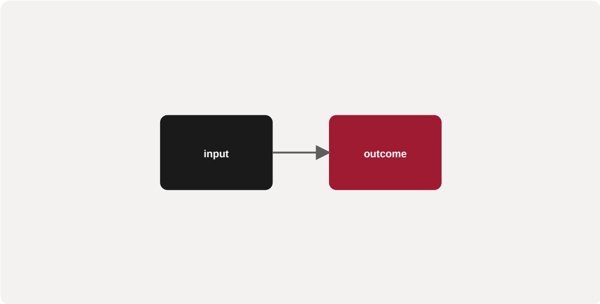

Figure 1. A schematic of the Build · Buy · Partner as applied in BARBM312 (illustrative).

HOW IT IS USED

A capability is scored on strategic value and on internal capability; the resulting position recommends build, buy or partner, with AI components assessed the same way.

WHEN IT IS USED

When sourcing AI models, data or tooling, and when deciding what to own versus rent in the AI stack.

BY WHOM IT IS USED

CTOs, procurement, and corporate-development teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Applied to AI, it clarifies that calling a commodity model is 'buy' while a proprietary data advantage is worth 'build' — preventing undifferentiated in-house effort.

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- **Legacy Modernisation Curve** · Slibrary 3
- **Value Proposition Canvas** · Slibrary 3
- **Data Network Effects** · Slibrary 3

SOURCES (HARVARD REFERENCING STYLE)

Williamson, O.E. (1985) The Economic Institutions of Capitalism. New York: Free Press.

Capron, L. and Mitchell, W. (2012) Build, Borrow, or Buy. Boston: HBR Press.

Legacy Modernisation Curve

Optimise old systems vs. rebuild

PROFESSIONAL DEFINITION

The Legacy Modernisation Curve depicts the trade-off between continuing to optimise an existing system and rebuilding it, showing how returns to optimisation flatten while a rebuild offers a higher but riskier trajectory.

WHO DEVELOPED IT

A practitioner model from enterprise-architecture and digital-transformation practice, related to technical-debt and application-portfolio theory.

WHY IT WAS DEVELOPED

It was developed to help organisations recognise when incremental improvement of a legacy estate has exhausted its returns and a step change is warranted.

FIGURE 1 · LEGACY MODERNISATION CURVE

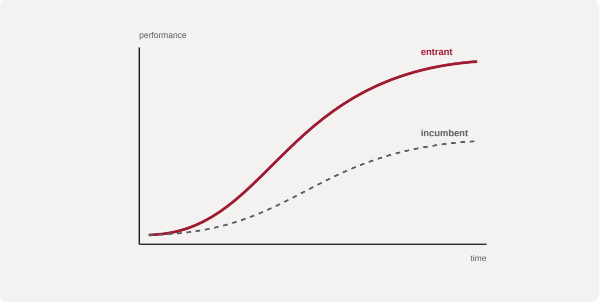

Figure 1. A schematic of the Legacy Modernisation Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

Current systems are plotted on the curve; diminishing optimisation returns are weighed against the cost, risk and upside of modernisation or AI-native rebuild.

WHEN IT IS USED

When deciding whether to layer AI onto legacy systems or re-platform, and during transformation planning.

BY WHOM IT IS USED

Enterprise architects, CIOs and transformation leads.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the optimise-versus-rebuild decision explicit and time-dependent, countering the inertia of perpetual incremental patching.

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- **Value Proposition Canvas** · Slibrary 3
- **Data Network Effects** · Slibrary 3

SOURCES (HARVARD REFERENCING STYLE)

Gartner (2019) '7 Options to Modernize Legacy Systems'.

Ross, J.W., Weill, P. and Robertson, D. (2006) Enterprise Architecture as Strategy. Boston: HBR Press.

Value Proposition Canvas

Gains, pains, jobs vs. products & services

PROFESSIONAL DEFINITION

The Value Proposition Canvas is a tool that maps a customer profile — jobs, pains and gains — against a value map of products, pain relievers and gain creators, to test the fit between what is offered and what is needed.

WHO DEVELOPED IT

Created by Alexander Osterwalder, Yves Pigneur and colleagues at Strategyzer, building on the Business Model Canvas (2010) and detailed in Value Proposition Design (2014).

WHY IT WAS DEVELOPED

It was developed to make the often-implicit logic of a value proposition explicit and testable, reducing the risk of building things customers do not want.

FIGURE 1 · VALUE PROPOSITION CANVAS

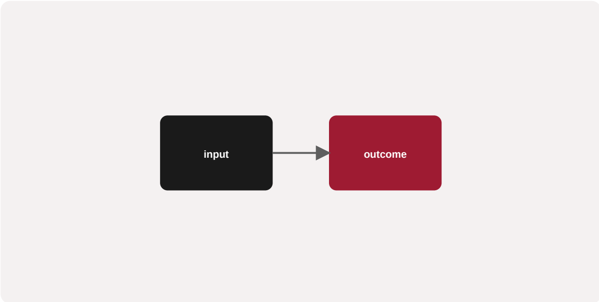

Figure 1. A schematic of the Value Proposition Canvas as applied in BARBM312 (illustrative).

HOW IT IS USED

The customer profile is completed first; the value map is then designed to relieve the most severe pains and create the most wanted gains, and fit is checked.

WHEN IT IS USED

During product design, opportunity validation and when shaping AI features around real customer jobs.

BY WHOM IT IS USED

Product, innovation and marketing teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It operationalises Jobs To Be Done into a visual, testable fit between offer and need, widely adopted as a standard innovation tool.

RELATED FRAMEWORKS IN THIS COURSE

- **Embedded vs. AI-Native 2×2** · Slibrary 3
- **Build · Buy · Partner** · Slibrary 3
- **Legacy Modernisation Curve** · Slibrary 3
- **Data Network Effects** · Slibrary 3

SOURCES (HARVARD REFERENCING STYLE)

Osterwalder, A. et al. (2014) Value Proposition Design. Hoboken: Wiley.

Osterwalder, A. and Pigneur, Y. (2010) Business Model Generation. Hoboken: Wiley.

Data Network Effects

More use → more data → better product

PROFESSIONAL DEFINITION

Data Network Effects describe a dynamic in which a product improves as more usage generates more data, which improves the product and attracts more users — a compounding, data-driven moat.

WHO DEVELOPED IT

Articulated by venture investors and platform scholars, notably Andrei Hagiu and Julian Wright, who cautioned that not all data improves products.

WHY IT WAS DEVELOPED

The concept was developed to distinguish genuine, defensible data advantages from the loose claim that 'more data wins', which often fails to compound.

FIGURE 1 · DATA NETWORK EFFECTS

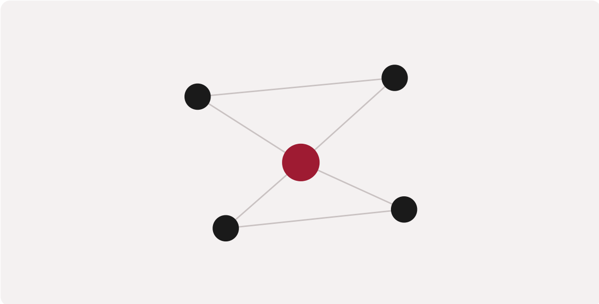

Figure 1. A schematic of the Data Network Effects as applied in BARBM312 (illustrative).

HOW IT IS USED

The team tests whether additional data actually improves the product and whether that improvement is hard for rivals to replicate; only then is a data moat real.

WHEN IT IS USED

When assessing AI product defensibility and investor due diligence on data-based advantage.

BY WHOM IT IS USED

Founders, product leaders and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It refines the network-effects canon by isolating when data — not just users — creates compounding advantage, puncturing naïve 'big data' moats.

RELATED FRAMEWORKS IN THIS COURSE

- **Embedded vs. AI-Native 2×2** · Slibrary 3
- **Build · Buy · Partner** · Slibrary 3
- **Legacy Modernisation Curve** · Slibrary 3
- **Value Proposition Canvas** · Slibrary 3

SOURCES (HARVARD REFERENCING STYLE)

Hagiu, A. and Wright, J. (2020) 'When data creates competitive advantage', Harvard Business Review, 98(1).

Parker, G., Van Alstyne, M. and Choudary, S.P. (2016) Platform Revolution. New York: Norton.

White-Space Map

Unmet needs between existing offers

PROFESSIONAL DEFINITION

A White-Space Map plots customer needs against served segments to expose unmet needs — the 'white space' between existing offerings where new value can be created.

WHO DEVELOPED IT

A strategy-and-innovation technique associated with growth strategy and with Mark Johnson's work on white space and business-model innovation.

WHY IT WAS DEVELOPED

It was developed to direct innovation toward genuinely unserved demand rather than crowded, well-contested territory.

FIGURE 1 · WHITE-SPACE MAP

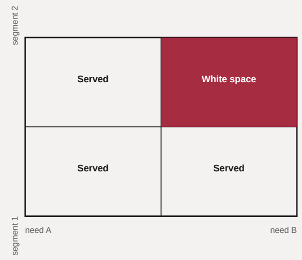

Figure 1. A schematic of the White-Space Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Needs and segments are crossed in a grid; cells that are important to customers yet poorly served are flagged as white-space opportunities for AI-enabled offerings.

WHEN IT IS USED

During opportunity sensing, portfolio planning and market-entry decisions.

BY WHOM IT IS USED

Strategy, innovation and corporate-development teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It systematises the search for non-obvious opportunity, complementing Blue Ocean thinking with a structured needs-by-segment lens.

RELATED FRAMEWORKS IN THIS COURSE

- **Demand-Signal Radar** · Slibrary 4
- **Blue Ocean Strategy Canvas** · Slibrary 4
- **Trend → Opportunity Funnel** · Slibrary 4
- **Three Horizons** · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Johnson, M.W. (2010) Seizing the White Space. Boston: HBR Press.
Christensen, C.M. and Raynor, M. (2003) The Innovator's Solution. Boston: HBS Press.

Demand-Signal Radar

Reading emerging consumer needs from data

PROFESSIONAL DEFINITION

The Demand-Signal Radar is a visual scan of emerging consumer needs detected from data signals across multiple dimensions, helping an organisation sense shifts in demand before they become obvious.

WHO DEVELOPED IT

A practitioner construct from demand-sensing and market-intelligence practice, related to environmental-scanning and weak-signal detection methods.

WHY IT WAS DEVELOPED

It was developed to help firms detect early, faint indicators of changing demand that aggregate metrics miss until it is too late to respond.

FIGURE 1 · DEMAND-SIGNAL RADAR

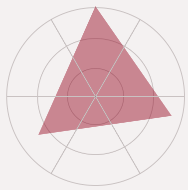

Figure 1. A schematic of the Demand-Signal Radar as applied in BARBM312 (illustrative).

HOW IT IS USED

Signal dimensions are arranged on a radar; AI-processed data raises or lowers each, and clusters of rising signals point to emerging opportunities.

WHEN IT IS USED

During trend analysis, product roadmapping and competitive sensing.

BY WHOM IT IS USED

Market-intelligence, strategy and product teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It turns scattered weak signals into a single, scannable picture, exploiting AI's ability to process diverse data into early demand indicators.

RELATED FRAMEWORKS IN THIS COURSE

- **White-Space Map** · Slibrary 4
- **Blue Ocean Strategy Canvas** · Slibrary 4
- **Trend → Opportunity Funnel** · Slibrary 4
- **Three Horizons** · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Ansoff, H.I. (1975) 'Managing strategic surprise by response to weak signals', California Management Review, 18(2).

Day, G.S. and Schoemaker, P.J.H. (2006) Peripheral Vision. Boston: HBR Press.

Blue Ocean Strategy Canvas

Create uncontested market space

PROFESSIONAL DEFINITION

The Strategy Canvas is the central diagnostic of Blue Ocean Strategy, plotting how competitors invest across the factors an industry competes on, to reveal opportunities to create uncontested 'blue ocean' market space.

WHO DEVELOPED IT

Developed by W. Chan Kim and Renée Mauborgne of INSEAD in Blue Ocean Strategy (2005).

WHY IT WAS DEVELOPED

It was developed to move firms away from bloody 'red ocean' competition on the same factors toward value innovation that makes the competition irrelevant.

FIGURE 1 · BLUE OCEAN STRATEGY CANVAS

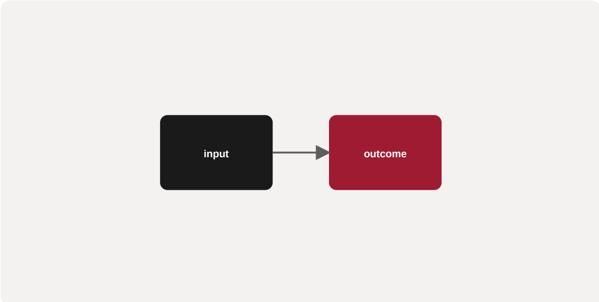

Figure 1. A schematic of the Blue Ocean Strategy Canvas as applied in BARBM312 (illustrative).

HOW IT IS USED

Competing factors are plotted on the horizontal axis and investment on the vertical; the firm then applies the eliminate-reduce-raise-create grid to redraw its value curve.

WHEN IT IS USED

During strategy formulation, market creation and when escaping commoditised competition with AI-enabled offerings.

BY WHOM IT IS USED

Senior strategists and executive teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes strategy from beating rivals to making them irrelevant through value innovation, supported by a distinctive visual diagnostic.

RELATED FRAMEWORKS IN THIS COURSE

- **White-Space Map** · Slibrary 4
- **Demand-Signal Radar** · Slibrary 4
- **Trend → Opportunity Funnel** · Slibrary 4
- **Three Horizons** · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Kim, W.C. and Mauborgne, R. (2005) Blue Ocean Strategy. Boston: HBS Press.

Kim, W.C. and Mauborgne, R. (2017) Blue Ocean Shift. New York: Hachette.

Trend → Opportunity Funnel

From weak signal to validated bet

PROFESSIONAL DEFINITION

The Trend-to-Opportunity Funnel is a staged filter that narrows a broad set of observed trends and weak signals down to a small set of validated opportunities worth investment.

WHO DEVELOPED IT

A foresight-practice construct combining environmental scanning with stage-gate filtering, used in corporate-foresight and innovation pipelines.

WHY IT WAS DEVELOPED

It was developed to convert the noise of trend-watching into disciplined opportunity selection, avoiding both trend-chasing and paralysis.

FIGURE 1 · TREND → OPPORTUNITY FUNNEL

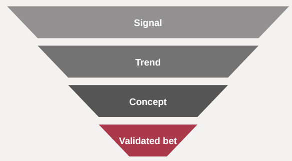

Figure 1. A schematic of the Trend → Opportunity Funnel as applied in BARBM312 (illustrative).

HOW IT IS USED

Trends enter the wide mouth; each stage applies tests of relevance, feasibility and strategic fit until a few validated bets remain.

WHEN IT IS USED

During innovation pipeline management and foresight-to-strategy translation.

BY WHOM IT IS USED

Foresight, innovation and strategy teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It imposes selection discipline on foresight, linking trend detection directly to a fundable opportunity portfolio.

RELATED FRAMEWORKS IN THIS COURSE

- **White-Space Map** · Slibrary 4
- **Demand-Signal Radar** · Slibrary 4
- **Blue Ocean Strategy Canvas** · Slibrary 4
- **Three Horizons** · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Rohrbeck, R. (2011) Corporate Foresight. Heidelberg: Physica-Verlag.

Cooper, R.G. (2008) 'Perspective: The Stage-Gate idea-to-launch process', Journal of Product Innovation Management, 25(3).

Three Horizons

Core, emerging and future growth horizons

PROFESSIONAL DEFINITION

The Three Horizons model frames growth as the simultaneous management of three time horizons: defending and extending the core (H1), building emerging businesses (H2), and creating options for the future (H3).

WHO DEVELOPED IT

Introduced by Mehrdad Baghai, Stephen Coley and David White of McKinsey in The Alchemy of Growth (1999).

WHY IT WAS DEVELOPED

It was developed to stop firms from over-investing in today's core at the expense of tomorrow's growth, balancing exploitation and exploration.

FIGURE 1 · THREE HORIZONS

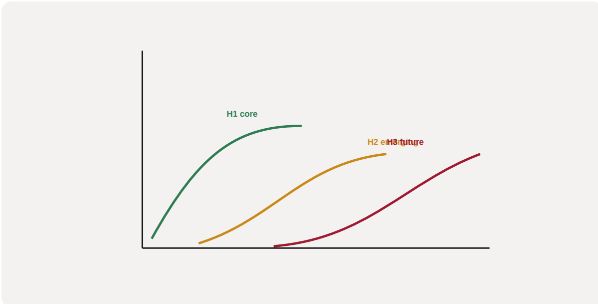

Figure 1. A schematic of the Three Horizons as applied in BARBM312 (illustrative).

HOW IT IS USED

Initiatives are sorted into the three horizons and resourced so that future growth is funded even while the core is optimised, with AI bets often spanning H2 and H3.

WHEN IT IS USED

During portfolio strategy, innovation governance and AI investment sequencing.

BY WHOM IT IS USED

Executives, strategy teams and innovation boards.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It institutionalises the parallel pursuit of present and future growth, countering the short-termism that lets disruptors win H3.

RELATED FRAMEWORKS IN THIS COURSE

- **White-Space Map** · Slibrary 4
- **Demand-Signal Radar** · Slibrary 4
- **Blue Ocean Strategy Canvas** · Slibrary 4
- **Trend → Opportunity Funnel** · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Baghai, M., Coley, S. and White, D. (1999) The Alchemy of Growth. Reading: Perseus.

McKinsey & Company (2009) 'Enduring Ideas: The three horizons of growth'.

Productivity J-Curve

Value dips before it climbs

PROFESSIONAL DEFINITION

The Productivity J-Curve describes how a general-purpose technology first depresses measured productivity — as firms invest in intangible complements — before delivering a steep, delayed payoff.

WHO DEVELOPED IT

Formalised by Erik Brynjolfsson, Daniel Rock and Chad Syverson (2021), building on Brynjolfsson's long study of the IT productivity paradox.

WHY IT WAS DEVELOPED

It was developed to explain why transformative technologies appear to fail in the statistics for years, reassuring decision-makers that the dip is part of the path.

FIGURE 1 · PRODUCTIVITY J-CURVE

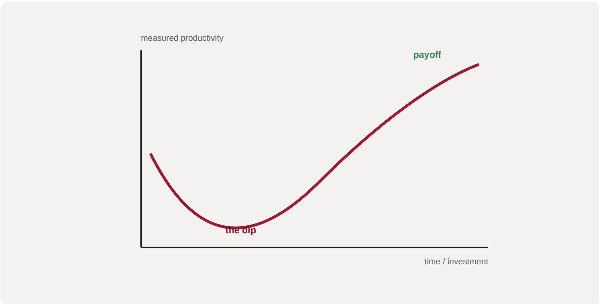

Figure 1. A schematic of the Productivity J-Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

An AI programme's measured productivity is read against the curve; investment in complementary intangibles is tracked to confirm a slope is being built.

WHEN IT IS USED

When boards lose patience with AI ROI, and when distinguishing a genuine lag from a genuine absence of value.

BY WHOM IT IS USED

CFOs, transformation leaders and investment committees.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It supplies a rigorous, evidence-based reason not to abandon AI programmes at the bottom of the dip, reframing the cost of complements as investment, not failure.

RELATED FRAMEWORKS IN THIS COURSE

- **Solow Paradox Revisited** · Slibrary 5
- **Adoption → Impact Lag** · Slibrary 5
- **Intangible Capital Iceberg** · Slibrary 5
- **Diffusion S-Curve** · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E., Rock, D. and Syverson, C. (2021) 'The Productivity J-Curve', American Economic Journal: Macroeconomics, 13(1), pp. 333–372.

Brynjolfsson, E. and McAfee, A. (2014) The Second Machine Age. New York: Norton.

Solow Paradox Revisited

AI everywhere except the statistics

PROFESSIONAL DEFINITION

The Solow Paradox is the observation that the computer age was 'everywhere except in the productivity statistics' — revisited today as AI investment outpaces measurable economic impact.

WHO DEVELOPED IT

Named for economist Robert Solow's 1987 remark; revisited by Brynjolfsson and others studying the AI era.

WHY IT WAS DEVELOPED

It was developed as a caution that visible technological adoption does not guarantee measured productivity gains, prompting investigation of why.

FIGURE 1 · SOLOW PARADOX REVISITED

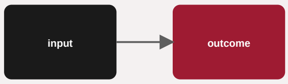

Figure 1. A schematic of the Solow Paradox Revisited as applied in BARBM312 (illustrative).

HOW IT IS USED

Apparent AI ubiquity is contrasted with productivity data; explanations — mismeasurement, lags, redistribution, mismanagement — are tested.

WHEN IT IS USED

When reconciling enthusiastic AI adoption with flat productivity figures.

BY WHOM IT IS USED

Economists, strategists and policy-minded executives.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames a recurring puzzle that disciplines hype, pushing analysts to look for lags and complements rather than assume immediate gains.

RELATED FRAMEWORKS IN THIS COURSE

- **Productivity J-Curve** · Slibrary 5
- **Adoption → Impact Lag** · Slibrary 5
- **Intangible Capital Iceberg** · Slibrary 5
- **Diffusion S-Curve** · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

Solow, R. (1987) 'We'd better watch out', New York Times Book Review, 12 July.

Brynjolfsson, E., Rock, D. and Syverson, C. (2019) 'Artificial intelligence and the modern productivity paradox', in The Economics of AI. Chicago: UCP.

Adoption → Impact Lag

Time between deploy and measurable gain

PROFESSIONAL DEFINITION

The Adoption-to-Impact Lag is the time delay between deploying a technology and observing its measurable economic impact, during which value accrues invisibly through complementary change.

WHO DEVELOPED IT

A concept grounded in technology-diffusion and productivity research, associated with Paul David's analysis of the electric dynamo and with Brynjolfsson's AI work.

WHY IT WAS DEVELOPED

It was developed to set realistic expectations: transformative technologies historically take years to show up in performance after adoption.

FIGURE 1 · ADOPTION → IMPACT LAG

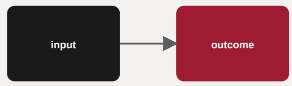

Figure 1. A schematic of the Adoption → Impact Lag as applied in BARBM312 (illustrative).

HOW IT IS USED

The expected lag is estimated and built into the business case so that early flat results are interpreted correctly rather than as failure.

WHEN IT IS USED

During AI business-case design and post-deployment review.

BY WHOM IT IS USED

Finance, strategy and programme leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the historically observed delay an explicit planning parameter, protecting good investments from premature judgment.

RELATED FRAMEWORKS IN THIS COURSE

- **Productivity J-Curve** · Slibrary 5
- **Solow Paradox Revisited** · Slibrary 5
- **Intangible Capital Iceberg** · Slibrary 5
- **Diffusion S-Curve** · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

David, P.A. (1990) 'The dynamo and the computer', American Economic Review, 80(2).

Brynjolfsson, E. and Hitt, L. (2003) 'Computing productivity', Review of Economics and Statistics, 85(4).

Intangible Capital Iceberg

Process & skill investment hidden below ROI

PROFESSIONAL DEFINITION

The Intangible Capital Iceberg depicts how the visible cost of a technology sits above a much larger submerged investment in intangibles — process redesign, skills, data and organisational change — required to realise its value.

WHO DEVELOPED IT

Drawn from the intangible-capital research of Brynjolfsson, Rock and Syverson and from Corrado, Hulten and Sichel's measurement of intangibles.

WHY IT WAS DEVELOPED

It was developed to explain why AI returns lag: the complementary intangible investment dwarfs the visible technology spend and is often unaccounted for.

FIGURE 1 · INTANGIBLE CAPITAL ICEBERG

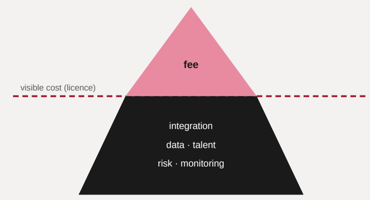

Figure 1. A schematic of the Intangible Capital Iceberg as applied in BARBM312 (illustrative).

HOW IT IS USED

The technology's visible cost is placed above the waterline; the required intangible investment below it is named and budgeted.

WHEN IT IS USED

When building AI business cases and explaining why adoption alone yields little.

BY WHOM IT IS USED

CFOs, transformation leaders and analysts.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It visualises the hidden majority of value-creating investment, countering the assumption that buying the technology is the main cost.

RELATED FRAMEWORKS IN THIS COURSE

- **Productivity J-Curve** · Slibrary 5
- **Solow Paradox Revisited** · Slibrary 5
- **Adoption → Impact Lag** · Slibrary 5
- **Diffusion S-Curve** · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

Corrado, C., Hulten, C. and Sichel, D. (2009) 'Intangible capital and U.S. economic growth', Review of Income and Wealth, 55(3).

Brynjolfsson, E., Rock, D. and Syverson, C. (2021) 'The Productivity J-Curve', AEJ: Macro, 13(1).

Diffusion S-Curve

Innovators to laggards over time

PROFESSIONAL DEFINITION

The Diffusion S-Curve charts how an innovation is adopted over time, from a slow start among innovators, through rapid mainstream uptake, to saturation among laggards, tracing an S-shaped cumulative curve.

WHO DEVELOPED IT

Formalised by Everett Rogers in Diffusion of Innovations (1962), building on earlier rural-sociology adoption studies.

WHY IT WAS DEVELOPED

It was developed to explain the social process of adoption and to identify the distinct adopter categories a new technology must win in sequence.

FIGURE 1 · DIFFUSION S-CURVE

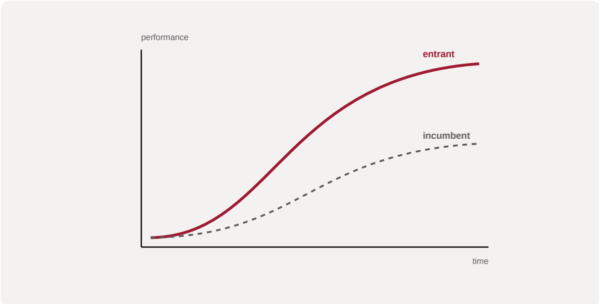

Figure 1. A schematic of the Diffusion S-Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

An innovation's current adoption is located on the curve; strategy is tailored to the next adopter group rather than treating the market as homogeneous.

WHEN IT IS USED

When forecasting AI adoption, timing market entry and segmenting go-to-market.

BY WHOM IT IS USED

Product marketers, strategists and forecasters.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames adoption as a staged social process rather than a switch, underpinning later models such as Crossing the Chasm.

RELATED FRAMEWORKS IN THIS COURSE

- **Productivity J-Curve** · Slibrary 5
- **Solow Paradox Revisited** · Slibrary 5
- **Adoption → Impact Lag** · Slibrary 5
- **Intangible Capital Iceberg** · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

Rogers, E.M. (1962) Diffusion of Innovations. New York: Free Press.

Bass, F.M. (1969) 'A new product growth model for consumer durables', Management Science, 15(5).

Value-Chain Heatmap

Where AI lifts each activity

PROFESSIONAL DEFINITION

A Value-Chain Heatmap overlays the intensity of a technology's potential impact onto Porter's value-chain activities, showing where AI can lift performance across primary and support functions.

WHO DEVELOPED IT

A practitioner application of Porter's value chain (1985) used by consultancies to prioritise AI investment across the enterprise.

WHY IT WAS DEVELOPED

It was developed to move AI investment from scattered experiments to a deliberate map of where the technology adds most value in the operating model.

FIGURE 1 · VALUE-CHAIN HEATMAP

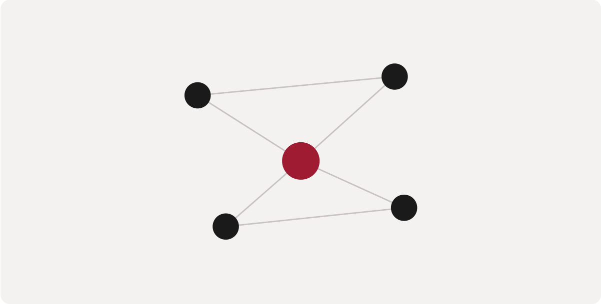

Figure 1. A schematic of the Value-Chain Heatmap as applied in BARBM312 (illustrative).

HOW IT IS USED

Each value-chain activity is shaded by AI opportunity; hotspots guide where to concentrate pilots and scaling.

WHEN IT IS USED

During enterprise AI strategy and investment prioritisation.

BY WHOM IT IS USED

Strategy, operations and transformation leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It combines a classic strategy map with AI-opportunity scoring, giving a single enterprise-wide view of where to act first.

RELATED FRAMEWORKS IN THIS COURSE

- **Function Benefit Matrix** · Slibrary 6
- **Automation Potential Grid** · Slibrary 6
- **AI Use-Case Portfolio** · Slibrary 6
- **Capability Uplift Map** · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

Porter, M.E. (1985) Competitive Advantage. New York: Free Press.
Bughin, J. et al. (2017) Artificial Intelligence: The Next Digital Frontier? McKinsey Global Institute.

Function Benefit Matrix

Marketing, ops, finance, HR gains mapped

PROFESSIONAL DEFINITION

The Function Benefit Matrix maps business functions against the type and size of benefit AI can deliver, distinguishing where AI mainly reduces cost from where it primarily drives revenue.

WHO DEVELOPED IT

A consulting construct used in enterprise AI benefit assessment, drawing on function-level studies such as McKinsey's State of AI surveys.

WHY IT WAS DEVELOPED

It was developed to give leaders a structured view of which functions to target first and what kind of value to expect from each.

FIGURE 1 · FUNCTION BENEFIT MATRIX

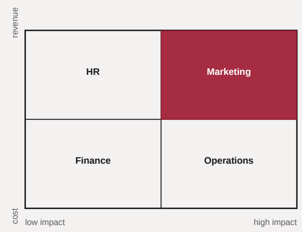

Figure 1. A schematic of the Function Benefit Matrix as applied in BARBM312 (illustrative).

HOW IT IS USED

Functions are scored on benefit type and magnitude; the matrix surfaces priority functions and clarifies the value narrative.

WHEN IT IS USED

During AI portfolio planning and benefit forecasting.

BY WHOM IT IS USED

Executives, function heads and transformation teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It separates cost benefits from revenue benefits by function, preventing a one-size-fits-all benefit story.

RELATED FRAMEWORKS IN THIS COURSE

- **Value-Chain Heatmap** · Slibrary 6
- **Automation Potential Grid** · Slibrary 6
- **AI Use-Case Portfolio** · Slibrary 6
- **Capability Uplift Map** · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

McKinsey & Company (2025) The State of AI in 2025. McKinsey Global Survey.

Davenport, T.H. and Ronanki, R. (2018) 'Artificial intelligence for the real world', Harvard Business Review, 96(1).

Automation Potential Grid

Task automatability vs. value

PROFESSIONAL DEFINITION

The Automation Potential Grid plots tasks by how automatable they are against how much value automating them would create, to prioritise where AI automation should be applied first.

WHO DEVELOPED IT

Grounded in the task-based automation research of Frey and Osborne and of Brynjolfsson, Mitchell and Rock on suitability for machine learning.

WHY IT WAS DEVELOPED

It was developed to move automation decisions from the whole-job level to the task level, where the real opportunity and risk lie.

FIGURE 1 · AUTOMATION POTENTIAL GRID

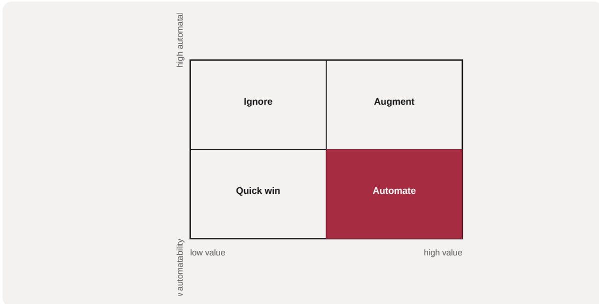

Figure 1. A schematic of the Automation Potential Grid as applied in BARBM312 (illustrative).

HOW IT IS USED

Tasks are scored on automatability and value; high-value, highly automatable tasks are prioritised, while high-value, low-automatability tasks are targeted for augmentation.

WHEN IT IS USED

During workforce and process redesign and AI use-case selection.

BY WHOM IT IS USED

Operations leaders, process owners and workforce planners.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes automation around tasks rather than jobs, aligning with evidence that AI augments far more tasks than it fully automates.

RELATED FRAMEWORKS IN THIS COURSE

Value-Chain Heatmap · Slibrary 6 | **Function Benefit Matrix** · Slibrary 6 | **AI Use-Case Portfolio** · Slibrary 6 | **Capability Uplift Map** · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E., Mitchell, T. and Rock, D. (2018) 'What can machines learn and what does it mean for occupations?', AEA Papers and Proceedings, 108.

Frey, C.B. and Osborne, M.A. (2017) 'The future of employment', Technological Forecasting and Social Change, 114.

AI Use-Case Portfolio

Quick wins vs. strategic bets

PROFESSIONAL DEFINITION

The AI Use-Case Portfolio arrays candidate AI initiatives by value and effort (or risk), separating quick wins from strategic bets and from initiatives to defer or avoid.

WHO DEVELOPED IT

A standard portfolio-prioritisation technique adapted to AI by consultancies and AI-strategy practitioners.

WHY IT WAS DEVELOPED

It was developed to impose discipline on a flood of AI ideas, balancing near-term wins with longer-term strategic investment.

FIGURE 1 · AI USE-CASE PORTFOLIO

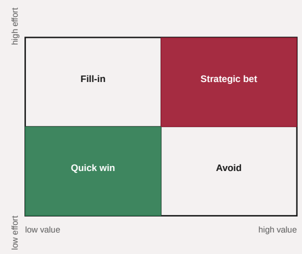

Figure 1. A schematic of the AI Use-Case Portfolio as applied in BARBM312 (illustrative).

HOW IT IS USED

Each use case is scored on value and effort; the resulting quadrants guide sequencing and resource allocation.

WHEN IT IS USED

During AI roadmap planning and governance review.

BY WHOM IT IS USED

AI programme leaders, product owners and steering committees.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It applies established portfolio logic to AI specifically, balancing momentum-building quick wins against transformational bets.

RELATED FRAMEWORKS IN THIS COURSE

- **Value-Chain Heatmap** · Slibrary 6
- **Function Benefit Matrix** · Slibrary 6
- **Automation Potential Grid** · Slibrary 6
- **Capability Uplift Map** · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

McKinsey & Company (2023) 'The economic potential of generative AI'.

Cooper, R.G., Edgett, S.J. and Kleinschmidt, E.J. (2001) Portfolio Management for New Products. Cambridge: Perseus.

Capability Uplift Map

Augmenting human capability per role

PROFESSIONAL DEFINITION

The Capability Uplift Map shows, role by role, how AI augments human capability — raising the performance or reach of a worker rather than replacing the role outright.

WHO DEVELOPED IT

Associated with the augmentation research of Wilson and Daugherty and the 'centaur' literature on human–AI collaboration.

WHY IT WAS DEVELOPED

It was developed to shift the conversation from replacement to augmentation, identifying where AI most amplifies human capability.

FIGURE 1 · CAPABILITY UPLIFT MAP

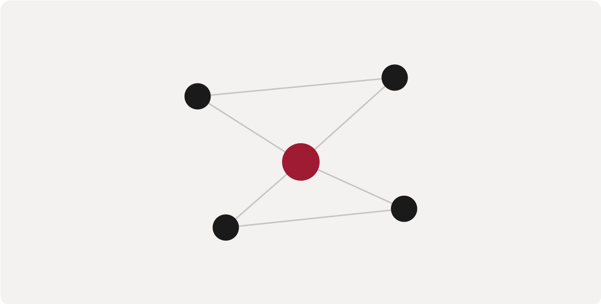

Figure 1. A schematic of the Capability Uplift Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Roles are mapped to the specific capabilities AI can uplift; investment targets augmentation that raises output and job quality together.

WHEN IT IS USED

During workforce strategy and AI-enabled role redesign.

BY WHOM IT IS USED

HR, operations and transformation leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It centres the augmentation thesis — humans plus AI outperform either alone — against a purely substitutive view of automation.

RELATED FRAMEWORKS IN THIS COURSE

- **Value-Chain Heatmap** · Slibrary 6
- **Function Benefit Matrix** · Slibrary 6
- **Automation Potential Grid** · Slibrary 6
- **AI Use-Case Portfolio** · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

Wilson, H.J. and Daugherty, P.R. (2018) 'Collaborative intelligence', Harvard Business Review, 96(4).

Daugherty, P.R. and Wilson, H.J. (2018) Human + Machine. Boston: HBR Press.

Cost-to-Serve Curve

Unit cost falling with automation

PROFESSIONAL DEFINITION

The Cost-to-Serve Curve shows how the unit cost of serving a customer or transaction falls as automation and AI absorb increasing volume, revealing scale economics.

WHO DEVELOPED IT

Rooted in cost-accounting and operations practice, with cost-to-serve analysis popularised in supply-chain and service-economics literature.

WHY IT WAS DEVELOPED

It was developed to quantify how automation changes the economics of service delivery, distinguishing fixed investment from falling marginal cost.

FIGURE 1 · COST-TO-SERVE CURVE

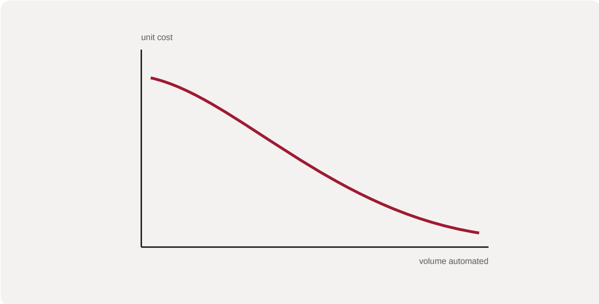

Figure 1. A schematic of the Cost-to-Serve Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

Unit cost is plotted against automated volume; the curve's shape informs pricing, capacity and the payback on automation investment.

WHEN IT IS USED

When evaluating automation ROI and designing scalable service models.

BY WHOM IT IS USED

Operations, finance and pricing leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the scale economics of AI-enabled service explicit, linking automation investment to declining unit cost.

RELATED FRAMEWORKS IN THIS COURSE

- **Process Cycle Efficiency** · Slibrary 7
- **Lean Waste Lens (AI)** · Slibrary 7
- **Throughput Accounting** · Slibrary 7
- **Effort-Value Reallocation** · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Cokins, G. (2001) Activity-Based Cost Management. Hoboken: Wiley.

Anderson, S.W. (2007) 'Managing costs and cost structure', Handbook of Management Accounting Research, 2.

Process Cycle Efficiency

Value-add time / total time

PROFESSIONAL DEFINITION

Process Cycle Efficiency is the ratio of value-added time to total cycle time in a process, quantifying how much of a process actually creates value versus waiting and rework.

WHO DEVELOPED IT

A core Lean metric, formalised in Lean and Six Sigma practice building on the Toyota Production System.

WHY IT WAS DEVELOPED

It was developed to expose hidden waste — the large share of cycle time spent waiting — that conventional efficiency measures miss.

FIGURE 1 · PROCESS CYCLE EFFICIENCY

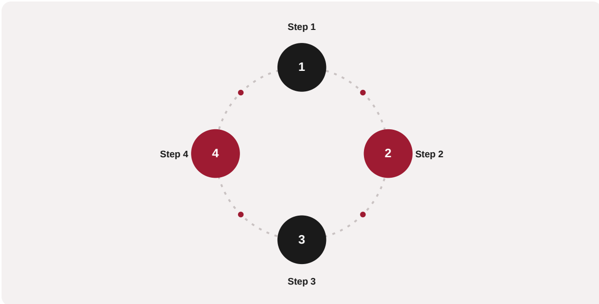

Figure 1. A schematic of the Process Cycle Efficiency as applied in BARBM312 (illustrative).

HOW IT IS USED

Value-added time is divided by total lead time; AI interventions are assessed by how much they raise the ratio rather than merely speed isolated steps.

WHEN IT IS USED

During process improvement and when targeting AI at genuine bottlenecks.

BY WHOM IT IS USED

Operations and continuous-improvement teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It focuses improvement on flow efficiency rather than local speed, ensuring AI is applied where it reduces total cycle time.

RELATED FRAMEWORKS IN THIS COURSE

- **Cost-to-Serve Curve** · Slibrary 7
- **Lean Waste Lens (AI)** · Slibrary 7
- **Throughput Accounting** · Slibrary 7
- **Effort-Value Reallocation** · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

George, M.L. (2002) Lean Six Sigma. New York: McGraw-Hill.

Womack, J.P. and Jones, D.T. (1996) Lean Thinking. New York: Simon & Schuster.

Lean Waste Lens (AI)

Eliminating the eight wastes with AI

PROFESSIONAL DEFINITION

The Lean Waste Lens applies the classic categories of waste (the eight 'mudas') to identify where AI can eliminate non-value-adding activity such as overprocessing, waiting, defects and unused talent.

WHO DEVELOPED IT

Built on the Toyota Production System and Lean thinking codified by Ohno, Womack and Jones, extended to AI-enabled operations.

WHY IT WAS DEVELOPED

It was developed to direct improvement effort at systematic categories of waste rather than ad hoc fixes.

FIGURE 1 · LEAN WASTE LENS (AI)

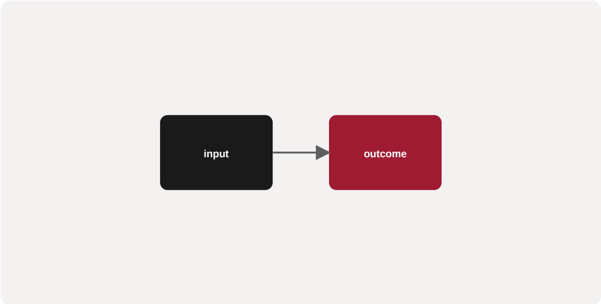

Figure 1. A schematic of the Lean Waste Lens (AI) as applied in BARBM312 (illustrative).

HOW IT IS USED

Each waste category is examined for AI remedies; AI is deployed where it removes waste rather than merely accelerating it.

WHEN IT IS USED

During operational improvement and AI use-case identification in processes.

BY WHOM IT IS USED

Operations, quality and continuous-improvement teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It channels AI toward proven waste categories, guarding against automating waste instead of eliminating it.

RELATED FRAMEWORKS IN THIS COURSE

- **Cost-to-Serve Curve** · Slibrary 7
- **Process Cycle Efficiency** · Slibrary 7
- **Throughput Accounting** · Slibrary 7
- **Effort-Value Reallocation** · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Ohno, T. (1988) Toyota Production System. Cambridge: Productivity Press.

Womack, J.P. and Jones, D.T. (1996) Lean Thinking. New York: Simon & Schuster.

Throughput Accounting

Bottleneck-driven gain measurement

PROFESSIONAL DEFINITION

Throughput Accounting evaluates decisions by their effect on throughput, inventory and operating expense, prioritising the system's constraint rather than local efficiencies.

WHO DEVELOPED IT

Developed by Eliyahu Goldratt as part of the Theory of Constraints, set out in The Goal (1984) and Throughput Accounting literature.

WHY IT WAS DEVELOPED

It was developed to counter cost-accounting distortions that reward local efficiency while ignoring the bottleneck that governs total output.

FIGURE 1 · THROUGHPUT ACCOUNTING

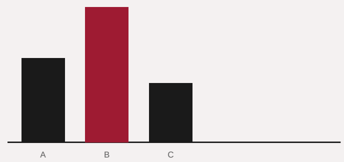

Figure 1. A schematic of the Throughput Accounting as applied in BARBM312 (illustrative).

HOW IT IS USED

Decisions, including AI investments, are judged by their effect on the constraint; improving a non-bottleneck is treated as illusory gain.

WHEN IT IS USED

When prioritising AI investment in operations and avoiding optimisation of non-bottlenecks.

BY WHOM IT IS USED

Operations and finance leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes performance around the system constraint, ensuring AI effort flows to the bottleneck that actually limits throughput.

RELATED FRAMEWORKS IN THIS COURSE

- **Cost-to-Serve Curve** · Slibrary 7
- **Process Cycle Efficiency** · Slibrary 7
- **Lean Waste Lens (AI)** · Slibrary 7
- **Effort-Value Reallocation** · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Goldratt, E.M. (1984) The Goal. Great Barrington: North River Press.

Corbett, T. (1998) Throughput Accounting. Great Barrington: North River Press.

Effort-Value Reallocation

Freed hours redeployed to high-value work

PROFESSIONAL DEFINITION

Effort-Value Reallocation tracks how time freed by automation is redeployed, distinguishing AI programmes that reinvest freed hours into high-value work from those that simply cut.

WHO DEVELOPED IT

A management construct emerging from the future-of-work and augmentation literature, related to job-redesign research.

WHY IT WAS DEVELOPED

It was developed to ensure that productivity gains from AI translate into higher-value output rather than being lost or merely banked as headcount cuts.

FIGURE 1 · EFFORT-VALUE REALLOCATION

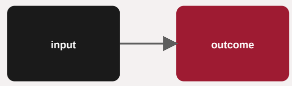

Figure 1. A schematic of the Effort-Value Reallocation as applied in BARBM312 (illustrative).

HOW IT IS USED

Freed effort is measured and explicitly reallocated to higher-value tasks; the value of reallocation is tracked alongside the cost saved.

WHEN IT IS USED

During AI-enabled workforce redesign and benefit realisation.

BY WHOM IT IS USED

HR, operations and transformation leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It treats reallocation of freed capacity as a managed decision, not a by-product, capturing the augmentation upside of automation.

RELATED FRAMEWORKS IN THIS COURSE

- **Cost-to-Serve Curve** · Slibrary 7
- **Process Cycle Efficiency** · Slibrary 7
- **Lean Waste Lens (AI)** · Slibrary 7
- **Throughput Accounting** · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Daugherty, P.R. and Wilson, H.J. (2018) Human + Machine. Boston: HBR Press.

Autor, D. (2015) 'Why are there still so many jobs?', Journal of Economic Perspectives, 29(3).

CAC : LTV Ratio

Acquisition cost vs. lifetime value

PROFESSIONAL DEFINITION

The CAC:LTV ratio compares the cost of acquiring a customer with the lifetime value that customer generates, a core test of whether a growth model is economically sustainable.

WHO DEVELOPED IT

A staple of SaaS and subscription economics, popularised by investors such as David Skok and by unit-economics practice.

WHY IT WAS DEVELOPED

It was developed to discipline growth spending, ensuring that acquisition cost is justified by the value a customer ultimately delivers.

FIGURE 1 · CAC : LTV RATIO

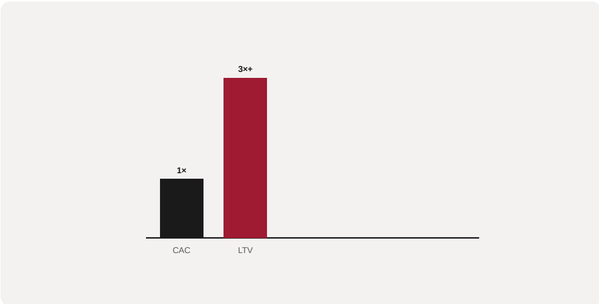

Figure 1. A schematic of the CAC : LTV Ratio as applied in BARBM312 (illustrative).

HOW IT IS USED

Lifetime value is divided by acquisition cost; a healthy ratio (commonly 3:1 or better) signals a fundable model, while AI is assessed by its effect on both terms.

WHEN IT IS USED

During growth-strategy and investment decisions and when evaluating AI's effect on acquisition or retention.

BY WHOM IT IS USED

Founders, growth leaders, finance and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It anchors growth in unit economics rather than vanity metrics, providing a single sustainability test for customer acquisition.

RELATED FRAMEWORKS IN THIS COURSE

- **Retention / Churn Cohort** · Slibrary 8
- **Pirate Metrics (AARRR)** · Slibrary 8
- **Net Revenue Retention** · Slibrary 8
- **Growth Loop** · Slibrary 8

SOURCES (HARVARD REFERENCING STYLE)

Skok, D. (2013) 'SaaS metrics 2.0', forEntrepreneurs.

Gupta, S., Lehmann, D.R. and Stuart, J.A. (2004) 'Valuing customers', Journal of Marketing Research, 41(1).

Retention / Churn Cohort

AI lift on customer retention over time

PROFESSIONAL DEFINITION

A Retention/Churn Cohort analysis tracks groups of customers acquired in the same period over time, revealing how retention decays and where AI-driven interventions lift it.

WHO DEVELOPED IT

A standard analytics technique in subscription and product analytics, formalised through cohort-analysis and customer-lifetime-value practice.

WHY IT WAS DEVELOPED

It was developed to expose retention dynamics hidden by aggregate metrics, isolating the effect of changes over the customer lifecycle.

FIGURE 1 · RETENTION / CHURN COHORT

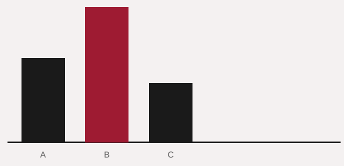

Figure 1. A schematic of the Retention / Churn Cohort as applied in BARBM312 (illustrative).

HOW IT IS USED

Cohorts are plotted over time; AI interventions (personalisation, churn prediction) are evaluated by their lift on later-period retention.

WHEN IT IS USED

When measuring AI's impact on retention and diagnosing churn.

BY WHOM IT IS USED

Growth, CX and analytics teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It provides a clean, time-aware view of retention that attributes change to interventions rather than noise.

RELATED FRAMEWORKS IN THIS COURSE

CAC : LTV Ratio · Slibrary 8
Pirate Metrics (AARRR) · Slibrary 8
Net Revenue Retention · Slibrary 8
Growth Loop · Slibrary 8

SOURCES (HARVARD REFERENCING STYLE)

Fader, P.S. and Hardie, B.G.S. (2009) 'Probability models for customer-base analysis', Journal of Interactive Marketing, 23(1).

Reichheld, F.F. (1996) The Loyalty Effect. Boston: HBS Press.

Pirate Metrics (AARRR)

Acquisition to revenue funnel

PROFESSIONAL DEFINITION

Pirate Metrics (AARRR) is a funnel of five growth stages — Acquisition, Activation, Retention, Referral and Revenue — that frames where to focus measurement and improvement in a growth model.

WHO DEVELOPED IT

Created by venture investor Dave McClure in 2007 to give startups a simple, actionable metrics framework.

WHY IT WAS DEVELOPED

It was developed to cut through vanity metrics and focus founders on the few stage-level numbers that actually drive growth.

FIGURE 1 · PIRATE METRICS (AARRR)

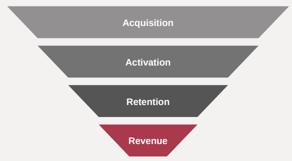

Figure 1. A schematic of the Pirate Metrics (AARRR) as applied in BARBM312 (illustrative).

HOW IT IS USED

Each stage is measured; the weakest stage becomes the focus of experimentation, with AI applied where it most improves stage conversion.

WHEN IT IS USED

During growth measurement and prioritisation for products and AI-enabled services.

BY WHOM IT IS USED

Founders, growth and product teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It distilled growth measurement into five memorable, actionable stages, becoming a default startup metrics language.

RELATED FRAMEWORKS IN THIS COURSE

CAC : LTV Ratio · Slibrary 8
Retention / Churn Cohort · Slibrary 8
Net Revenue Retention · Slibrary 8
Growth Loop · Slibrary 8

SOURCES (HARVARD REFERENCING STYLE)

McClure, D. (2007) 'Startup metrics for pirates', 500 Startups.

Croll, A. and Yoskovitz, B. (2013) Lean Analytics. Sebastopol: O'Reilly.

Net Revenue Retention

Expansion minus churn from AI features

PROFESSIONAL DEFINITION

Net Revenue Retention (NRR) measures the revenue retained and expanded from existing customers over a period, net of churn and contraction — a key indicator of compounding, durable growth.

WHO DEVELOPED IT

A metric central to modern SaaS and subscription analysis, emphasised by growth investors and operators.

WHY IT WAS DEVELOPED

It was developed to capture whether a customer base grows on its own through expansion, independent of new acquisition.

FIGURE 1 · NET REVENUE RETENTION

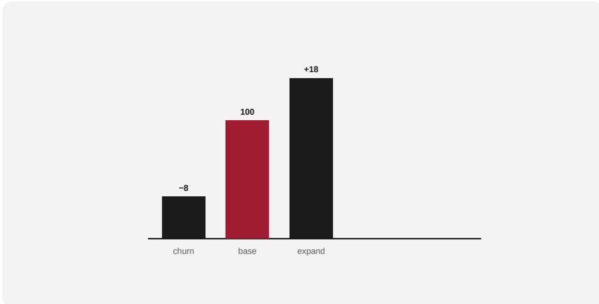

Figure 1. A schematic of the Net Revenue Retention as applied in BARBM312 (illustrative).

HOW IT IS USED

Starting revenue is adjusted for expansion, contraction and churn; NRR above 100% indicates self-expanding revenue, and AI features are assessed by their NRR contribution.

WHEN IT IS USED

When evaluating durable growth, AI feature monetisation and investor readiness.

BY WHOM IT IS USED

Finance, growth leaders and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It isolates the compounding power of the existing base, a stronger signal of durable AI-product value than gross growth.

RELATED FRAMEWORKS IN THIS COURSE

- **CAC : LTV Ratio** · Slibrary 8
- **Retention / Churn Cohort** · Slibrary 8
- **Pirate Metrics (AARRR)** · Slibrary 8
- **Growth Loop** · Slibrary 8

SOURCES (HARVARD REFERENCING STYLE)

Skok, D. (2017) 'SaaS metrics that matter', forEntrepreneurs.

Bain & Company (2021) 'The economics of net revenue retention'.

Growth Loop

Output that feeds the next acquisition

PROFESSIONAL DEFINITION

A Growth Loop is a closed, self-reinforcing system in which the output of one cycle of usage becomes the input that drives the next, producing compounding rather than linear growth.

WHO DEVELOPED IT

Articulated by growth practitioners such as those at Reforge (notably Brian Balfour) as a successor to funnel thinking.

WHY IT WAS DEVELOPED

It was developed to capture how durable products grow through reinvested output — content, data, referrals — rather than one-off acquisition.

FIGURE 1 · GROWTH LOOP

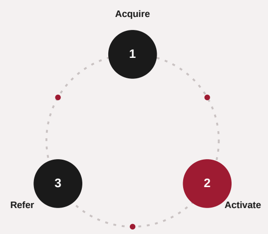

Figure 1. A schematic of the Growth Loop as applied in BARBM312 (illustrative).

HOW IT IS USED

The loop's steps are mapped and the reinvestment mechanism identified; AI is applied where it strengthens the reinvested output.

WHEN IT IS USED

During growth-model design for products with compounding dynamics.

BY WHOM IT IS USED

Growth, product and strategy leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes growth as a reinforcing loop rather than a leaky funnel, focusing effort on the compounding mechanism.

RELATED FRAMEWORKS IN THIS COURSE

- **CAC : LTV Ratio** · Slibrary 8
- **Retention / Churn Cohort** · Slibrary 8
- **Pirate Metrics (AARRR)** · Slibrary 8
- **Net Revenue Retention** · Slibrary 8

SOURCES (HARVARD REFERENCING STYLE)

Balfour, B. (2017) 'Growth loops are the new funnels', Reforge.
Ellis, S. and Brown, M. (2017) Hacking Growth. New York: Currency.

AI Value Stack

Chips → cloud → models → apps

PROFESSIONAL DEFINITION

The AI Value Stack is a layered map of the AI economy — semiconductors, cloud infrastructure, foundation models, and applications — showing where cost is incurred and where value and margin accrue.

WHO DEVELOPED IT

A synthesis from venture and industry analysis of the generative-AI supply chain, discussed by investors such as Andreessen Horowitz and Sequoia.

WHY IT WAS DEVELOPED

It was developed to help firms see that AI cost and value are distributed across a stack, and to locate where they are exposed or where they capture margin.

FIGURE 1 · AI VALUE STACK

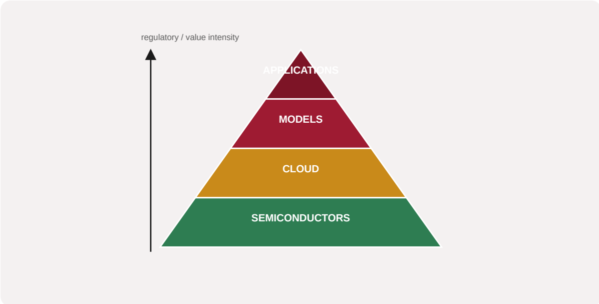

Figure 1. A schematic of the AI Value Stack as applied in BARBM312 (illustrative).

HOW IT IS USED

A firm maps its dependencies and spend across the layers; concentration in any layer signals cost or lock-in risk to be managed.

WHEN IT IS USED

When designing AI sourcing strategy and analysing cost structure and dependency.

BY WHOM IT IS USED

CTOs, strategists and finance leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It gives a single picture of the AI economy's layers, clarifying that most application-layer firms rent the layers beneath them.

RELATED FRAMEWORKS IN THIS COURSE

- **Compute Cost Pyramid** · Slibrary 9
- **Token Economics** · Slibrary 9
- **Hyperscaler Dependency Map** · Slibrary 9
- **Capex vs. Opex Shift** · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

a16z (2023) 'Who owns the generative AI platform?'.

Sequoia Capital (2023) 'Generative AI: A creative new world'.

Compute Cost Pyramid

Training vs. inference vs. fine-tune

PROFESSIONAL DEFINITION

The Compute Cost Pyramid distinguishes the cost profiles of training, fine-tuning and inference, showing that while training is rare and expensive, inference is continuous and scales with usage.

WHO DEVELOPED IT

A practitioner construct from machine-learning operations and cloud-economics analysis of large-model lifecycles.

WHY IT WAS DEVELOPED

It was developed to correct the assumption that training dominates AI cost; for deployed systems, inference at scale is frequently the larger bill.

FIGURE 1 · COMPUTE COST PYRAMID

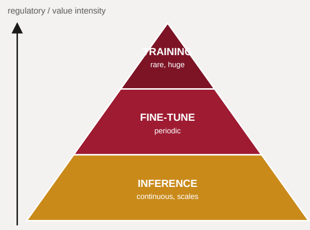

Figure 1. A schematic of the Compute Cost Pyramid as applied in BARBM312 (illustrative).

HOW IT IS USED

Costs are allocated across training, fine-tuning and inference; planning focuses on the layer that dominates the firm's actual usage pattern.

WHEN IT IS USED

During AI cost forecasting and architecture decisions.

BY WHOM IT IS USED

ML engineering, finance and platform teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes AI cost around lifecycle stage, highlighting inference economics that one-off training-cost framing obscures.

RELATED FRAMEWORKS IN THIS COURSE

- **AI Value Stack** · Slibrary 9
- **Token Economics** · Slibrary 9
- **Hyperscaler Dependency Map** · Slibrary 9
- **Capex vs. Opex Shift** · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

Patterson, D. et al. (2021) 'Carbon emissions and large neural network training', arXiv:2104.10350.

Sevilla, J. et al. (2022) 'Compute trends across three eras of machine learning', IJCNN.

Token Economics

Cost per token at scale

PROFESSIONAL DEFINITION

Token Economics analyses the cost of operating large language models on a per-token basis, revealing how prompt design, context length and volume drive operating cost at scale.

WHO DEVELOPED IT

An operational framing from LLM-application practice, formalised as providers priced models per input and output token.

WHY IT WAS DEVELOPED

It was developed to make the recurring cost of LLM applications visible and manageable, since small per-token costs compound enormously at scale.

FIGURE 1 · TOKEN ECONOMICS

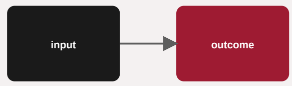

Figure 1. A schematic of the Token Economics as applied in BARBM312 (illustrative).

HOW IT IS USED

Cost per token is multiplied across realistic volumes; prompt and context efficiency are optimised to control spend.

WHEN IT IS USED

When budgeting and optimising LLM-based products.

BY WHOM IT IS USED

Product, engineering and finance teams running LLM applications.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It translates abstract model usage into a concrete, controllable unit cost, enabling FinOps discipline for generative AI.

RELATED FRAMEWORKS IN THIS COURSE

- **AI Value Stack** · Slibrary 9
- **Compute Cost Pyramid** · Slibrary 9
- **Hyperscaler Dependency Map** · Slibrary 9
- **Capex vs. Opex Shift** · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

OpenAI (2023) 'Pricing' documentation.

Chen, L. et al. (2023) 'FrugalGPT: How to use large language models while reducing cost', arXiv:2305.05176.

Hyperscaler Dependency Map

Concentration and lock-in risk

PROFESSIONAL DEFINITION

The Hyperscaler Dependency Map visualises an organisation's reliance on a small number of cloud and model providers, exposing concentration and vendor lock-in risk.

WHO DEVELOPED IT

A risk-management construct from cloud-strategy and concentration-risk practice, echoed by regulators concerned with cloud concentration.

WHY IT WAS DEVELOPED

It was developed to surface the strategic risk of depending on one or two hyperscalers for compute, models and data.

FIGURE 1 · HYPERSCALER DEPENDENCY MAP

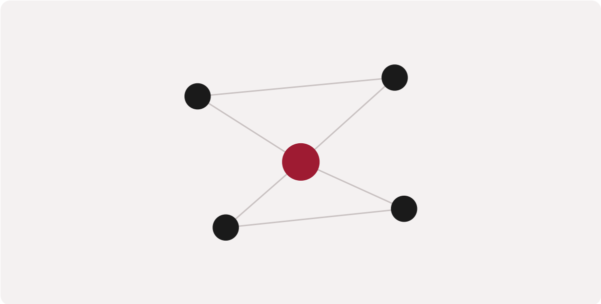

Figure 1. A schematic of the Hyperscaler Dependency Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Dependencies are mapped as a network; single points of failure and lock-in are identified and mitigation (multi-cloud, portability) considered.

WHEN IT IS USED

During cloud and AI sourcing strategy and resilience planning.

BY WHOM IT IS USED

CTOs, risk officers and procurement.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes provider concentration explicit and actionable, balancing the convenience of hyperscalers against strategic dependency.

RELATED FRAMEWORKS IN THIS COURSE

- **AI Value Stack** · Slibrary 9
- **Compute Cost Pyramid** · Slibrary 9
- **Token Economics** · Slibrary 9
- **Capex vs. Opex Shift** · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

Bank of England (2022) 'Financial Stability in Focus: Cloud services'.

Armbrust, M. et al. (2010) 'A view of cloud computing', Communications of the ACM, 53(4).

Capex vs. Opex Shift

Owning vs. renting intelligence

PROFESSIONAL DEFINITION

The Capex-vs-Opex Shift describes how cloud and AI move IT spending from upfront capital investment in owned assets to ongoing operating expense for rented capability.

WHO DEVELOPED IT

A cloud-economics principle long discussed in IT-strategy and finance, central to the cloud value proposition.

WHY IT WAS DEVELOPED

It was developed to clarify the financial trade-off between owning infrastructure and renting it, with implications for flexibility, cost control and risk.

FIGURE 1 · CAPEX VS. OPEX SHIFT

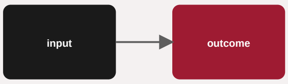

Figure 1. A schematic of the Capex vs. Opex Shift as applied in BARBM312 (illustrative).

HOW IT IS USED

Owning versus renting intelligence is evaluated on cost, flexibility and risk; the shift to opex is weighed against long-run cost at scale.

WHEN IT IS USED

When deciding AI infrastructure ownership and modelling total cost.

BY WHOM IT IS USED

CFOs, CIOs and procurement.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames the financial structure of AI investment, highlighting that opex flexibility can become a large recurring cost at scale.

RELATED FRAMEWORKS IN THIS COURSE

- **AI Value Stack** · Slibrary 9
- **Compute Cost Pyramid** · Slibrary 9
- **Token Economics** · Slibrary 9
- **Hyperscaler Dependency Map** · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

Carr, N. (2008) The Big Switch. New York: Norton.

Armbrust, M. et al. (2010) 'A view of cloud computing', CACM, 53(4).

Private · Hybrid · Hyperscaler 2×2

Control vs. cost of hosting

PROFESSIONAL DEFINITION

This matrix compares private, hybrid and hyperscaler hosting approaches for AI, trading the control of self-hosting against the cost and convenience of public cloud.

WHO DEVELOPED IT

A cloud-architecture decision construct from enterprise IT and AI-infrastructure practice.

WHY IT WAS DEVELOPED

It was developed to structure the hosting decision, which carries large consequences for cost, control, compliance and speed.

FIGURE 1 · PRIVATE · HYBRID · HYPERSCALER 2×2

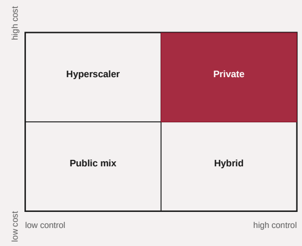

Figure 1. A schematic of the Private · Hybrid · Hyperscaler 2×2 as applied in BARBM312 (illustrative).

HOW IT IS USED

Workloads are placed by their control and cost requirements; sensitive or steady workloads may favour private, variable workloads the hyperscaler.

WHEN IT IS USED

When choosing AI hosting and balancing compliance against cost.

BY WHOM IT IS USED

Cloud architects, CTOs and compliance leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It clarifies that hosting is not binary but a portfolio, matching each workload to the right control–cost trade-off.

RELATED FRAMEWORKS IN THIS COURSE

- **Cloud Cost Waterfall** · Slibrary 10
- **Latency-Cost Frontier** · Slibrary 10
- **Data Gravity Map** · Slibrary 10
- **FinOps Loop** · Slibrary 10

SOURCES (HARVARD REFERENCING STYLE)

NIST (2011) The NIST Definition of Cloud Computing, SP 800-145.

Gartner (2022) 'Cloud strategy decision framework'.

Cloud Cost Waterfall

Where the monthly bill actually goes

PROFESSIONAL DEFINITION

A Cloud Cost Waterfall decomposes a cloud or AI bill into its components — compute, storage, networking, data transfer and managed services — to reveal where spend actually goes.

WHO DEVELOPED IT

A FinOps-practice construct for cost transparency, codified by the FinOps Foundation.

WHY IT WAS DEVELOPED

It was developed to counter opaque cloud billing, exposing the often-surprising drivers of spend such as data egress.

FIGURE 1 · CLOUD COST WATERFALL

Figure 1. A schematic of the Cloud Cost Waterfall as applied in BARBM312 (illustrative).

HOW IT IS USED

The total bill is broken into a waterfall of components; the largest and least-expected drivers are targeted for optimisation.

WHEN IT IS USED

During cloud cost optimisation and AI operating-cost analysis.

BY WHOM IT IS USED

FinOps, finance and platform teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes cloud spend legible component by component, enabling targeted optimisation rather than blunt cuts.

RELATED FRAMEWORKS IN THIS COURSE

Private · Hybrid · Hyperscaler 2×2 · Slibrary 10 | **Latency-Cost Frontier** · Slibrary 10 | **Data Gravity Map** · Slibrary 10 | **FinOps Loop** · Slibrary 10

SOURCES (HARVARD REFERENCING STYLE)

FinOps Foundation (2020) FinOps Framework.

Storment, J.R. and Fuller, M. (2019) Cloud FinOps. Sebastopol: O'Reilly.

Latency-Cost Frontier

Speed you pay for vs. speed you need

PROFESSIONAL DEFINITION

The Latency-Cost Frontier expresses the trade-off between the speed of an AI system and its cost, showing that lower latency generally commands higher cost and that not all use cases need the fastest tier.

WHO DEVELOPED IT

An engineering-economics construct from systems and ML-serving practice.

WHY IT WAS DEVELOPED

It was developed to prevent over-spending on speed that the use case does not require, matching latency to genuine need.

FIGURE 1 · LATENCY-COST FRONTIER

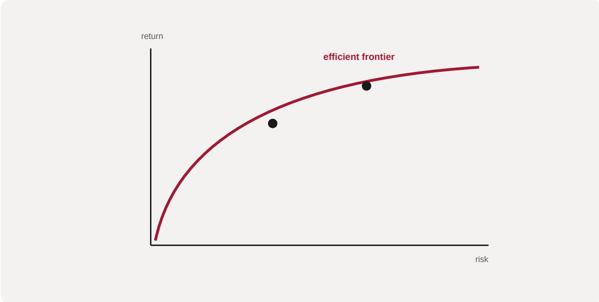

Figure 1. A schematic of the Latency-Cost Frontier as applied in BARBM312 (illustrative).

HOW IT IS USED

Use cases are placed on the frontier; each is served at the cheapest tier that meets its real latency requirement.

WHEN IT IS USED

When designing AI serving architecture and controlling inference cost.

BY WHOM IT IS USED

ML engineering, product and finance teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the speed-versus-cost trade-off explicit per use case, avoiding paying for latency that delivers no value.

RELATED FRAMEWORKS IN THIS COURSE

- **Private · Hybrid · Hyperscaler 2×2** · Slibrary 10
- **Cloud Cost Waterfall** · Slibrary 10
- **Data Gravity Map** · Slibrary 10
- **FinOps Loop** · Slibrary 10

SOURCES (HARVARD REFERENCING STYLE)

Dean, J. and Barroso, L.A. (2013) 'The tail at scale', Communications of the ACM, 56(2).

Crankshaw, D. et al. (2017) 'Clipper: A low-latency online prediction serving system', NSDI.

Data Gravity Map

Compute follows where data lives

PROFESSIONAL DEFINITION

A Data Gravity Map illustrates how applications and compute are drawn toward where large data sets reside, because moving data is costly and slow — so 'data gravity' shapes architecture.

WHO DEVELOPED IT

Coined by Dave McCrory in 2010 to describe the pull that accumulated data exerts on services and infrastructure.

WHY IT WAS DEVELOPED

It was developed to explain why workloads cluster around data and to inform decisions about where to process AI workloads.

FIGURE 1 · DATA GRAVITY MAP

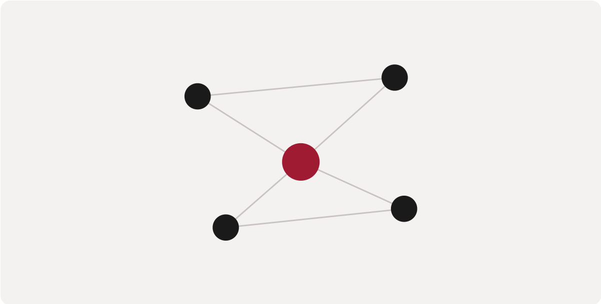

Figure 1. A schematic of the Data Gravity Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Major data stores are mapped; compute and applications are located to minimise costly data movement, influencing cloud and edge choices.

WHEN IT IS USED

During data and AI architecture and multi-cloud planning.

BY WHOM IT IS USED

Architects, data engineers and infrastructure leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It names a force that quietly governs architecture and cost, guiding where AI processing should physically occur.

RELATED FRAMEWORKS IN THIS COURSE

- **Private · Hybrid · Hyperscaler 2×2** · Slibrary 10
- **Cloud Cost Waterfall** · Slibrary 10
- **Latency-Cost Frontier** · Slibrary 10
- **FinOps Loop** · Slibrary 10

SOURCES (HARVARD REFERENCING STYLE)

McCrory, D. (2010) 'Data gravity in the clouds', blog.

Cloud Security Alliance (2021) 'Data gravity and cloud architecture'.

FinOps Loop

Inform → optimise → operate cloud spend

PROFESSIONAL DEFINITION

The FinOps Loop is a continuous cycle — Inform, Optimise, Operate — for managing cloud and AI spend collaboratively across finance, engineering and business teams.

WHO DEVELOPED IT

Codified by the FinOps Foundation, building on cloud-financial-management practice.

WHY IT WAS DEVELOPED

It was developed to bring financial accountability to variable cloud and AI consumption without slowing engineering.

FIGURE 1 · FINOPS LOOP

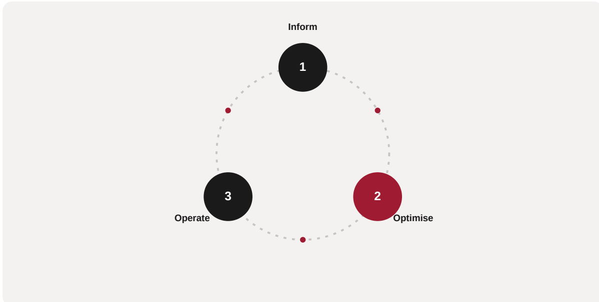

Figure 1. A schematic of the FinOps Loop as applied in BARBM312 (illustrative).

HOW IT IS USED

Teams cycle through visibility (Inform), efficiency action (Optimise) and embedding practices (Operate), iterating continuously.

WHEN IT IS USED

When establishing ongoing AI and cloud cost governance.

BY WHOM IT IS USED

FinOps practitioners, finance and engineering leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It operationalises cost management as a shared, continuous practice rather than a periodic finance exercise.

RELATED FRAMEWORKS IN THIS COURSE

- **Private · Hybrid · Hyperscaler 2×2** · Slibrary 10
- **Cloud Cost Waterfall** · Slibrary 10
- **Latency-Cost Frontier** · Slibrary 10
- **Data Gravity Map** · Slibrary 10

SOURCES (HARVARD REFERENCING STYLE)

FinOps Foundation (2020) FinOps Framework.

Storment, J.R. and Fuller, M. (2019) Cloud FinOps. Sebastopol: O'Reilly.

Open vs. Proprietary LLM 2×2

Control & cost vs. capability & speed

PROFESSIONAL DEFINITION

This matrix contrasts open-source and proprietary large language models, trading the control and marginal-cost advantages of open models against the capability and speed of frontier proprietary APIs.

WHO DEVELOPED IT

A strategy framing from the 2020s open-versus-closed model debate, discussed across AI-engineering and policy communities.

WHY IT WAS DEVELOPED

It was developed to structure the model-sourcing decision, which shapes cost, control, capability and dependency.

FIGURE 1 · OPEN VS. PROPRIETARY LLM 2×2

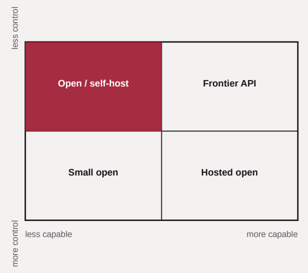

Figure 1. A schematic of the Open vs. Proprietary LLM 2×2 as applied in BARBM312 (illustrative).

HOW IT IS USED

Each use case is placed by its capability needs and control requirements; the matrix recommends open self-hosting, hosted open, or frontier API.

WHEN IT IS USED

When selecting models and balancing control, cost and capability.

BY WHOM IT IS USED

ML engineering, product and strategy leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It clarifies a fast-moving trade-off, showing open models can lower marginal cost and increase control while demanding more in-house capability.

RELATED FRAMEWORKS IN THIS COURSE

- **Make-vs-Buy Decision Tree** · Slibrary 11
- **Talent Scarcity Heatmap** · Slibrary 11
- **Skill-Premium Curve** · Slibrary 11
- **MLOps Maturity Ladder** · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Bommasani, R. et al. (2021) 'On the opportunities and risks of foundation models', arXiv:2108.07258.

Liesenfeld, A. and Dingemanse, M. (2024) 'Rethinking open source generative AI', FAccT.

Make-vs-Buy Decision Tree

When to fine-tune, when to call an API

PROFESSIONAL DEFINITION

The Make-vs-Buy Decision Tree guides whether to build, fine-tune or simply call an external model or component, by working through questions of differentiation, capability and cost.

WHO DEVELOPED IT

An adaptation of classic make-or-buy and transaction-cost reasoning to the AI model lifecycle.

WHY IT WAS DEVELOPED

It was developed to prevent undifferentiated in-house model effort and to reserve build capacity for genuine sources of advantage.

FIGURE 1 · MAKE-VS-BUY DECISION TREE

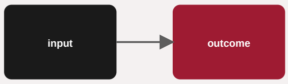

Figure 1. A schematic of the Make-vs-Buy Decision Tree as applied in BARBM312 (illustrative).

HOW IT IS USED

The tree asks whether the capability is strategic, whether data confers advantage, and whether an API suffices, terminating in build, fine-tune or buy.

WHEN IT IS USED

When deciding how to source AI capability for a specific use case.

BY WHOM IT IS USED

CTOs, ML leads and product owners.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It turns the abstract build-or-buy question into a structured, repeatable decision for AI components.

RELATED FRAMEWORKS IN THIS COURSE

Open vs. Proprietary LLM 2×2 · Slibrary 11
Talent Scarcity Heatmap · Slibrary 11
Skill-Premium Curve · Slibrary 11
MLOps Maturity Ladder · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Williamson, O.E. (1985) The Economic Institutions of Capitalism. New York: Free Press.
Bommasani, R. et al. (2021) 'On the opportunities and risks of foundation models', arXiv:2108.07258.

Talent Scarcity Heatmap

Where AI/MLOps skills cost most

PROFESSIONAL DEFINITION

The Talent Scarcity Heatmap maps where specialist AI, data and MLOps skills are most scarce and expensive, informing build-versus-buy, location and partnership decisions.

WHO DEVELOPED IT

A workforce-planning construct drawing on labour-market analysis of AI skills, including OECD and industry skills studies.

WHY IT WAS DEVELOPED

It was developed to make the scarcity and cost of AI talent a visible input to strategy rather than an after-the-fact constraint.

FIGURE 1 · TALENT SCARCITY HEATMAP

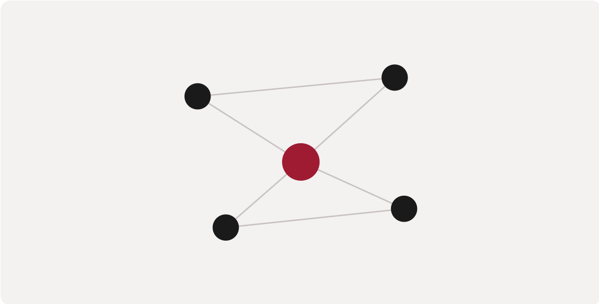

Figure 1. A schematic of the Talent Scarcity Heatmap as applied in BARBM312 (illustrative).

HOW IT IS USED

Skills are shaded by scarcity and cost; the map informs whether to build internal capability, outsource, or partner, and where to locate teams.

WHEN IT IS USED

During AI workforce and sourcing strategy.

BY WHOM IT IS USED

HR, technology and transformation leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It elevates talent scarcity to a first-order strategic variable, recognising that the wage premium can dwarf software spend.

RELATED FRAMEWORKS IN THIS COURSE

- **Open vs. Proprietary LLM 2×2** · Slibrary 11
- **Make-vs-Buy Decision Tree** · Slibrary 11
- **Skill-Premium Curve** · Slibrary 11
- **MLOps Maturity Ladder** · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

OECD (2023) OECD Employment Outlook 2023: Artificial Intelligence and the Labour Market. Paris: OECD.

Stanford HAI (2024) AI Index Report.

Skill-Premium Curve

Wage premium for scarce AI roles

PROFESSIONAL DEFINITION

The Skill-Premium Curve shows how scarce AI-related skills command a rising wage premium, and how that premium changes as supply catches up or technology shifts demand.

WHO DEVELOPED IT

Grounded in labour-economics analysis of skill-biased technological change, associated with economists such as David Autor and Lawrence Katz.

WHY IT WAS DEVELOPED

It was developed to explain and anticipate the cost of scarce skills, informing whether to hire, train or wait.

FIGURE 1 · SKILL-PREMIUM CURVE

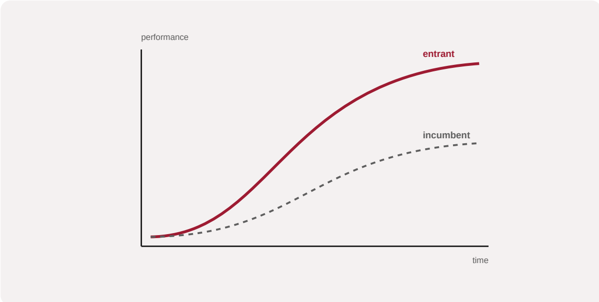

Figure 1. A schematic of the Skill-Premium Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

The premium for target skills is tracked over time; strategy weighs paying the premium now against building supply through training.

WHEN IT IS USED

When planning AI hiring and compensation strategy.

BY WHOM IT IS USED

HR, finance and workforce-planning leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames AI talent cost dynamically, recognising premiums shift with supply and with the technology itself.

RELATED FRAMEWORKS IN THIS COURSE

- **Open vs. Proprietary LLM 2×2** · Slibrary 11
- **Make-vs-Buy Decision Tree** · Slibrary 11
- **Talent Scarcity Heatmap** · Slibrary 11
- **MLOps Maturity Ladder** · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Autor, D.H., Katz, L.F. and Krueger, A.B. (1998) 'Computing inequality', Quarterly Journal of Economics, 113(4).

Acemoglu, D. and Autor, D. (2011) 'Skills, tasks and technologies', Handbook of Labor Economics, 4.

MLOps Maturity Ladder

Ad hoc to fully automated pipelines

PROFESSIONAL DEFINITION

The MLOps Maturity Ladder describes stages of operational capability for machine learning, from ad hoc manual work to fully automated, self-healing pipelines, indicating the cost and reliability at each stage.

WHO DEVELOPED IT

Formalised in industry MLOps guidance, including Google Cloud's MLOps maturity levels and Microsoft's MLOps practices.

WHY IT WAS DEVELOPED

It was developed to help organisations locate their ML operational maturity and the investment required to scale models reliably.

FIGURE 1 · MLOPS MATURITY LADDER

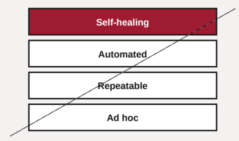

Figure 1. A schematic of the MLOps Maturity Ladder as applied in BARBM312 (illustrative).

HOW IT IS USED

Current practice is placed on a rung; the next rung defines the automation, monitoring and governance investment needed to scale.

WHEN IT IS USED

When scaling AI from pilot to production and planning MLOps investment.

BY WHOM IT IS USED

ML engineering, platform and operations leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It turns reliable ML operations into a staged capability, exposing the often-underestimated cost of moving beyond prototypes.

RELATED FRAMEWORKS IN THIS COURSE

- **Open vs. Proprietary LLM 2×2** · Slibrary 11
- **Make-vs-Buy Decision Tree** · Slibrary 11
- **Talent Scarcity Heatmap** · Slibrary 11
- **Skill-Premium Curve** · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Google Cloud (2020) 'MLOps: Continuous delivery and automation pipelines in machine learning'.

Kreuzberger, D., Kühl, N. and Hirschl, S. (2023) 'Machine learning operations (MLOps)', IEEE Access, 11.

Cost-Benefit Trade-off Matrix

Value vs. total risk-adjusted cost

PROFESSIONAL DEFINITION

The Cost-Benefit Trade-off Matrix arrays AI initiatives by their expected value against their total, risk-adjusted cost, supporting disciplined prioritisation of where to invest.

WHO DEVELOPED IT

A general decision-analysis construct adapted to AI investment appraisal.

WHY IT WAS DEVELOPED

It was developed to ensure AI investment decisions weigh full risk-adjusted cost against realistic benefit, not headline capability.

FIGURE 1 · COST-BENEFIT TRADE-OFF MATRIX

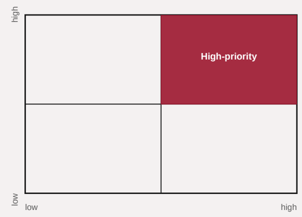

Figure 1. A schematic of the Cost-Benefit Trade-off Matrix as applied in BARBM312 (illustrative).

HOW IT IS USED

Each initiative is scored on value and total risk-adjusted cost; the matrix highlights high-value, justified-cost initiatives.

WHEN IT IS USED

During AI portfolio prioritisation and investment governance.

BY WHOM IT IS USED

Executives, finance and AI programme leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It integrates the cost lessons of the unit — hidden costs and risk — into a single prioritisation view.

RELATED FRAMEWORKS IN THIS COURSE

- **AI Risk Register** · Slibrary 12
- **Bias & Failure Modes Map** · Slibrary 12
- **Black Swan Radar** · Slibrary 12
- **Mitigation Cost Curve** · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

Boardman, A.E. et al. (2017) Cost-Benefit Analysis: Concepts and Practice. Cambridge: CUP.
McKinsey & Company (2023) 'The economic potential of generative AI'.

AI Risk Register

Likelihood × impact × mitigation

PROFESSIONAL DEFINITION

An AI Risk Register catalogues the risks of an AI initiative by likelihood and impact, with owners and mitigations, providing a structured basis for governance and de-risking.

WHO DEVELOPED IT

An adaptation of standard enterprise risk-management practice to AI, aligned with frameworks such as the NIST AI Risk Management Framework.

WHY IT WAS DEVELOPED

It was developed to make AI-specific risks — bias, drift, security, compliance — explicit, owned and managed rather than discovered after harm.

FIGURE 1 · AI RISK REGISTER

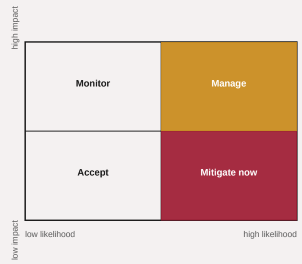

Figure 1. A schematic of the AI Risk Register as applied in BARBM312 (illustrative).

HOW IT IS USED

Risks are scored on likelihood and impact, assigned owners and mitigations, and reviewed as the system evolves.

WHEN IT IS USED

Throughout the AI lifecycle, from design through operation.

BY WHOM IT IS USED

Risk officers, AI governance teams and project leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It applies disciplined risk management to AI specifically, supporting accountable, auditable deployment.

RELATED FRAMEWORKS IN THIS COURSE

- **Cost-Benefit Trade-off Matrix** · Slibrary 12
- **Bias & Failure Modes Map** · Slibrary 12
- **Black Swan Radar** · Slibrary 12
- **Mitigation Cost Curve** · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

NIST (2023) AI Risk Management Framework (AI RMF 1.0).

ISO/IEC (2023) ISO/IEC 23894: AI Risk Management.

Bias & Failure Modes Map

Where models break and who pays

PROFESSIONAL DEFINITION

The Bias & Failure Modes Map identifies where an AI model can produce biased, unfair or otherwise harmful outputs and who bears the consequences, guiding mitigation and monitoring.

WHO DEVELOPED IT

Drawn from algorithmic-fairness and responsible-AI research, notably the work of Cathy O'Neil and the FAccT community.

WHY IT WAS DEVELOPED

It was developed to anticipate the specific ways models fail and discriminate, since such failures carry reputational, legal and human cost.

FIGURE 1 · BIAS & FAILURE MODES MAP

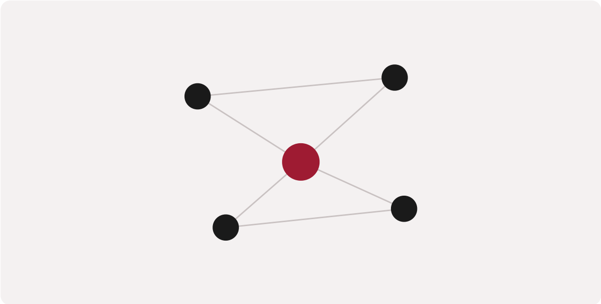

Figure 1. A schematic of the Bias & Failure Modes Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Potential failure and bias modes are enumerated per use case, with detection and mitigation assigned to each.

WHEN IT IS USED

During responsible-AI design and pre-deployment review.

BY WHOM IT IS USED

Responsible-AI, risk and ML teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It systematises the search for harm, treating bias and failure as design concerns rather than post-hoc surprises.

RELATED FRAMEWORKS IN THIS COURSE

- **Cost-Benefit Trade-off Matrix** · Slibrary 12
- **AI Risk Register** · Slibrary 12
- **Black Swan Radar** · Slibrary 12
- **Mitigation Cost Curve** · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

O'Neil, C. (2016) Weapons of Math Destruction. New York: Crown.

Mehrabi, N. et al. (2021) 'A survey on bias and fairness in machine learning', ACM Computing Surveys, 54(6).

Black Swan Radar

Taleb's tail-risk early warning

PROFESSIONAL DEFINITION

The Black Swan Radar is a lens for surfacing rare, high-impact, hard-to-predict events — Taleb's 'black swans' — that conventional risk analysis underweights, so that fragility to them can be reduced.

WHO DEVELOPED IT

Based on Nassim Nicholas Taleb's work in The Black Swan (2007) and his broader writing on uncertainty and fragility.

WHY IT WAS DEVELOPED

It was developed to counter the tendency to plan for the average case and ignore the rare event that actually determines survival.

FIGURE 1 · BLACK SWAN RADAR

Figure 1. A schematic of the Black Swan Radar as applied in BARBM312 (illustrative).

HOW IT IS USED

Plausible tail events are surfaced and the system's fragility to them assessed; mitigation favours robustness and optionality over precise prediction.

WHEN IT IS USED

During risk strategy, especially for AI systems exposed to rare but catastrophic failure.

BY WHOM IT IS USED

Risk officers, strategists and senior leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It shifts risk thinking from forecasting the likely to building resilience against the consequential rare event.

RELATED FRAMEWORKS IN THIS COURSE

- **Cost-Benefit Trade-off Matrix** · Slibrary 12
- **AI Risk Register** · Slibrary 12
- **Bias & Failure Modes Map** · Slibrary 12
- **Mitigation Cost Curve** · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

Taleb, N.N. (2007) The Black Swan. London: Penguin.
Taleb, N.N. (2012) Antifragile. New York: Random House.

Mitigation Cost Curve

Diminishing returns on de-risking

PROFESSIONAL DEFINITION

The Mitigation Cost Curve shows how the cost of reducing a risk rises while the marginal risk removed falls, revealing the point beyond which further de-risking costs more than the risk it removes.

WHO DEVELOPED IT

A risk-economics construct related to the economics of safety and diminishing returns to control.

WHY IT WAS DEVELOPED

It was developed to identify the optimal level of mitigation, avoiding both reckless under-investment and uneconomic over-control.

FIGURE 1 · MITIGATION COST CURVE

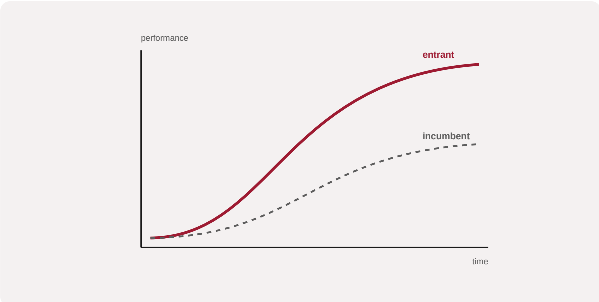

Figure 1. A schematic of the Mitigation Cost Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

Mitigation cost is plotted against residual risk; investment stops where marginal cost exceeds marginal risk reduction.

WHEN IT IS USED

When deciding how much to invest in AI risk controls.

BY WHOM IT IS USED

Risk, finance and governance leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It introduces economic discipline to de-risking, recognising that perfect safety is neither attainable nor optimal.

RELATED FRAMEWORKS IN THIS COURSE

- **Cost-Benefit Trade-off Matrix** · Slibrary 12
- **AI Risk Register** · Slibrary 12
- **Bias & Failure Modes Map** · Slibrary 12
- **Black Swan Radar** · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

Viscusi, W.K. (1996) 'Economic foundations of the current regulatory reform efforts', Journal of Economic Perspectives, 10(3).

Hubbard, D.W. (2020) The Failure of Risk Management. Hoboken: Wiley.

IP Quadrant

Patent · copyright · trademark · trade secret

PROFESSIONAL DEFINITION

The IP Quadrant organises the four main forms of intellectual property — patent, copyright, trademark and trade secret — clarifying what each protects and how each applies to AI assets.

WHO DEVELOPED IT

A foundational construct of intellectual-property law and management, taught across IP-strategy texts.

WHY IT WAS DEVELOPED

It was developed to help managers match each asset to the appropriate protection rather than defaulting to one form.

FIGURE 1 · IP QUADRANT

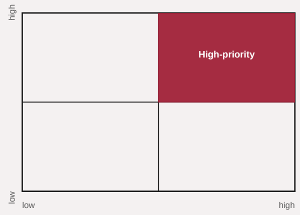

Figure 1. A schematic of the IP Quadrant as applied in BARBM312 (illustrative).

HOW IT IS USED

Each AI-related asset is classified into the quadrant best suited to it, and a layered protection strategy is built across forms.

WHEN IT IS USED

When designing IP protection for AI products, data and models.

BY WHOM IT IS USED

Legal, IP and product-strategy teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It clarifies that AI assets often need a combination of protections, since no single form covers models, data and brand together.

RELATED FRAMEWORKS IN THIS COURSE

- **Authorship Decision Tree** · Slibrary 13
- **Patentability Funnel** · Slibrary 13
- **Inventorship Test** · Slibrary 13
- **Trademark Distinctiveness Spectrum** · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

WIPO (2020) WIPO Technology Trends 2019: Artificial Intelligence. Geneva: WIPO.

Cornish, W., Llewelyn, D. and Aplin, T. (2019) Intellectual Property. London: Sweet & Maxwell.

Authorship Decision Tree

Human, machine or joint authorship?

PROFESSIONAL DEFINITION

The Authorship Decision Tree guides whether an AI-assisted output is human-authored, machine-generated, or jointly created, with different copyright and ownership consequences for each path.

WHO DEVELOPED IT

Associated with the legal scholarship of Jyh-An Lee and colleagues on AI, authorship and intellectual property.

WHY IT WAS DEVELOPED

It was developed because most jurisdictions require a human author for copyright, leaving AI-generated output in a contested space that must be navigated.

FIGURE 1 · AUTHORSHIP DECISION TREE

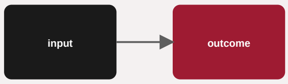

Figure 1. A schematic of the Authorship Decision Tree as applied in BARBM312 (illustrative).

HOW IT IS USED

The tree asks about the human's creative contribution and control, classifying the output and its likely ownership and protectability.

WHEN IT IS USED

When determining rights in AI-generated content and code.

BY WHOM IT IS USED

Legal, IP and product leaders working with generative AI.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It operationalises an unsettled legal question into a usable decision aid, central to value capture in generative AI.

RELATED FRAMEWORKS IN THIS COURSE

- **IP Quadrant** · Slibrary 13
- **Patentability Funnel** · Slibrary 13
- **Inventorship Test** · Slibrary 13
- **Trademark Distinctiveness Spectrum** · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

Lee, J.-A., Hilty, R.M. and Liu, K.-C. (eds.) (2021) Artificial Intelligence and Intellectual Property. Oxford: OUP.

US Copyright Office (2023) 'Copyright registration guidance: Works containing material generated by AI'.

Patentability Funnel

Novel · non-obvious · useful filter

PROFESSIONAL DEFINITION

The Patentability Funnel filters an invention through the core tests of patentability — novelty, non-obviousness (inventive step) and utility — to assess whether AI-related innovation can be patented.

WHO DEVELOPED IT

A representation of established patent-law doctrine, applied to AI by patent practitioners and offices such as the EPO and USPTO.

WHY IT WAS DEVELOPED

It was developed to help innovators assess, early and cheaply, whether an AI invention is likely to clear the bar for patent protection.

FIGURE 1 · PATENTABILITY FUNNEL

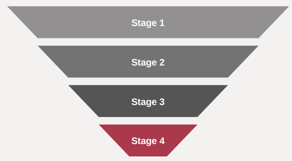

Figure 1. A schematic of the Patentability Funnel as applied in BARBM312 (illustrative).

HOW IT IS USED

The invention passes through each test; failure at any stage indicates an alternative protection (e.g. trade secret) may be preferable.

WHEN IT IS USED

When deciding whether and how to protect AI inventions.

BY WHOM IT IS USED

Patent attorneys, R&D and IP-strategy leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames patentability as a sequential filter, clarifying where AI inventions commonly fail (often obviousness or eligible subject matter).

RELATED FRAMEWORKS IN THIS COURSE

- **IP Quadrant** · Slibrary 13
- **Authorship Decision Tree** · Slibrary 13
- **Inventorship Test** · Slibrary 13
- **Trademark Distinctiveness Spectrum** · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

European Patent Office (2021) Guidelines for Examination, Part G.

Abbott, R. (2020) The Reasonable Robot. Cambridge: CUP.

Inventorship Test

Can an AI be named an inventor?

PROFESSIONAL DEFINITION

The Inventorship Test addresses whether an AI system can be named as an inventor on a patent, a question that courts and offices have largely answered by requiring a human inventor.

WHO DEVELOPED IT

Crystallised through the DABUS litigation brought by Stephen Thaler across multiple jurisdictions, and analysed by scholars such as Ryan Abbott.

WHY IT WAS DEVELOPED

It was developed to resolve who, if anyone, can hold rights in inventions generated with substantial AI contribution.

FIGURE 1 · INVENTORSHIP TEST

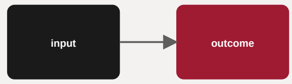

Figure 1. A schematic of the Inventorship Test as applied in BARBM312 (illustrative).

HOW IT IS USED

The contribution of human and machine is examined against the legal requirement of a natural-person inventor; rights strategy follows the conclusion.

WHEN IT IS USED

When AI materially contributes to an invention and inventorship must be determined.

BY WHOM IT IS USED

Patent attorneys, R&D leaders and IP strategists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It marks the current legal boundary — that inventorship requires a human — with significant consequences for AI-driven R&D.

RELATED FRAMEWORKS IN THIS COURSE

IP Quadrant · Slibrary 13
Authorship Decision Tree · Slibrary 13
Patentability Funnel · Slibrary 13
Trademark Distinctiveness Spectrum · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

Thaler v Comptroller-General of Patents (UK Supreme Court, 2023) UKSC 49.
Abbott, R. (2020) The Reasonable Robot. Cambridge: CUP.

Trademark Distinctiveness Spectrum

Generic → fanciful marks

PROFESSIONAL DEFINITION

The Trademark Distinctiveness Spectrum ranks marks from generic and descriptive (weak or unprotectable) through suggestive to arbitrary and fanciful (strongest), guiding brand and naming strategy.

WHO DEVELOPED IT

A doctrine of trademark law crystallised in the United States by the Abercrombie case and reflected in trademark practice generally.

WHY IT WAS DEVELOPED

It was developed to explain why some marks receive strong protection and others none, informing how brands should be named.

FIGURE 1 · TRADEMARK DISTINCTIVENESS SPECTRUM

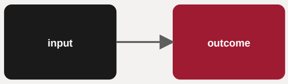

Figure 1. A schematic of the Trademark Distinctiveness Spectrum as applied in BARBM312 (illustrative).

HOW IT IS USED

A proposed mark is located on the spectrum; naming favours the distinctive end to secure stronger, more defensible protection.

WHEN IT IS USED

When naming AI products and building brand IP.

BY WHOM IT IS USED

Brand, marketing and legal teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It links naming choices directly to legal strength, a consideration easily overlooked in fast-moving AI product launches.

RELATED FRAMEWORKS IN THIS COURSE

- **IP Quadrant** · Slibrary 13
- **Authorship Decision Tree** · Slibrary 13
- **Patentability Funnel** · Slibrary 13
- **Inventorship Test** · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

Abercrombie & Fitch Co. v Hunting World, Inc. (1976) 537 F.2d 4.

Cornish, W., Llewelyn, D. and Aplin, T. (2019) Intellectual Property. London: Sweet & Maxwell.

Wrapper vs. Moat Ladder

Thin layer to defensible system

PROFESSIONAL DEFINITION

The Wrapper-vs-Moat Ladder distinguishes degrees of defensibility for AI products, from a thin wrapper over a third-party model, through proprietary workflow and data, to a durable system-level moat.

WHO DEVELOPED IT

A contemporary venture-strategy framing from the generative-AI investment debate.

WHY IT WAS DEVELOPED

It was developed to answer the investor's question of whether an AI product owns anything defensible beneath a thin application layer.

FIGURE 1 · WRAPPER VS. MOAT LADDER

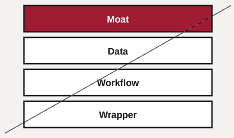

Figure 1. A schematic of the Wrapper vs. Moat Ladder as applied in BARBM312 (illustrative).

HOW IT IS USED

A product is placed on the ladder by its sources of defensibility; strategy aims to climb from wrapper toward moat.

WHEN IT IS USED

During AI product strategy and investment due diligence.

BY WHOM IT IS USED

Founders, product leaders and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It names the central defensibility question of the AI era — a wrapper is a feature, not a moat — and charts the path to durability.

RELATED FRAMEWORKS IN THIS COURSE

- **7 Powers** · Slibrary 14
- **Defensibility Matrix** · Slibrary 14
- **Proprietary Data Flywheel** · Slibrary 14
- **Value Capture Map** · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Helmer, H. (2016) 7 Powers. Deep Strategy.
a16z (2023) 'Who owns the generative AI platform?'.

7 Powers

Hamilton Helmer's durable advantages

PROFESSIONAL DEFINITION

7 Powers identifies the seven sources of durable competitive advantage — scale economies, network economies, counter-positioning, switching costs, branding, cornered resource and process power — that allow a firm to earn persistent returns.

WHO DEVELOPED IT

Developed by strategist and investor Hamilton Helmer in 7 Powers (2016), distilling decades of strategy practice.

WHY IT WAS DEVELOPED

It was developed to define precisely what 'power' means — the potential to earn persistent differential returns — and how it is created.

FIGURE 1 · 7 POWERS

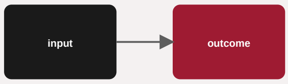

Figure 1. A schematic of the 7 Powers as applied in BARBM312 (illustrative).

HOW IT IS USED

A business is tested for which of the seven powers it holds or could build; AI advantages are mapped to specific powers rather than vague differentiation.

WHEN IT IS USED

During strategy formulation and assessment of AI-business defensibility.

BY WHOM IT IS USED

Founders, strategists and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It provides a rigorous, exhaustive taxonomy of durable advantage, sharpening loose talk of 'moats' into specific, buildable powers.

RELATED FRAMEWORKS IN THIS COURSE

- **Wrapper vs. Moat Ladder** · Slibrary 14
- **Defensibility Matrix** · Slibrary 14
- **Proprietary Data Flywheel** · Slibrary 14
- **Value Capture Map** · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Helmer, H. (2016) 7 Powers: The Foundations of Business Strategy. Deep Strategy.

Porter, M.E. (1980) Competitive Strategy. New York: Free Press.

Defensibility Matrix

Differentiation × switching cost

PROFESSIONAL DEFINITION

The Defensibility Matrix maps an AI offering by its degree of differentiation against the switching cost it creates, distinguishing commodity features from genuine moats.

WHO DEVELOPED IT

A synthesis from competitive-strategy practice applied to AI products.

WHY IT WAS DEVELOPED

It was developed to test quickly whether an AI product is defensible or merely a replicable feature.

FIGURE 1 · DEFENSIBILITY MATRIX

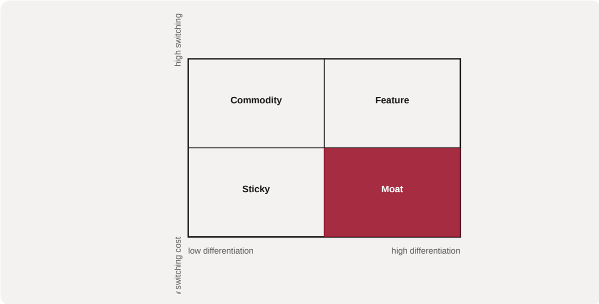

Figure 1. A schematic of the Defensibility Matrix as applied in BARBM312 (illustrative).

HOW IT IS USED

An offering is placed by differentiation and switching cost; only the high-high quadrant constitutes a moat worth defending.

WHEN IT IS USED

During product and investment strategy for AI.

BY WHOM IT IS USED

Founders, product and investment leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It distils defensibility into two decisive axes, complementing the fuller 7 Powers analysis with a quick diagnostic.

RELATED FRAMEWORKS IN THIS COURSE

- **Wrapper vs. Moat Ladder** · Slibrary 14
- **7 Powers** · Slibrary 14
- **Proprietary Data Flywheel** · Slibrary 14
- **Value Capture Map** · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Porter, M.E. (1985) Competitive Advantage. New York: Free Press.

Helmer, H. (2016) 7 Powers. Deep Strategy.

Proprietary Data Flywheel

Data advantage that compounds

PROFESSIONAL DEFINITION

A Proprietary Data Flywheel is a self-reinforcing loop in which a firm's unique data improves its product, which attracts more usage and generates more unique data, compounding advantage over time.

WHO DEVELOPED IT

A construct from data-strategy and platform practice, related to data network effects and the work of Hagiu and Wright.

WHY IT WAS DEVELOPED

It was developed to identify when data genuinely compounds into advantage, as opposed to merely accumulating.

FIGURE 1 · PROPRIETARY DATA FLYWHEEL

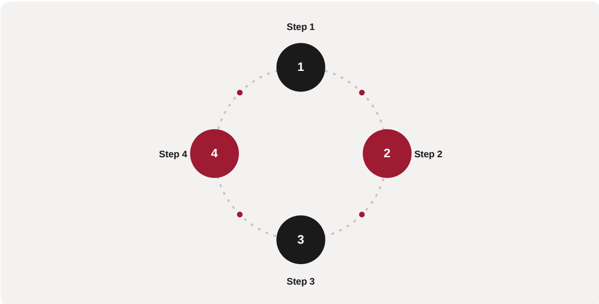

Figure 1. A schematic of the Proprietary Data Flywheel as applied in BARBM312 (illustrative).

HOW IT IS USED

The loop is tested for whether new data is unique, improves the product, and is hard to replicate; only then does it constitute a flywheel.

WHEN IT IS USED

When assessing data-based defensibility of AI products.

BY WHOM IT IS USED

Founders, product and data leaders, investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It distinguishes a compounding data moat from inert data accumulation, a key test of durable AI advantage.

RELATED FRAMEWORKS IN THIS COURSE

- **Wrapper vs. Moat Ladder** · Slibrary 14
- **7 Powers** · Slibrary 14
- **Defensibility Matrix** · Slibrary 14
- **Value Capture Map** · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Hagiu, A. and Wright, J. (2020) 'When data creates competitive advantage', HBR, 98(1).

Gregory, R.W. et al. (2021) 'The role of artificial intelligence and data network effects', Academy of Management Review, 46(3).

Value Capture Map

Where margin actually accrues

PROFESSIONAL DEFINITION

A Value Capture Map traces where, along a value chain or stack, margin actually accrues, distinguishing where value is created from where it is captured.

WHO DEVELOPED IT

Grounded in value-chain and value-capture theory, associated with strategy scholars including Porter and Brandenburger and Stuart.

WHY IT WAS DEVELOPED

It was developed to ensure firms position themselves where value is captured, not merely created — a frequent failing in technology supply chains.

FIGURE 1 · VALUE CAPTURE MAP

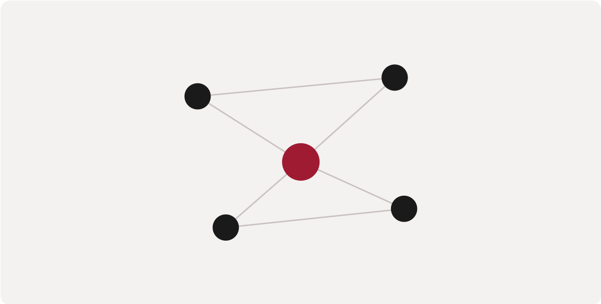

Figure 1. A schematic of the Value Capture Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Value creation and capture are mapped across the chain; strategy moves the firm toward the points of durable capture.

WHEN IT IS USED

During strategy and AI-stack positioning decisions.

BY WHOM IT IS USED

Strategists, founders and corporate-development teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It separates value creation from value capture, explaining why some AI layers earn margin while others are commoditised.

RELATED FRAMEWORKS IN THIS COURSE

- **Wrapper vs. Moat Ladder** · Slibrary 14
- **7 Powers** · Slibrary 14
- **Defensibility Matrix** · Slibrary 14
- **Proprietary Data Flywheel** · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Brandenburger, A.M. and Stuart, H.W. (1996) 'Value-based business strategy', Journal of Economics & Management Strategy, 5(1).

Porter, M.E. (1985) Competitive Advantage. New York: Free Press.

Data Asset Classification

Owned · licensed · public · derived

PROFESSIONAL DEFINITION

Data Asset Classification organises a firm's data by its origin and rights — owned, licensed, public or derived — clarifying what value can be captured from each and what obligations attach.

WHO DEVELOPED IT

A data-governance construct from information-management and data-strategy practice.

WHY IT WAS DEVELOPED

It was developed to make explicit which data a firm may freely exploit, which carries licence or privacy constraints, and which is truly proprietary.

FIGURE 1 · DATA ASSET CLASSIFICATION

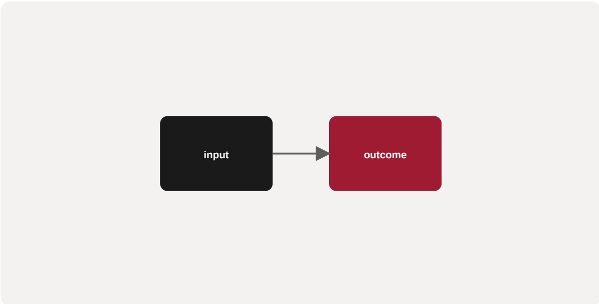

Figure 1. A schematic of the Data Asset Classification as applied in BARBM312 (illustrative).

HOW IT IS USED

Each data set is classified by origin and rights; value strategy and compliance follow from the classification.

WHEN IT IS USED

When treating data as IP and assessing data-based value and risk.

BY WHOM IT IS USED

Data governance, legal and strategy teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It links data value directly to rights and provenance, preventing over-claiming of value from data the firm does not fully control.

RELATED FRAMEWORKS IN THIS COURSE

- **Data Rights Stack** · Slibrary 15
- **Data Moat Strength** · Slibrary 15
- **Privacy-Value Frontier** · Slibrary 15
- **Synthetic Data Trade-off** · Slibrary 15

SOURCES (HARVARD REFERENCING STYLE)

DAMA International (2017) DAMA-DMBOK: Data Management Body of Knowledge. 2nd edn. Basking Ridge: Technics.

OECD (2019) Enhancing Access to and Sharing of Data. Paris: OECD.

Data Rights Stack

Collection → use → licensing rights

PROFESSIONAL DEFINITION

The Data Rights Stack layers the rights associated with data — collection, use, sharing and licensing — clarifying the chain of permissions that governs how data can be exploited.

WHO DEVELOPED IT

A construct from data-governance and privacy practice, reflecting data-protection regimes such as the GDPR.

WHY IT WAS DEVELOPED

It was developed to ensure that downstream uses of data, including AI training, rest on a valid chain of rights.

FIGURE 1 · DATA RIGHTS STACK

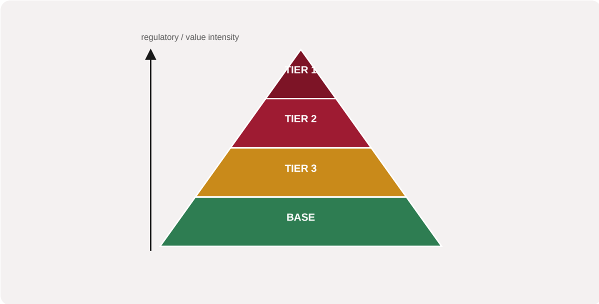

Figure 1. A schematic of the Data Rights Stack as applied in BARBM312 (illustrative).

HOW IT IS USED

Each layer of rights is verified for a given data set before it is used; gaps in the chain block or constrain exploitation.

WHEN IT IS USED

Before using data for AI training or productisation.

BY WHOM IT IS USED

Legal, privacy and data teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the often-implicit chain of data permissions explicit, reducing legal exposure in data-driven AI.

RELATED FRAMEWORKS IN THIS COURSE

Data Asset Classification · Slibrary 15
Data Moat Strength · Slibrary 15
Privacy-Value Frontier · Slibrary 15
Synthetic Data Trade-off · Slibrary 15

SOURCES (HARVARD REFERENCING STYLE)

European Parliament and Council (2016) Regulation (EU) 2016/679 (GDPR).
OECD (2021) 'Mapping data portability initiatives, opportunities and challenges'.

Data Moat Strength

Uniqueness × scale × freshness

PROFESSIONAL DEFINITION

Data Moat Strength assesses how defensible a data advantage is, by its uniqueness, scale and freshness, distinguishing fleeting data from a durable moat.

WHO DEVELOPED IT

A synthesis from data-strategy and competitive-advantage analysis of data assets.

WHY IT WAS DEVELOPED

It was developed to test whether a data advantage will persist or be eroded as rivals acquire comparable data.

FIGURE 1 · DATA MOAT STRENGTH

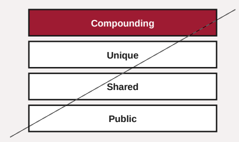

Figure 1. A schematic of the Data Moat Strength as applied in BARBM312 (illustrative).

HOW IT IS USED

A data asset is scored on uniqueness, scale and freshness; only data strong on all three constitutes a durable moat.

WHEN IT IS USED

When evaluating data-based defensibility.

BY WHOM IT IS USED

Strategy, data and investment leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It refines the notion of a 'data moat' into measurable attributes, avoiding overstated claims of data advantage.

RELATED FRAMEWORKS IN THIS COURSE

- **Data Asset Classification** · Slibrary 15
- **Data Rights Stack** · Slibrary 15
- **Privacy-Value Frontier** · Slibrary 15
- **Synthetic Data Trade-off** · Slibrary 15

SOURCES (HARVARD REFERENCING STYLE)

Hagiu, A. and Wright, J. (2020) 'When data creates competitive advantage', HBR, 98(1).

Gregory, R.W. et al. (2021) 'AI and data network effects', AMR, 46(3).

Privacy-Value Frontier

Compliance limits on data value

PROFESSIONAL DEFINITION

The Privacy-Value Frontier expresses the trade-off between the value extractable from personal data and the privacy and compliance constraints that limit its use, identifying the efficient boundary.

WHO DEVELOPED IT

A construct from privacy-economics and responsible-data practice, informed by data-protection law.

WHY IT WAS DEVELOPED

It was developed to help firms maximise legitimate data value without breaching privacy norms or law.

FIGURE 1 · PRIVACY-VALUE FRONTIER

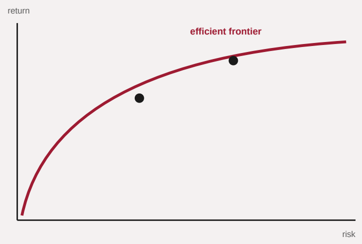

Figure 1. A schematic of the Privacy-Value Frontier as applied in BARBM312 (illustrative).

HOW IT IS USED

Uses of data are placed against the frontier; the firm operates at the boundary that maximises value within compliance.

WHEN IT IS USED

When designing data-driven AI products subject to privacy regulation.

BY WHOM IT IS USED

Privacy, legal and product teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames privacy not only as a constraint but as a boundary defining the efficient, lawful use of data for value.

RELATED FRAMEWORKS IN THIS COURSE

- **Data Asset Classification** · Slibrary 15
- **Data Rights Stack** · Slibrary 15
- **Data Moat Strength** · Slibrary 15
- **Synthetic Data Trade-off** · Slibrary 15

SOURCES (HARVARD REFERENCING STYLE)

Acquisti, A., Taylor, C. and Wagman, L. (2016) 'The economics of privacy', Journal of Economic Literature, 54(2).
European Parliament and Council (2016) Regulation (EU) 2016/679 (GDPR).

Synthetic Data Trade-off

Cheap data vs. real-world fidelity

PROFESSIONAL DEFINITION

The Synthetic Data Trade-off weighs the low cost and privacy benefits of artificially generated data against its potential loss of fidelity to real-world distributions.

WHO DEVELOPED IT

A construct from machine-learning practice as synthetic data became a common alternative to scarce or sensitive real data.

WHY IT WAS DEVELOPED

It was developed to guide when synthetic data is a sound substitute and when it risks degrading model performance.

FIGURE 1 · SYNTHETIC DATA TRADE-OFF

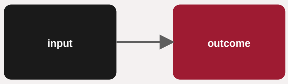

Figure 1. A schematic of the Synthetic Data Trade-off as applied in BARBM312 (illustrative).

HOW IT IS USED

For a use case, the cost and privacy gains of synthetic data are weighed against fidelity risk and validated against real data where possible.

WHEN IT IS USED

When sourcing training data under cost or privacy constraints.

BY WHOM IT IS USED

ML engineering, data and privacy teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames synthetic data as a deliberate trade-off rather than a free substitute, guarding against silent model degradation.

RELATED FRAMEWORKS IN THIS COURSE

- **Data Asset Classification** · Slibrary 15
- **Data Rights Stack** · Slibrary 15
- **Data Moat Strength** · Slibrary 15
- **Privacy-Value Frontier** · Slibrary 15

SOURCES (HARVARD REFERENCING STYLE)

Jordon, J. et al. (2022) 'Synthetic data — what, why and how?', Royal Society / The Alan Turing Institute.

Nikolenko, S.I. (2021) Synthetic Data for Deep Learning. Cham: Springer.

IP Protection Portfolio

Layered patent + secret + brand defence

PROFESSIONAL DEFINITION

An IP Protection Portfolio combines patents, trade secrets, trademarks and other rights into a layered defence, recognising that no single instrument fully protects an AI business.

WHO DEVELOPED IT

A construct from intellectual-property strategy and management practice.

WHY IT WAS DEVELOPED

It was developed because AI value spans models, data, methods and brand, each best protected by a different instrument.

FIGURE 1 · IP PROTECTION PORTFOLIO

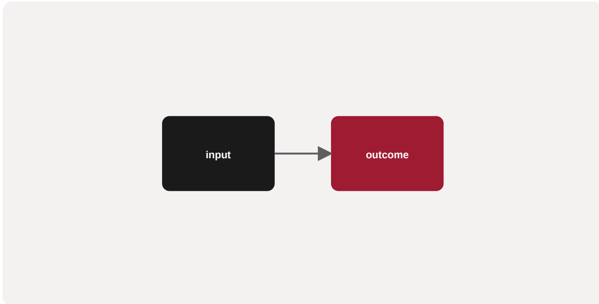

Figure 1. A schematic of the IP Protection Portfolio as applied in BARBM312 (illustrative).

HOW IT IS USED

Each asset is matched to the most effective protection and the portfolio assembled to cover gaps and reinforce defensibility.

WHEN IT IS USED

When designing the overall IP strategy for an AI venture.

BY WHOM IT IS USED

Legal, IP and executive leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It treats protection as a coordinated portfolio rather than isolated filings, fitting the multi-faceted nature of AI value.

RELATED FRAMEWORKS IN THIS COURSE

- **Licensing Models Map** · Slibrary 16
- **Freedom-to-Operate Check** · Slibrary 16
- **Open-Source Compliance Ladder** · Slibrary 16
- **AI Output Ownership Flow** · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

WIPO (2020) WIPO Technology Trends 2019: Artificial Intelligence. Geneva: WIPO.

Reitzig, M. (2004) 'Strategic management of intellectual property', MIT Sloan Management Review, 45(3).

Licensing Models Map

Open · permissive · restrictive · proprietary

PROFESSIONAL DEFINITION

The Licensing Models Map arranges software and data licences from fully open and permissive through to restrictive and proprietary, clarifying the rights and obligations each entails.

WHO DEVELOPED IT

A construct from open-source and IP-licensing practice, reflecting licence taxonomies maintained by the Open Source Initiative.

WHY IT WAS DEVELOPED

It was developed to help firms choose and comply with licences, since obligations differ sharply across the spectrum.

FIGURE 1 · LICENSING MODELS MAP

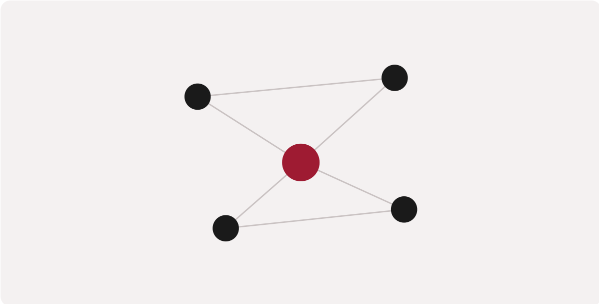

Figure 1. A schematic of the Licensing Models Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Components and outputs are mapped to their licences; obligations (e.g. copyleft) are identified and managed before distribution.

WHEN IT IS USED

When using or distributing AI models, code and data.

BY WHOM IT IS USED

Legal, engineering and compliance teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the consequences of licence choice explicit, particularly important where AI code may inherit obligations from training data.

RELATED FRAMEWORKS IN THIS COURSE

- **IP Protection Portfolio** · Slibrary 16
- **Freedom-to-Operate Check** · Slibrary 16
- **Open-Source Compliance Ladder** · Slibrary 16
- **AI Output Ownership Flow** · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Open Source Initiative (2024) 'Licenses & Standards'.

Rosen, L. (2004) Open Source Licensing. Upper Saddle River: Prentice Hall.

Freedom-to-Operate Check

Are you infringing someone else?

PROFESSIONAL DEFINITION

A Freedom-to-Operate (FTO) check assesses whether a product can be commercialised without infringing third-party intellectual-property rights, reducing the risk of litigation.

WHO DEVELOPED IT

A standard IP-management practice used widely in technology and pharmaceutical commercialisation.

WHY IT WAS DEVELOPED

It was developed to surface infringement risk before launch, when remedies are cheaper than after.

FIGURE 1 · FREEDOM-TO-OPERATE CHECK

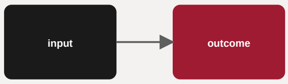

Figure 1. A schematic of the Freedom-to-Operate Check as applied in BARBM312 (illustrative).

HOW IT IS USED

Relevant third-party rights are searched and analysed; the product is cleared, designed around obstacles, or licences are sought.

WHEN IT IS USED

Before launching or scaling an AI product.

BY WHOM IT IS USED

Legal, IP and product-launch teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It brings disciplined infringement diligence to fast-moving AI launches, where training data and methods create novel exposure.

RELATED FRAMEWORKS IN THIS COURSE

- **IP Protection Portfolio** · Slibrary 16
- **Licensing Models Map** · Slibrary 16
- **Open-Source Compliance Ladder** · Slibrary 16
- **AI Output Ownership Flow** · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

WIPO (2020) 'Freedom to operate', WIPO guidance.

Cornish, W., Llewelyn, D. and Aplin, T. (2019) Intellectual Property. London: Sweet & Maxwell.

Open-Source Compliance Ladder

Obligations by licence type

PROFESSIONAL DEFINITION

The Open-Source Compliance Ladder orders open-source obligations by stringency — from permissive notice requirements to strong copyleft — guiding the compliance effort each licence demands.

WHO DEVELOPED IT

A construct from open-source-compliance practice, reflecting work by bodies such as the Linux Foundation's OpenChain project.

WHY IT WAS DEVELOPED

It was developed to help organisations comply with open-source obligations, which scale with licence type and can affect proprietary code.

FIGURE 1 · OPEN-SOURCE COMPLIANCE LADDER

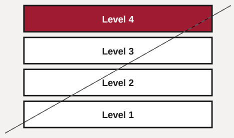

Figure 1. A schematic of the Open-Source Compliance Ladder as applied in BARBM312 (illustrative).

HOW IT IS USED

Each component's licence is placed on the ladder and the corresponding obligations met before distribution.

WHEN IT IS USED

When incorporating open-source AI components into products.

BY WHOM IT IS USED

Legal, engineering and compliance teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It clarifies the graduated nature of open-source obligations, helping avoid inadvertent copyleft exposure in AI products.

RELATED FRAMEWORKS IN THIS COURSE

- **IP Protection Portfolio** · Slibrary 16
- **Licensing Models Map** · Slibrary 16
- **Freedom-to-Operate Check** · Slibrary 16
- **AI Output Ownership Flow** · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

OpenChain Project (2022) ISO/IEC 5230: Open Source Compliance.
Fontana, R. et al. (2021) Open Source Compliance in the Enterprise. Linux Foundation.

AI Output Ownership Flow

Who owns model-generated code/content

PROFESSIONAL DEFINITION

The AI Output Ownership Flow traces who owns content or code generated by an AI system, following the chain from training data and model terms through to user and provider agreements.

WHO DEVELOPED IT

A construct from technology-law practice responding to the ownership uncertainties of generative AI outputs.

WHY IT WAS DEVELOPED

It was developed because ownership of AI-generated output is unsettled and depends on a chain of contracts and copyright rules.

FIGURE 1 · AI OUTPUT OWNERSHIP FLOW

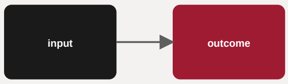

Figure 1. A schematic of the AI Output Ownership Flow as applied in BARBM312 (illustrative).

HOW IT IS USED

The flow follows model terms, input rights and user agreements to determine likely ownership and any encumbrances on output.

WHEN IT IS USED

When commercialising AI-generated content or code.

BY WHOM IT IS USED

Legal, product and commercial teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It maps an emerging, contested area into a navigable flow, essential for capturing value from generative outputs safely.

RELATED FRAMEWORKS IN THIS COURSE

- **IP Protection Portfolio** · Slibrary 16
- **Licensing Models Map** · Slibrary 16
- **Freedom-to-Operate Check** · Slibrary 16
- **Open-Source Compliance Ladder** · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

US Copyright Office (2023) 'Copyright registration guidance: Works containing AI-generated material'.

Lee, J.-A., Hilty, R.M. and Liu, K.-C. (eds.) (2021) Artificial Intelligence and Intellectual Property. Oxford: OUP.

OODA Loop (Enterprise)

Decide faster than the market moves

PROFESSIONAL DEFINITION

The Enterprise OODA Loop applies Boyd's Observe–Orient–Decide–Act cycle at organisational scale, treating the speed of corporate re-decision as a source of competitive advantage in fast-changing markets.

WHO DEVELOPED IT

Extends John Boyd's military OODA loop to organisations, as taken up in agile-strategy and business-agility literature.

WHY IT WAS DEVELOPED

It was developed to argue that, under disruption, the organisation that re-decides fastest can outmanoeuvre larger, slower rivals.

FIGURE 1 · OODA LOOP (ENTERPRISE)

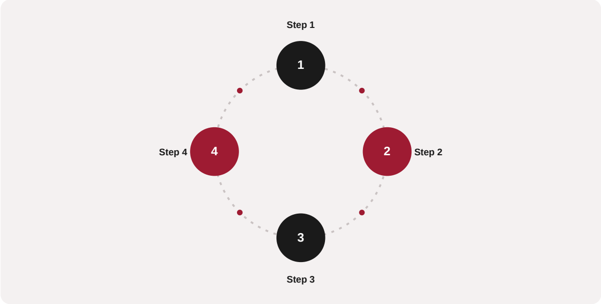

Figure 1. A schematic of the OODA Loop (Enterprise) as applied in BARBM312 (illustrative).

HOW IT IS USED

Corporate decision cycles are mapped and accelerated; AI compresses observation and orientation while humans retain fast, accountable decision and action.

WHEN IT IS USED

During disruption, rapid market shifts and agile-transformation efforts.

BY WHOM IT IS USED

Senior leaders, strategists and transformation teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes agility as decision tempo, making speed of re-decision — not size — the moat under AI-accelerated change.

RELATED FRAMEWORKS IN THIS COURSE

- **Cynefin Framework** · Slibrary 17
- **Agile Operating Model** · Slibrary 17
- **Sense-Respond Loop** · Slibrary 17
- **Real Options Lens** · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

Boyd, J. (1986) 'Patterns of Conflict' (briefing).

Reeves, M., Haanaes, K. and Sinha, J. (2015) Your Strategy Needs a Strategy. Boston: HBR Press.

Cynefin Framework

Match response to problem complexity

PROFESSIONAL DEFINITION

The Cynefin Framework sorts situations into domains — clear, complicated, complex and chaotic — so that leaders match their response to the kind of problem they face rather than applying one approach to all.

WHO DEVELOPED IT

Created by Dave Snowden while at IBM, set out with Mary Boone in a 2007 Harvard Business Review article.

WHY IT WAS DEVELOPED

It was developed to counter the error of treating every problem as merely complicated, when complex problems require probing and adaptation.

FIGURE 1 · CYNEFIN FRAMEWORK

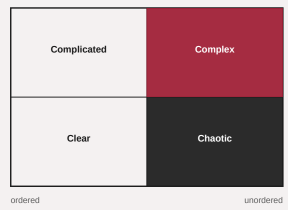

Figure 1. A schematic of the Cynefin Framework as applied in BARBM312 (illustrative).

HOW IT IS USED

A situation is placed in a domain; complex problems are met with safe-to-fail experiments and amplification of what works, not detailed up-front planning.

WHEN IT IS USED

When deciding how to respond to disruption and uncertainty.

BY WHOM IT IS USED

Leaders, strategists and decision-makers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It introduces a sense-making typology that prevents misapplied management approaches, especially in the complex domain where disruption lives.

RELATED FRAMEWORKS IN THIS COURSE

- **OODA Loop (Enterprise)** · Slibrary 17
- **Agile Operating Model** · Slibrary 17
- **Sense-Respond Loop** · Slibrary 17
- **Real Options Lens** · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

Snowden, D.J. and Boone, M.E. (2007) 'A leader's framework for decision making', Harvard Business Review, 85(11).

Kurtz, C.F. and Snowden, D.J. (2003) 'The new dynamics of strategy', IBM Systems Journal, 42(3).

Agile Operating Model

Squads, tribes and fast feedback

PROFESSIONAL DEFINITION

An Agile Operating Model organises work around small, cross-functional teams with fast feedback loops, enabling rapid responsiveness in place of slow, hierarchical decision-making.

WHO DEVELOPED IT

Synthesised from agile software practice and scaled-agile thinking, popularised for whole organisations by writers such as Aghina and colleagues at McKinsey.

WHY IT WAS DEVELOPED

It was developed to give organisations the responsiveness that fast-changing, AI-shaped markets demand.

FIGURE 1 · AGILE OPERATING MODEL

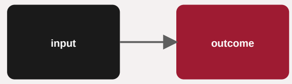

Figure 1. A schematic of the Agile Operating Model as applied in BARBM312 (illustrative).

HOW IT IS USED

Work is reorganised into empowered teams with clear missions and fast feedback; governance lightens to allow speed within guardrails.

WHEN IT IS USED

During agile transformation and when responsiveness becomes a strategic priority.

BY WHOM IT IS USED

Executives, transformation leaders and team leads.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It scales agile principles from software to the enterprise, structurally enabling the decision speed agility requires.

RELATED FRAMEWORKS IN THIS COURSE

- **OODA Loop (Enterprise)** · Slibrary 17
- **Cynefin Framework** · Slibrary 17
- **Sense-Respond Loop** · Slibrary 17
- **Real Options Lens** · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

Aghina, W. et al. (2018) 'The five trademarks of agile organizations', McKinsey & Company.

Rigby, D.K., Sutherland, J. and Takeuchi, H. (2016) 'Embracing agile', Harvard Business Review, 94(5).

Sense-Respond Loop

Continuous adaptation to signals

PROFESSIONAL DEFINITION

The Sense-Respond Loop is a model of continuous adaptation in which an organisation senses signals from its environment and responds rapidly, iterating rather than executing a fixed long-range plan.

WHO DEVELOPED IT

Articulated by Stephan Haeckel in Adaptive Enterprise (1999) as an alternative to make-and-sell models.

WHY IT WAS DEVELOPED

It was developed for environments too turbulent for predictive planning, where responsiveness beats forecasting.

FIGURE 1 · SENSE-RESPOND LOOP

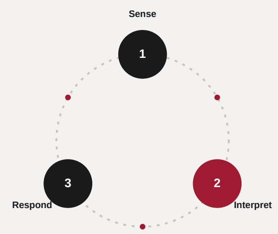

Figure 1. A schematic of the Sense-Respond Loop as applied in BARBM312 (illustrative).

HOW IT IS USED

The organisation builds capability to sense early and respond fast, treating strategy as continuous adaptation.

WHEN IT IS USED

In volatile markets and during AI-driven disruption.

BY WHOM IT IS USED

Strategists and operating leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes strategy from prediction to adaptation, aligning with agile and OODA thinking under uncertainty.

RELATED FRAMEWORKS IN THIS COURSE

- **OODA Loop (Enterprise)** · Slibrary 17
- **Cynefin Framework** · Slibrary 17
- **Agile Operating Model** · Slibrary 17
- **Real Options Lens** · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

Haeckel, S.H. (1999) Adaptive Enterprise: Creating and Leading Sense-and-Respond Organizations. Boston: HBS Press.

Reeves, M., Levin, S. and Ueda, D. (2016) 'The biology of corporate survival', Harvard Business Review, 94(1).

Real Options Lens

Small bets that buy future flexibility

PROFESSIONAL DEFINITION

The Real Options Lens treats strategic investments as options — small commitments that buy the right, but not the obligation, to invest further — valuing flexibility under uncertainty.

WHO DEVELOPED IT

Adapted from financial-options theory to strategy by scholars including Timothy Luehrman and the real-options literature.

WHY IT WAS DEVELOPED

It was developed to value managerial flexibility, which traditional discounted-cash-flow analysis undervalues in uncertain settings.

FIGURE 1 · REAL OPTIONS LENS

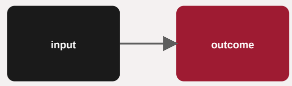

Figure 1. A schematic of the Real Options Lens as applied in BARBM312 (illustrative).

HOW IT IS USED

Initiatives are structured as staged options; small bets preserve the right to scale, abandon or pivot as uncertainty resolves.

WHEN IT IS USED

When investing under high uncertainty, including AI bets with unclear payoffs.

BY WHOM IT IS USED

Strategy, finance and innovation leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It values the flexibility to wait, expand or abandon — a natural fit for sequencing uncertain AI investments.

RELATED FRAMEWORKS IN THIS COURSE

- **OODA Loop (Enterprise)** · Slibrary 17
- **Cynefin Framework** · Slibrary 17
- **Agile Operating Model** · Slibrary 17
- **Sense-Respond Loop** · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

Luehrman, T.A. (1998) 'Strategy as a portfolio of real options', Harvard Business Review, 76(5).

Dixit, A.K. and Pindyck, R.S. (1994) Investment Under Uncertainty. Princeton: PUP.

Disruption S-Curves

Incumbent vs. entrant trajectories

PROFESSIONAL DEFINITION

Disruption S-Curves plot the performance trajectories of incumbent and entrant technologies over time, showing how an initially inferior entrant can overtake an incumbent as its curve steepens.

WHO DEVELOPED IT

Rooted in technology-S-curve analysis (Foster) and central to Clayton Christensen's theory of disruptive innovation.

WHY IT WAS DEVELOPED

It was developed to explain how disruptors that start below incumbents' performance can eventually surpass them, often unnoticed until too late.

FIGURE 1 · DISRUPTION S-CURVES

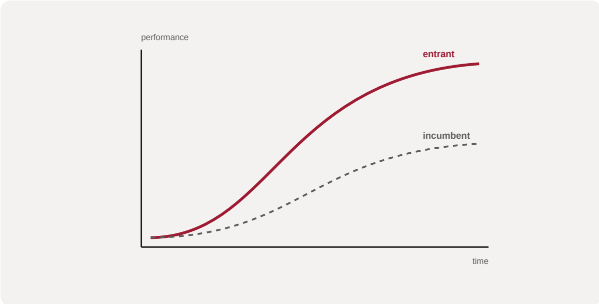

Figure 1. A schematic of the Disruption S-Curves as applied in BARBM312 (illustrative).

HOW IT IS USED

Incumbent and entrant trajectories are plotted; the crossover point and the incumbent's vulnerability are assessed.

WHEN IT IS USED

When analysing disruptive threats and timing strategic responses.

BY WHOM IT IS USED

Strategists, innovation leaders and executives.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It visualises why good firms miss disruption — they track the wrong curve — underpinning the Innovator's Dilemma.

RELATED FRAMEWORKS IN THIS COURSE

- **Innovator's Dilemma** · Slibrary 18
- **Three Horizons of Growth** · Slibrary 18
- **Adjacent Possible Map** · Slibrary 18
- **Disruption Vulnerability Audit** · Slibrary 18

SOURCES (HARVARD REFERENCING STYLE)

Foster, R.N. (1986) Innovation: The Attacker's Advantage. New York: Summit.

Christensen, C.M. (1997) The Innovator's Dilemma. Boston: HBS Press.

Innovator's Dilemma

Why good firms miss disruption

PROFESSIONAL DEFINITION

The Innovator's Dilemma is the finding that well-managed incumbents can fail precisely because they listen to their best customers and invest in sustaining innovations, leaving them exposed to disruptive entrants.

WHO DEVELOPED IT

Articulated by Clayton Christensen in The Innovator's Dilemma (1997).

WHY IT WAS DEVELOPED

It was developed to explain the paradox that doing everything 'right' by mainstream-customer logic can cause incumbents to lose to initially inferior disruptors.

FIGURE 1 · INNOVATOR'S DILEMMA

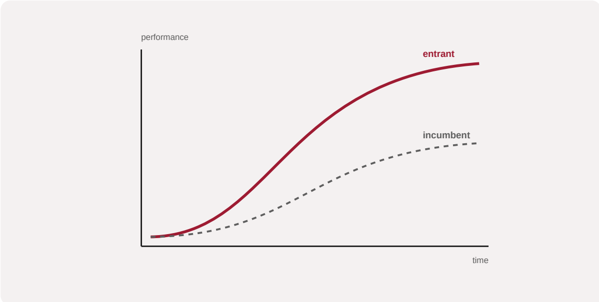

Figure 1. A schematic of the Innovator's Dilemma as applied in BARBM312 (illustrative).

HOW IT IS USED

The firm examines whether disruptive threats are dismissed because they serve undemanding or new segments, and whether it can pursue them without cannibalising the core.

WHEN IT IS USED

When facing potential disruption and deciding how to respond.

BY WHOM IT IS USED

Executives, strategists and innovation leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It overturned the assumption that incumbents fail through complacency, showing that competence at the core can itself be the vulnerability.

RELATED FRAMEWORKS IN THIS COURSE

- **Disruption S-Curves** · Slibrary 18
- **Three Horizons of Growth** · Slibrary 18
- **Adjacent Possible Map** · Slibrary 18
- **Disruption Vulnerability Audit** · Slibrary 18

SOURCES (HARVARD REFERENCING STYLE)

Christensen, C.M. (1997) The Innovator's Dilemma. Boston: HBS Press.

Christensen, C.M. and Raynor, M. (2003) The Innovator's Solution. Boston: HBS Press.

Three Horizons of Growth

Defend, build, create simultaneously

PROFESSIONAL DEFINITION

The Three Horizons of Growth frames the simultaneous management of defending the core (H1), building emerging businesses (H2) and creating future options (H3), so disruption does not catch the firm investing only in today.

WHO DEVELOPED IT

Introduced by Baghai, Coley and White of McKinsey in The Alchemy of Growth (1999).

WHY IT WAS DEVELOPED

It was developed to balance exploitation of the present with exploration of the future, the failure of which leaves incumbents exposed to disruptors.

FIGURE 1 · THREE HORIZONS OF GROWTH

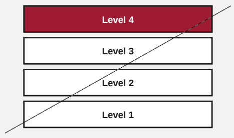

Figure 1. A schematic of the Three Horizons of Growth as applied in BARBM312 (illustrative).

HOW IT IS USED

Initiatives are sorted into horizons and funded in parallel, ensuring future growth is protected even while the core is optimised.

WHEN IT IS USED

During growth and innovation portfolio planning under disruption.

BY WHOM IT IS USED

Executives and innovation governance teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It institutionalises parallel investment across time horizons, countering the short-termism that disruption exploits.

RELATED FRAMEWORKS IN THIS COURSE

- **Disruption S-Curves** · Slibrary 18
- **Innovator's Dilemma** · Slibrary 18
- **Adjacent Possible Map** · Slibrary 18
- **Disruption Vulnerability Audit** · Slibrary 18

SOURCES (HARVARD REFERENCING STYLE)

Baghai, M., Coley, S. and White, D. (1999) The Alchemy of Growth. Reading: Perseus.

McKinsey & Company (2009) 'Enduring Ideas: The three horizons of growth'.

Adjacent Possible Map

Where the next move can realistically go

PROFESSIONAL DEFINITION

The Adjacent Possible Map identifies the set of moves realistically reachable from a firm's current position — the adjacent possibilities — guiding expansion that builds on existing capabilities.

WHO DEVELOPED IT

Drawing on Stuart Kauffman's biological concept of the 'adjacent possible', applied to strategy and innovation.

WHY IT WAS DEVELOPED

It was developed to focus expansion on feasible, capability-leveraging adjacencies rather than distant, unsupported leaps.

FIGURE 1 · ADJACENT POSSIBLE MAP

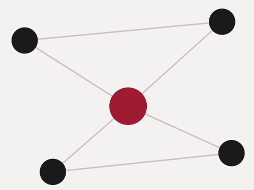

Figure 1. A schematic of the Adjacent Possible Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Current capabilities and assets define the reachable adjacent moves; expansion targets the most valuable feasible adjacency.

WHEN IT IS USED

When planning growth and diversification under change.

BY WHOM IT IS USED

Strategy and corporate-development teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It disciplines expansion by realistic reach, balancing ambition against capability in volatile markets.

RELATED FRAMEWORKS IN THIS COURSE

- **Disruption S-Curves** · Slibrary 18
- **Innovator's Dilemma** · Slibrary 18
- **Three Horizons of Growth** · Slibrary 18
- **Disruption Vulnerability Audit** · Slibrary 18

SOURCES (HARVARD REFERENCING STYLE)

Kauffman, S. (2000) Investigations. Oxford: OUP.
Johnson, S. (2010) Where Good Ideas Come From. New York: Riverhead.

Disruption Vulnerability Audit

How exposed is your model?

PROFESSIONAL DEFINITION

A Disruption Vulnerability Audit assesses how exposed a business model is to disruption, examining where value is created, how easily it could be unbundled, and where entrants might attack.

WHO DEVELOPED IT

A practitioner construct from disruption and business-model-innovation analysis.

WHY IT WAS DEVELOPED

It was developed to make incumbents honestly assess their exposure before a disruptor forces the issue.

FIGURE 1 · DISRUPTION VULNERABILITY AUDIT

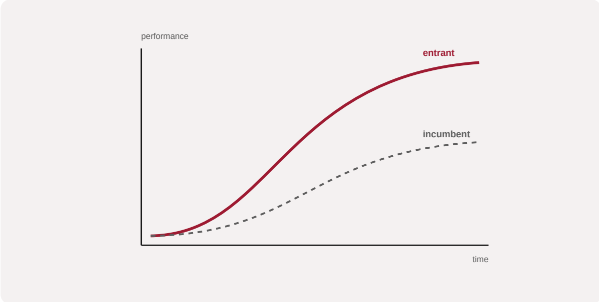

Figure 1. A schematic of the Disruption Vulnerability Audit as applied in BARBM312 (illustrative).

HOW IT IS USED

The model is stress-tested against disruptive moves — unbundling, new entrants, AI-native rivals — and weak points identified.

WHEN IT IS USED

During strategic risk review and disruption planning.

BY WHOM IT IS USED

Executives, strategists and risk leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It converts the abstract fear of disruption into a structured self-assessment of specific vulnerabilities.

RELATED FRAMEWORKS IN THIS COURSE

- **Disruption S-Curves** · Slibrary 18
- **Innovator's Dilemma** · Slibrary 18
- **Three Horizons of Growth** · Slibrary 18
- **Adjacent Possible Map** · Slibrary 18

SOURCES (HARVARD REFERENCING STYLE)

Christensen, C.M. (1997) The Innovator's Dilemma. Boston: HBS Press.

Wessel, M. and Christensen, C.M. (2012) 'Surviving disruption', Harvard Business Review, 90(12).

Brand Pyramid

Purpose → story → truths → offering

PROFESSIONAL DEFINITION

The Brand Pyramid layers a brand from its foundational context and offering up through truths and story to its overarching purpose, providing a structured view of what a brand stands for.

WHO DEVELOPED IT

A widely used brand-strategy model with variants from agencies and brand-management practice; related to Keller's brand-equity pyramid.

WHY IT WAS DEVELOPED

It was developed to give brand strategy a coherent structure, linking tangible offering to intangible purpose.

FIGURE 1 · BRAND PYRAMID

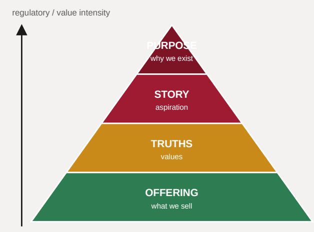

Figure 1. A schematic of the Brand Pyramid as applied in BARBM312 (illustrative).

HOW IT IS USED

Each layer is articulated from the base up; rebranding decisions are tested for consistency across the pyramid.

WHEN IT IS USED

During brand strategy and rebranding.

BY WHOM IT IS USED

Brand, marketing and executive leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It organises brand meaning hierarchically, ensuring rebrands change the right layer without fracturing the whole.

RELATED FRAMEWORKS IN THIS COURSE

- **12 Brand Archetypes** · Slibrary 19
- **Rebrand Decision Matrix** · Slibrary 19
- **StoryBrand Framework** · Slibrary 19
- **Brand Equity Bridge** · Slibrary 19

SOURCES (HARVARD REFERENCING STYLE)

Keller, K.L. (2001) 'Building customer-based brand equity', Marketing Management, 10(2).

Aaker, D.A. (1996) Building Strong Brands. New York: Free Press.

12 Brand Archetypes

Personality framework for repositioning

PROFESSIONAL DEFINITION

The 12 Brand Archetypes apply Jungian archetypes — such as the Hero, Sage, Creator and Outlaw — to brands, giving each a consistent personality that shapes voice, story and positioning.

WHO DEVELOPED IT

Adapted to branding by Margaret Mark and Carol Pearson in The Hero and the Outlaw (2001), drawing on Carl Jung's archetypes.

WHY IT WAS DEVELOPED

It was developed to give brands a coherent, resonant personality grounded in deep, universal patterns.

FIGURE 1 · 12 BRAND ARCHETYPES

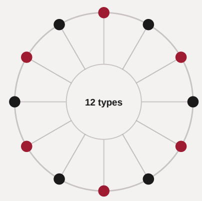

Figure 1. A schematic of the 12 Brand Archetypes as applied in BARBM312 (illustrative).

HOW IT IS USED

A brand is matched to an archetype that guides repositioning, tone and narrative consistently across touchpoints.

WHEN IT IS USED

During brand definition and rebranding.

BY WHOM IT IS USED

Brand and marketing strategists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It links brand personality to enduring psychological archetypes, giving rebrands a stable foundation for new expression.

RELATED FRAMEWORKS IN THIS COURSE

- **Brand Pyramid** · Slibrary 19
- **Rebrand Decision Matrix** · Slibrary 19
- **StoryBrand Framework** · Slibrary 19
- **Brand Equity Bridge** · Slibrary 19

SOURCES (HARVARD REFERENCING STYLE)

Mark, M. and Pearson, C.S. (2001) The Hero and the Outlaw. New York: McGraw-Hill.

Jung, C.G. (1968) The Archetypes and the Collective Unconscious. Princeton: PUP.

Rebrand Decision Matrix

Refresh vs. reposition vs. rebuild

PROFESSIONAL DEFINITION

The Rebrand Decision Matrix distinguishes when a brand needs a light refresh, a fuller reposition, or a complete rebuild, by the scale of change required and the state of existing brand equity.

WHO DEVELOPED IT

A practitioner construct from brand-strategy and rebranding practice.

WHY IT WAS DEVELOPED

It was developed to prevent both timid refreshes that fail to respond to disruption and reckless rebuilds that destroy equity.

FIGURE 1 · REBRAND DECISION MATRIX

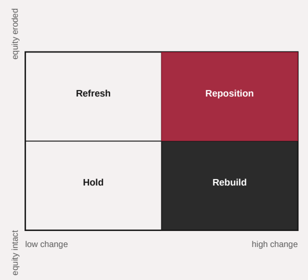

Figure 1. A schematic of the Rebrand Decision Matrix as applied in BARBM312 (illustrative).

HOW IT IS USED

The situation is placed by required change and equity health; the matrix recommends refresh, reposition or rebuild.

WHEN IT IS USED

When deciding the scope of a rebrand under disruption.

BY WHOM IT IS USED

Brand, marketing and executive leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It matches the scale of brand change to circumstance, guarding the equity that is often the firm's real moat.

RELATED FRAMEWORKS IN THIS COURSE

- **Brand Pyramid** · Slibrary 19
- **12 Brand Archetypes** · Slibrary 19
- **StoryBrand Framework** · Slibrary 19
- **Brand Equity Bridge** · Slibrary 19

SOURCES (HARVARD REFERENCING STYLE)

Aaker, D.A. (1996) Building Strong Brands. New York: Free Press.

Muzellec, L. and Lambkin, M. (2006) 'Corporate rebranding', European Journal of Marketing, 40(7/8).

StoryBrand Framework

Make the customer the hero

PROFESSIONAL DEFINITION

The StoryBrand Framework structures brand messaging as a story in which the customer is the hero, the brand is the guide, and a clear plan leads to success and away from failure.

WHO DEVELOPED IT

Developed by Donald Miller in Building a StoryBrand (2017), applying narrative structure to marketing.

WHY IT WAS DEVELOPED

It was developed to cut through unclear messaging by casting the customer, not the brand, as the hero of the story.

FIGURE 1 · STORYBRAND FRAMEWORK

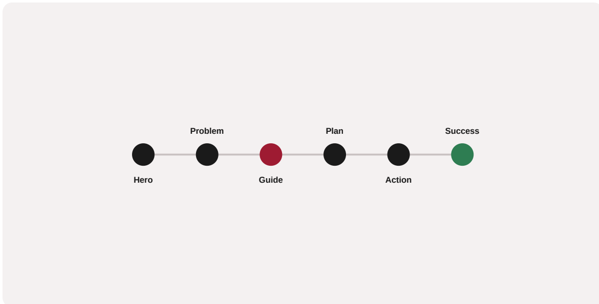

Figure 1. A schematic of the StoryBrand Framework as applied in BARBM312 (illustrative).

HOW IT IS USED

Messaging is rebuilt around the seven-part story structure, with the brand positioned as the guide offering a plan.

WHEN IT IS USED

During brand messaging, repositioning and rebranding.

BY WHOM IT IS USED

Marketing and brand teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the customer the hero — a discipline that clarifies messaging and resists brand self-absorption.

RELATED FRAMEWORKS IN THIS COURSE

- **Brand Pyramid** · Slibrary 19
- **12 Brand Archetypes** · Slibrary 19
- **Rebrand Decision Matrix** · Slibrary 19
- **Brand Equity Bridge** · Slibrary 19

SOURCES (HARVARD REFERENCING STYLE)

Miller, D. (2017) Building a StoryBrand. Nashville: HarperCollins Leadership.

Campbell, J. (1949) The Hero with a Thousand Faces. New York: Pantheon.

Brand Equity Bridge

Carrying value across a transition

PROFESSIONAL DEFINITION

The Brand Equity Bridge depicts how a rebrand can carry accumulated brand equity across a transition, preserving the value of recognition and trust while adopting a new identity.

WHO DEVELOPED IT

A construct from brand-management practice on managing brand transitions and migrations.

WHY IT WAS DEVELOPED

It was developed to prevent rebrands from discarding hard-won equity, a frequent and costly error.

FIGURE 1 · BRAND EQUITY BRIDGE

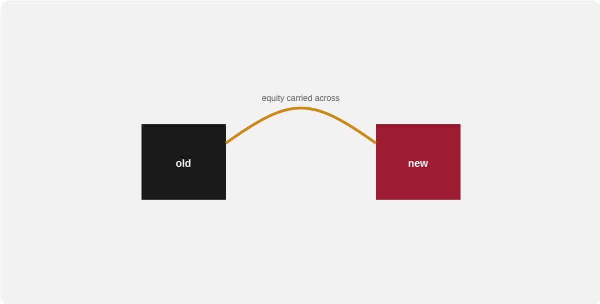

Figure 1. A schematic of the Brand Equity Bridge as applied in BARBM312 (illustrative).

HOW IT IS USED

Sources of equity are identified and deliberately carried across the transition through continuity cues and migration strategy.

WHEN IT IS USED

During rebranding, mergers and brand migrations.

BY WHOM IT IS USED

Brand, marketing and executive leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It treats equity transfer as a managed bridge, protecting intangible value through change.

RELATED FRAMEWORKS IN THIS COURSE

- **Brand Pyramid** · Slibrary 19
- **12 Brand Archetypes** · Slibrary 19
- **Rebrand Decision Matrix** · Slibrary 19
- **StoryBrand Framework** · Slibrary 19

SOURCES (HARVARD REFERENCING STYLE)

Aaker, D.A. (1991) Managing Brand Equity. New York: Free Press.

Muzellec, L. and Lambkin, M. (2006) 'Corporate rebranding', European Journal of Marketing, 40(7/8).

Kotter's 8 Steps

Leading change without losing people

PROFESSIONAL DEFINITION

Kotter's 8-Step Process for Leading Change sets out a sequence — from establishing urgency to anchoring change in the culture — for delivering organisational change without the common failure modes.

WHO DEVELOPED IT

Developed by John Kotter in Leading Change (1996), based on study of corporate change efforts.

WHY IT WAS DEVELOPED

It was developed because most change programmes fail, often for want of urgency or short-term wins, which the steps address.

FIGURE 1 · KOTTER'S 8 STEPS

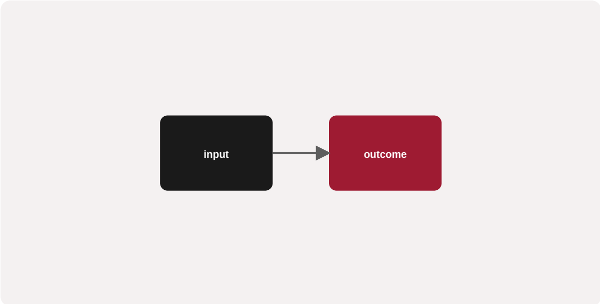

Figure 1. A schematic of the Kotter's 8 Steps as applied in BARBM312 (illustrative).

HOW IT IS USED

Change is led through the eight steps in sequence, ensuring urgency, coalition, vision, communication, wins and anchoring.

WHEN IT IS USED

During organisational change, including AI-driven transformation and rebranding.

BY WHOM IT IS USED

Change leaders, executives and transformation teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It distilled change leadership into a memorable, evidence-based sequence that remains a default playbook.

RELATED FRAMEWORKS IN THIS COURSE

- **ADKAR Model** · Slibrary 20
- **Burning Platform vs. Burning Ambition** · Slibrary 20
- **Transition Curve** · Slibrary 20
- **Change Readiness Radar** · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

Kotter, J.P. (1996) Leading Change. Boston: HBS Press.

Kotter, J.P. (1995) 'Leading change: Why transformation efforts fail', Harvard Business Review, 73(2).

ADKAR Model

Individual change as the unit of org change

PROFESSIONAL DEFINITION

The ADKAR Model frames individual change through five outcomes — Awareness, Desire, Knowledge, Ability and Reinforcement — treating organisational change as the sum of successful individual transitions.

WHO DEVELOPED IT

Developed by Jeff Hiatt of Prosci, set out in ADKAR: A Model for Change (2006).

WHY IT WAS DEVELOPED

It was developed to make change management concrete at the individual level, where organisational change ultimately succeeds or fails.

FIGURE 1 · ADKAR MODEL

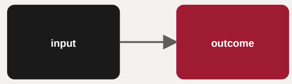

Figure 1. A schematic of the ADKAR Model as applied in BARBM312 (illustrative).

HOW IT IS USED

Each person's progress through the five outcomes is supported and tracked; blockages are addressed at the specific outcome that stalls.

WHEN IT IS USED

During change implementation, including AI adoption.

BY WHOM IT IS USED

Change managers, HR and team leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It locates change in the individual, complementing organisation-level models like Kotter's with a personal-transition lens.

RELATED FRAMEWORKS IN THIS COURSE

- **Kotter's 8 Steps** · Slibrary 20
- **Burning Platform vs. Burning Ambition** · Slibrary 20
- **Transition Curve** · Slibrary 20
- **Change Readiness Radar** · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

Hiatt, J. (2006) ADKAR: A Model for Change in Business, Government and Our Community. Loveland: Prosci.

Prosci (2018) Best Practices in Change Management.

Burning Platform vs. Burning Ambition

Two motives for change

PROFESSIONAL DEFINITION

This framing contrasts two motives for change — a 'burning platform' driven by threat and a 'burning ambition' driven by opportunity — recognising that change can be powered by fear or by aspiration.

WHO DEVELOPED IT

A change-leadership construct, with the burning-platform metaphor popularised in corporate-turnaround discourse and the ambition counterpoint added in later practice.

WHY IT WAS DEVELOPED

It was developed to broaden change motivation beyond crisis, since fear-driven change can be brittle and demoralising.

FIGURE 1 · BURNING PLATFORM VS. BURNING AMBITION

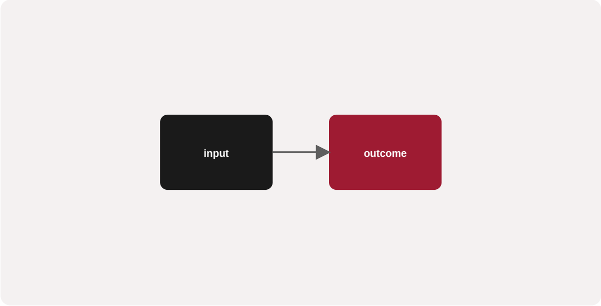

Figure 1. A schematic of the Burning Platform vs. Burning Ambition as applied in BARBM312 (illustrative).

HOW IT IS USED

Leaders choose and communicate the motive — threat, aspiration, or both — that best mobilises the organisation for the change at hand.

WHEN IT IS USED

When framing and launching change and transformation.

BY WHOM IT IS USED

Change leaders and executives.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It balances threat-driven urgency with opportunity-driven aspiration, offering a more sustainable basis for change.

RELATED FRAMEWORKS IN THIS COURSE

- **Kotter's 8 Steps** · Slibrary 20
- **ADKAR Model** · Slibrary 20
- **Transition Curve** · Slibrary 20
- **Change Readiness Radar** · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

Kotter, J.P. (1996) Leading Change. Boston: HBS Press.

Conner, D.R. (1992) Managing at the Speed of Change. New York: Villard.

Transition Curve

Emotional arc of disruption

PROFESSIONAL DEFINITION

The Transition Curve describes the emotional stages people pass through during disruptive change — from shock and resistance to exploration and commitment — so leaders can support each stage.

WHO DEVELOPED IT

Adapted from Elisabeth Kübler-Ross's stages of grief to organisational change by writers such as John Fisher.

WHY IT WAS DEVELOPED

It was developed to help leaders anticipate and support the human, emotional reality of change rather than treating it as purely rational.

FIGURE 1 · TRANSITION CURVE

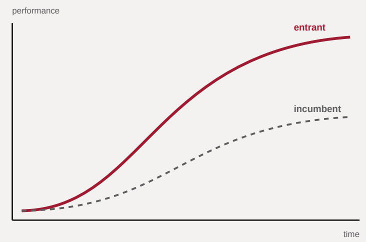

Figure 1. A schematic of the Transition Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

Individuals are met where they are on the curve, with support tailored to their current stage to ease the transition.

WHEN IT IS USED

During disruptive change and transformation.

BY WHOM IT IS USED

Change managers, HR and leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It centres the emotional experience of change, complementing process models with human dynamics.

RELATED FRAMEWORKS IN THIS COURSE

- **Kotter's 8 Steps** · Slibrary 20
- **ADKAR Model** · Slibrary 20
- **Burning Platform vs. Burning Ambition** · Slibrary 20
- **Change Readiness Radar** · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

Kübler-Ross, E. (1969) On Death and Dying. New York: Macmillan.

Fisher, J.M. (2005) 'A time for change', Human Resource Development International, 8(2).

Change Readiness Radar

Is the organisation ready to move?

PROFESSIONAL DEFINITION

A Change Readiness Radar assesses an organisation's preparedness for change across dimensions such as leadership, capability, culture and resources, identifying where readiness must be built first.

WHO DEVELOPED IT

A construct from change-management practice for diagnosing readiness before launching change.

WHY IT WAS DEVELOPED

It was developed to prevent change efforts from launching into unprepared organisations, a common cause of failure.

FIGURE 1 · CHANGE READINESS RADAR

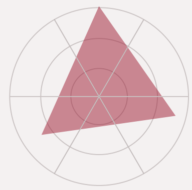

Figure 1. A schematic of the Change Readiness Radar as applied in BARBM312 (illustrative).

HOW IT IS USED

Readiness dimensions are scored on a radar; weak areas are strengthened before, or alongside, the change.

WHEN IT IS USED

Before and during change and transformation programmes.

BY WHOM IT IS USED

Change leaders, HR and executives.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes readiness an explicit, measured precondition rather than an assumption, improving the odds of change success.

RELATED FRAMEWORKS IN THIS COURSE

Kotter's 8 Steps · Slibrary 20 | **ADKAR Model** · Slibrary 20 | **Burning Platform vs. Burning Ambition** · Slibrary 20 | **Transition Curve** · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

Armenakis, A.A., Harris, S.G. and Mossholder, K.W. (1993) 'Creating readiness for organizational change', Human Relations, 46(6).

Prosci (2018) Best Practices in Change Management.

Modern Portfolio Theory

Efficient frontier of risk vs. return

PROFESSIONAL DEFINITION

Modern Portfolio Theory (MPT) shows how combining assets with imperfectly correlated returns reduces portfolio risk for a given expected return, defining an efficient frontier of optimal portfolios.

WHO DEVELOPED IT

Developed by Harry Markowitz in his 1952 paper 'Portfolio Selection', for which he later received the Nobel Prize.

WHY IT WAS DEVELOPED

It was developed to formalise diversification, showing mathematically that risk depends on covariance, not just individual asset risk.

FIGURE 1 · MODERN PORTFOLIO THEORY

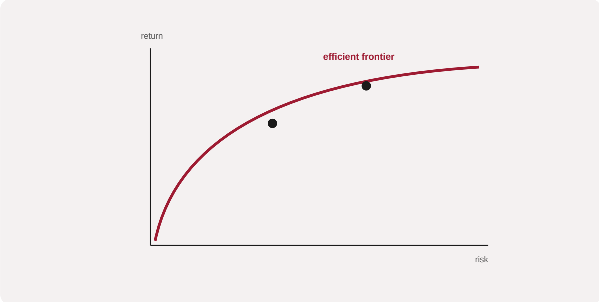

Figure 1. A schematic of the Modern Portfolio Theory as applied in BARBM312 (illustrative).

HOW IT IS USED

Assets are combined to minimise variance for a target return; portfolios are selected from the efficient frontier per the investor's risk appetite.

WHEN IT IS USED

When constructing portfolios and assessing diversification, including AI-driven allocation.

BY WHOM IT IS USED

Portfolio managers, investment analysts and advisers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It established diversification as the 'only free lunch' in investing and founded quantitative portfolio construction — a law AI accelerates but does not repeal.

RELATED FRAMEWORKS IN THIS COURSE

- **Risk-Return Spectrum** · Slibrary 21
- **Diversification Curve** · Slibrary 21
- **Capital Asset Pricing Model** · Slibrary 21
- **Black-Litterman Model** · Slibrary 21

SOURCES (HARVARD REFERENCING STYLE)

Markowitz, H. (1952) 'Portfolio selection', The Journal of Finance, 7(1), pp. 77–91.

Markowitz, H. (1959) Portfolio Selection: Efficient Diversification of Investments. New York: Wiley.

Risk-Return Spectrum

No return without bearing risk

PROFESSIONAL DEFINITION

The Risk-Return Spectrum expresses the foundational principle that higher expected returns require accepting higher risk, ordering asset classes along this trade-off.

WHO DEVELOPED IT

A foundational principle of finance, formalised through portfolio theory and the work of Markowitz, Sharpe and others.

WHY IT WAS DEVELOPED

It was developed to make explicit that return is compensation for bearing risk, disciplining expectations of 'risk-free' high returns.

FIGURE 1 · RISK-RETURN SPECTRUM

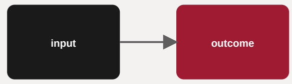

Figure 1. A schematic of the Risk-Return Spectrum as applied in BARBM312 (illustrative).

HOW IT IS USED

Investments are placed on the spectrum; choices are made consistent with the investor's risk tolerance and objectives.

WHEN IT IS USED

When setting investment strategy and evaluating AI-driven products promising high returns.

BY WHOM IT IS USED

Investors, advisers and analysts.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It anchors all investing in the risk-return trade-off, a discipline against AI-hype claims of high return without risk.

RELATED FRAMEWORKS IN THIS COURSE

- **Modern Portfolio Theory** · Slibrary 21
- **Diversification Curve** · Slibrary 21
- **Capital Asset Pricing Model** · Slibrary 21
- **Black-Litterman Model** · Slibrary 21

SOURCES (HARVARD REFERENCING STYLE)

Sharpe, W.F. (1964) 'Capital asset prices', The Journal of Finance, 19(3).

Bodie, Z., Kane, A. and Marcus, A.J. (2021) Investments. New York: McGraw-Hill.

Diversification Curve

Falling risk as assets decorrelate

PROFESSIONAL DEFINITION

The Diversification Curve shows how portfolio risk falls as more imperfectly correlated assets are added, approaching but never eliminating undiversifiable systematic risk.

WHO DEVELOPED IT

Derived from Markowitz's portfolio theory and standard in investment textbooks.

WHY IT WAS DEVELOPED

It was developed to quantify the benefit and the limit of diversification, distinguishing diversifiable from systematic risk.

FIGURE 1 · DIVERSIFICATION CURVE

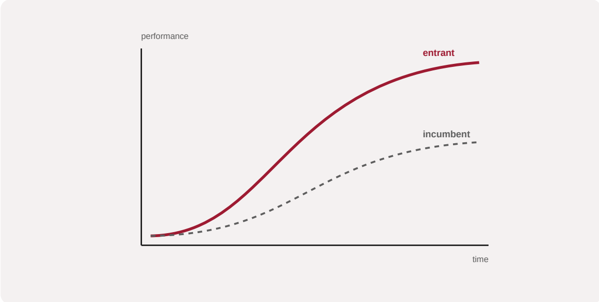

Figure 1. A schematic of the Diversification Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

Risk is plotted against the number of holdings; the curve shows the diminishing benefit of adding assets and the irreducible systematic floor.

WHEN IT IS USED

When designing diversified portfolios.

BY WHOM IT IS USED

Portfolio managers and advisers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It visualises both the power and the boundary of diversification, separating risk that can be diversified away from risk that cannot.

RELATED FRAMEWORKS IN THIS COURSE

- **Modern Portfolio Theory** · Slibrary 21
- **Risk-Return Spectrum** · Slibrary 21
- **Capital Asset Pricing Model** · Slibrary 21
- **Black-Litterman Model** · Slibrary 21

SOURCES (HARVARD REFERENCING STYLE)

Markowitz, H. (1952) 'Portfolio selection', The Journal of Finance, 7(1).

Statman, M. (1987) 'How many stocks make a diversified portfolio?', Journal of Financial and Quantitative Analysis, 22(3).

Capital Asset Pricing Model

Beta and the price of market risk

PROFESSIONAL DEFINITION

The Capital Asset Pricing Model (CAPM) relates an asset's expected return to its systematic risk (beta), pricing the market risk that cannot be diversified away.

WHO DEVELOPED IT

Developed independently by William Sharpe, John Lintner and Jan Mossin in the 1960s, building on Markowitz; Sharpe shared the Nobel Prize.

WHY IT WAS DEVELOPED

It was developed to price risk, providing a benchmark expected return for an asset given its exposure to market risk.

FIGURE 1 · CAPITAL ASSET PRICING MODEL

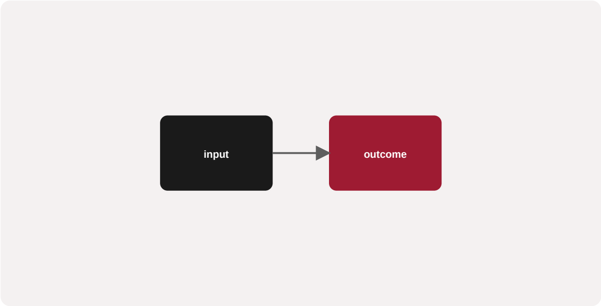

Figure 1. A schematic of the Capital Asset Pricing Model as applied in BARBM312 (illustrative).

HOW IT IS USED

Expected return is computed from the risk-free rate, beta and the market risk premium; assets are assessed against this benchmark.

WHEN IT IS USED

When pricing risk, estimating cost of capital and benchmarking returns.

BY WHOM IT IS USED

Analysts, portfolio managers and corporate-finance teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It introduced beta as the price of systematic risk, a cornerstone of modern asset pricing that frames AI-driven return claims.

RELATED FRAMEWORKS IN THIS COURSE

- **Modern Portfolio Theory** · Slibrary 21
- **Risk-Return Spectrum** · Slibrary 21
- **Diversification Curve** · Slibrary 21
- **Black-Litterman Model** · Slibrary 21

SOURCES (HARVARD REFERENCING STYLE)

Sharpe, W.F. (1964) 'Capital asset prices', The Journal of Finance, 19(3).

Lintner, J. (1965) 'The valuation of risk assets', Review of Economics and Statistics, 47(1).

Black-Litterman Model

Blending views with market equilibrium

PROFESSIONAL DEFINITION

The Black-Litterman Model blends an investor's subjective views with the market-equilibrium expected returns to produce stable, intuitive portfolio allocations, addressing the instability of naïve mean-variance optimisation.

WHO DEVELOPED IT

Developed by Fischer Black and Robert Litterman at Goldman Sachs in the early 1990s.

WHY IT WAS DEVELOPED

It was developed to fix the extreme, unstable portfolios that pure Markowitz optimisation produces when fed noisy return estimates.

FIGURE 1 · BLACK-LITTERMAN MODEL

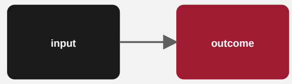

Figure 1. A schematic of the Black-Litterman Model as applied in BARBM312 (illustrative).

HOW IT IS USED

Market-implied returns are taken as a prior and adjusted by the investor's stated views with associated confidence, yielding balanced allocations.

WHEN IT IS USED

When constructing portfolios that combine market information with active views, including AI-generated signals.

BY WHOM IT IS USED

Quantitative portfolio managers and asset allocators.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It married Bayesian reasoning with portfolio theory, producing robust allocations and a principled way to incorporate AI-derived views.

RELATED FRAMEWORKS IN THIS COURSE

- **Modern Portfolio Theory** · Slibrary 21
- **Risk-Return Spectrum** · Slibrary 21
- **Diversification Curve** · Slibrary 21
- **Capital Asset Pricing Model** · Slibrary 21

SOURCES (HARVARD REFERENCING STYLE)

Black, F. and Litterman, R. (1992) 'Global portfolio optimization', Financial Analysts Journal, 48(5).

He, G. and Litterman, R. (1999) 'The intuition behind Black-Litterman model portfolios', Goldman Sachs.

Fintech Disruption Map

Where AI re-prices financial services

PROFESSIONAL DEFINITION

The Fintech Disruption Map locates where AI and digital technology are re-pricing or unbundling financial services, from payments and lending to advice and trading.

WHO DEVELOPED IT

A construct from fintech-strategy analysis, reflecting industry and consultancy mapping of financial-services disruption.

WHY IT WAS DEVELOPED

It was developed to help incumbents and entrants see where value is migrating as AI reshapes finance.

FIGURE 1 · FINTECH DISRUPTION MAP

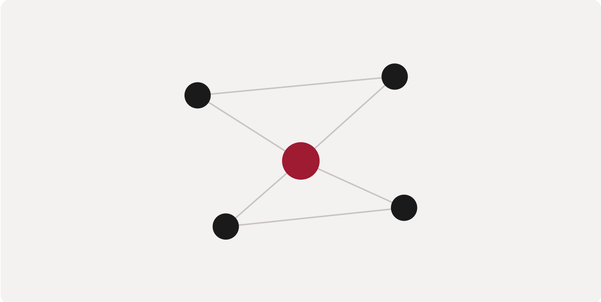

Figure 1. A schematic of the Fintech Disruption Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Financial-services activities are mapped by their exposure to AI-driven disruption and unbundling; strategy follows the migration of value.

WHEN IT IS USED

When analysing competitive threats and opportunities in financial services.

BY WHOM IT IS USED

Financial-services strategists, investors and product leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It gives a structured view of where AI is genuinely re-pricing finance rather than treating disruption as uniform.

RELATED FRAMEWORKS IN THIS COURSE

- **ML Adoption Curve (Investing)** · Slibrary 22
- **Robo-Advisory Stack** · Slibrary 22
- **Alt-Data Pyramid** · Slibrary 22
- **Blockchain Instrument Map** · Slibrary 22

SOURCES (HARVARD REFERENCING STYLE)

Philippon, T. (2016) 'The fintech opportunity', NBER Working Paper 22476.
World Economic Forum (2017) Beyond Fintech. Geneva: WEF.

ML Adoption Curve (Investing)

Muldowney's stages of ML in asset mgmt

PROFESSIONAL DEFINITION

The ML Adoption Curve for investment management describes the stages by which investment firms move from experimentation with machine learning to its full integration into the investment process.

WHO DEVELOPED IT

Associated with Peter Muldowney's analysis of machine-learning adoption in investment management.

WHY IT WAS DEVELOPED

It was developed to help investment firms locate their ML maturity and plan a realistic path to integration.

FIGURE 1 · ML ADOPTION CURVE (INVESTING)

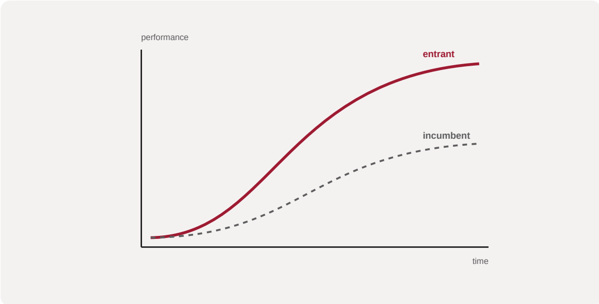

Figure 1. A schematic of the ML Adoption Curve (Investing) as applied in BARBM312 (illustrative).

HOW IT IS USED

The firm places itself on the curve and identifies the data, talent and governance needed to advance without over-reaching.

WHEN IT IS USED

When planning ML adoption in asset and investment management.

BY WHOM IT IS USED

CIOs, quantitative teams and investment-operations leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames ML adoption in investing as a staged journey, countering the assumption that capability can be acquired instantly.

RELATED FRAMEWORKS IN THIS COURSE

- **Fintech Disruption Map** · Slibrary 22
- **Robo-Advisory Stack** · Slibrary 22
- **Alt-Data Pyramid** · Slibrary 22
- **Blockchain Instrument Map** · Slibrary 22

SOURCES (HARVARD REFERENCING STYLE)

Muldowney, P. (2019) 'Machine learning in investment management', industry analysis.

López de Prado, M. (2019) 'Ten applications of financial machine learning', SSRN.

Robo-Advisory Stack

Profiling → allocation → rebalancing

PROFESSIONAL DEFINITION

The Robo-Advisory Stack layers the components of automated investment advice — client profiling, portfolio allocation, execution and rebalancing — showing how AI automates the advisory process.

WHO DEVELOPED IT

A construct from wealth-management practice as robo-advisers such as Betterment and Wealthfront emerged.

WHY IT WAS DEVELOPED

It was developed to clarify how automated advice is assembled and where human judgment and AI each contribute.

FIGURE 1 · ROBO-ADVISORY STACK

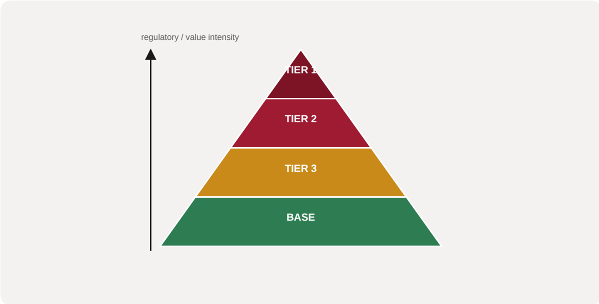

Figure 1. A schematic of the Robo-Advisory Stack as applied in BARBM312 (illustrative).

HOW IT IS USED

Each layer is examined for automation and oversight; the stack guides design and governance of automated advice.

WHEN IT IS USED

When designing or assessing automated and hybrid advisory services.

BY WHOM IT IS USED

Wealth-management, product and compliance leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It decomposes automated advice into governable layers, clarifying where AI adds value and where fiduciary judgment remains.

RELATED FRAMEWORKS IN THIS COURSE

- **Fintech Disruption Map** · Slibrary 22
- **ML Adoption Curve (Investing)** · Slibrary 22
- **Alt-Data Pyramid** · Slibrary 22
- **Blockchain Instrument Map** · Slibrary 22

SOURCES (HARVARD REFERENCING STYLE)

Philippon, T. (2019) 'On fintech and financial inclusion', NBER Working Paper 26330.

D'Acunto, F., Prabhala, N. and Rossi, A.G. (2019) 'The promises and pitfalls of robo-advising', Review of Financial Studies, 32(5).

Alt-Data Pyramid

Satellite, card, web data as alpha

PROFESSIONAL DEFINITION

The Alt-Data Pyramid organises alternative data sources — such as satellite imagery, card transactions and web traffic — by their rawness and the processing required to turn them into investment signals (alpha).

WHO DEVELOPED IT

A construct from quantitative-investing and alternative-data practice.

WHY IT WAS DEVELOPED

It was developed to show that alternative data must be processed and validated, not merely acquired, to yield genuine signal.

FIGURE 1 · ALT-DATA PYRAMID

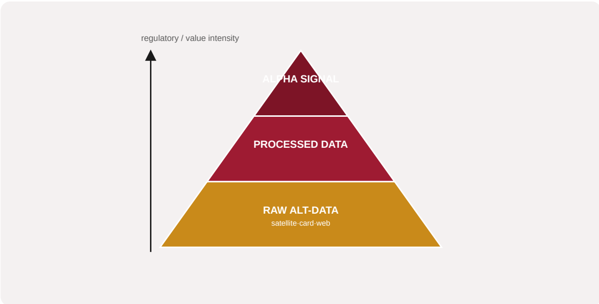

Figure 1. A schematic of the Alt-Data Pyramid as applied in BARBM312 (illustrative).

HOW IT IS USED

Sources are arranged from raw data up to processed alpha signals; investment in processing targets where signal can be extracted.

WHEN IT IS USED

When sourcing and evaluating alternative data for investment.

BY WHOM IT IS USED

Quantitative investors and data teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It clarifies that alternative data's value lies in processing and uniqueness, not mere possession, tempering alt-data hype.

RELATED FRAMEWORKS IN THIS COURSE

- **Fintech Disruption Map** · Slibrary 22
- **ML Adoption Curve (Investing)** · Slibrary 22
- **Robo-Advisory Stack** · Slibrary 22
- **Blockchain Instrument Map** · Slibrary 22

SOURCES (HARVARD REFERENCING STYLE)

Kolanovic, M. and Krishnamachari, R.T. (2017) Big Data and AI Strategies. J.P. Morgan.

Monk, A., Prins, M. and Rook, D. (2019) 'Rethinking alternative data in institutional investment', Journal of Financial Data Science, 1(1).

Blockchain Instrument Map

Tokenised and smart-contract assets

PROFESSIONAL DEFINITION

The Blockchain Instrument Map surveys tokenised and smart-contract-based financial instruments, clarifying how distributed-ledger technology creates new asset types and settlement mechanisms.

WHO DEVELOPED IT

A construct from digital-asset and decentralised-finance analysis.

WHY IT WAS DEVELOPED

It was developed to help investors understand the landscape of blockchain-based instruments and their distinct risks.

FIGURE 1 · BLOCKCHAIN INSTRUMENT MAP

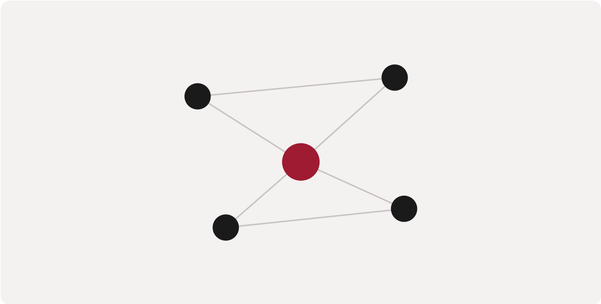

Figure 1. A schematic of the Blockchain Instrument Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Instruments are mapped by type and mechanism; their characteristics and risks are assessed against traditional analogues.

WHEN IT IS USED

When evaluating digital-asset and tokenised investments.

BY WHOM IT IS USED

Investors, product and risk teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It organises a fast-evolving instrument landscape, supporting disciplined assessment rather than speculation.

RELATED FRAMEWORKS IN THIS COURSE

- **Fintech Disruption Map** · Slibrary 22
- **ML Adoption Curve (Investing)** · Slibrary 22
- **Robo-Advisory Stack** · Slibrary 22
- **Alt-Data Pyramid** · Slibrary 22

SOURCES (HARVARD REFERENCING STYLE)

Catalini, C. and Gans, J.S. (2020) 'Some simple economics of the blockchain', Communications of the ACM, 63(7).
BIS (2023) 'The crypto ecosystem: key elements and risks', Bank for International Settlements.

Fama-French Factors

Size, value and beyond-market returns

PROFESSIONAL DEFINITION

The Fama-French model explains asset returns using multiple factors beyond market risk — notably size and value, later extended to profitability and investment — showing returns have more than one systematic source.

WHO DEVELOPED IT

Developed by Eugene Fama and Kenneth French, beginning with their 1992–1993 three-factor model; Fama shared the Nobel Prize.

WHY IT WAS DEVELOPED

It was developed because the single-factor CAPM failed to explain observed return patterns associated with firm size and value.

FIGURE 1 · FAMA-FRENCH FACTORS

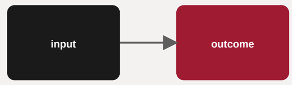

Figure 1. A schematic of the Fama-French Factors as applied in BARBM312 (illustrative).

HOW IT IS USED

Returns are decomposed into exposures to the factors; portfolios are constructed and evaluated by their factor loadings.

WHEN IT IS USED

When explaining returns, constructing factor portfolios and evaluating AI-generated strategies.

BY WHOM IT IS USED

Quantitative investors, analysts and academics.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It established multi-factor asset pricing, showing that returns reward several systematic exposures, not market risk alone.

RELATED FRAMEWORKS IN THIS COURSE

- **Risk-Profiling Matrix** · Slibrary 23
- **Prediction Machines Lens** · Slibrary 23
- **Backtesting Pitfalls Map** · Slibrary 23
- **Monte Carlo Cone** · Slibrary 23

SOURCES (HARVARD REFERENCING STYLE)

Fama, E.F. and French, K.R. (1993) 'Common risk factors in the returns on stocks and bonds', Journal of Financial Economics, 33(1).

Fama, E.F. and French, K.R. (2015) 'A five-factor asset pricing model', Journal of Financial Economics, 116(1).

Risk-Profiling Matrix

Capacity × tolerance for risk

PROFESSIONAL DEFINITION

The Risk-Profiling Matrix assesses an investor by both risk capacity (ability to bear loss) and risk tolerance (willingness to bear it), guiding suitable portfolio risk.

WHO DEVELOPED IT

A construct from financial-advice and suitability practice, reflected in regulatory suitability requirements.

WHY IT WAS DEVELOPED

It was developed to ensure portfolios match both the investor's financial ability and psychological willingness to bear risk.

FIGURE 1 · RISK-PROFILING MATRIX

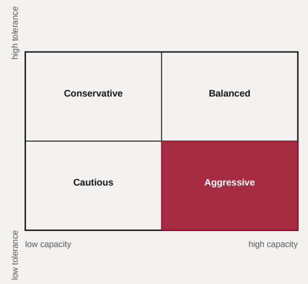

Figure 1. A schematic of the Risk-Profiling Matrix as applied in BARBM312 (illustrative).

HOW IT IS USED

The investor is placed by capacity and tolerance; portfolio risk is set to the lower of the two to ensure suitability.

WHEN IT IS USED

When determining suitable investment risk, including in robo-advice.

BY WHOM IT IS USED

Advisers, wealth managers and compliance teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It separates capacity from tolerance, preventing mismatches that pure questionnaire scores miss.

RELATED FRAMEWORKS IN THIS COURSE

- **Fama-French Factors** · Slibrary 23
- **Prediction Machines Lens** · Slibrary 23
- **Backtesting Pitfalls Map** · Slibrary 23
- **Monte Carlo Cone** · Slibrary 23

SOURCES (HARVARD REFERENCING STYLE)

Grable, J.E. and Lytton, R.H. (1999) 'Financial risk tolerance revisited', Financial Services Review, 8(3).

FCA (2011) 'Assessing suitability', Financial Conduct Authority.

Prediction Machines Lens

AI cheapens prediction, not judgment

PROFESSIONAL DEFINITION

The Prediction Machines Lens frames AI economically as a drop in the cost of prediction, clarifying that AI makes prediction cheap while judgment, data and action remain complementary and human.

WHO DEVELOPED IT

Developed by economists Ajay Agrawal, Joshua Gans and Avi Goldfarb in Prediction Machines (2018).

WHY IT WAS DEVELOPED

It was developed to give a clear economic account of what AI does — lower the cost of prediction — and what it does not replace.

FIGURE 1 · PREDICTION MACHINES LENS

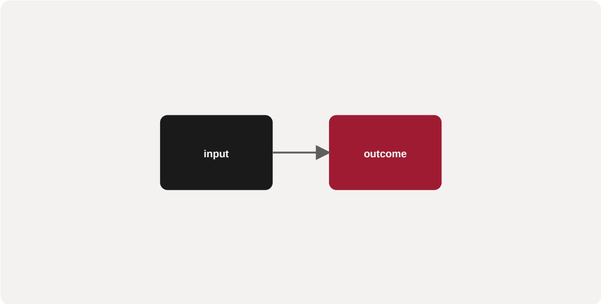

Figure 1. A schematic of the Prediction Machines Lens as applied in BARBM312 (illustrative).

HOW IT IS USED

AI opportunities are analysed by identifying where cheaper prediction creates value and where human judgment must still be supplied.

WHEN IT IS USED

When assessing where AI adds value in investment and decision processes.

BY WHOM IT IS USED

Investors, strategists and decision-makers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes AI as a prediction technology, separating cheap prediction from the judgment that remains decisive — vital where AI meets finance.

RELATED FRAMEWORKS IN THIS COURSE

- **Fama-French Factors** · Slibrary 23
- **Risk-Profiling Matrix** · Slibrary 23
- **Backtesting Pitfalls Map** · Slibrary 23
- **Monte Carlo Cone** · Slibrary 23

SOURCES (HARVARD REFERENCING STYLE)

Agrawal, A., Gans, J. and Goldfarb, A. (2018) Prediction Machines. Boston: HBR Press.

Agrawal, A., Gans, J. and Goldfarb, A. (2022) Power and Prediction. Boston: HBR Press.

Backtesting Pitfalls Map

Overfitting and look-ahead bias

PROFESSIONAL DEFINITION

The Backtesting Pitfalls Map catalogues the ways historical strategy tests mislead — overfitting, look-ahead bias, survivorship bias and multiple testing — so apparent edges can be scrutinised.

WHO DEVELOPED IT

Associated with Marcos López de Prado's work on the dangers of backtesting in financial machine learning.

WHY IT WAS DEVELOPED

It was developed because AI makes it trivially easy to find spurious patterns, and most spectacular backtests are false discoveries.

FIGURE 1 · BACKTESTING PITFALLS MAP

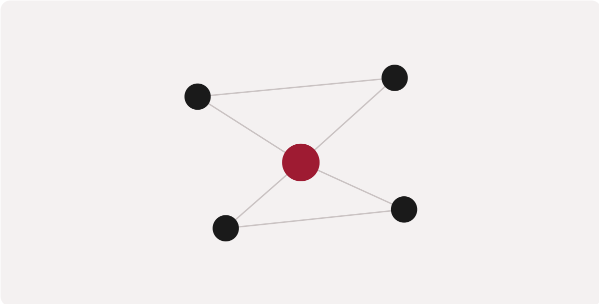

Figure 1. A schematic of the Backtesting Pitfalls Map as applied in BARBM312 (illustrative).

HOW IT IS USED

A backtest is examined against each pitfall; only strategies that survive scrutiny and out-of-sample testing are trusted.

WHEN IT IS USED

When evaluating quantitative and AI-driven investment strategies.

BY WHOM IT IS USED

Quantitative investors, risk and research teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It systematises scepticism toward backtests, a crucial discipline as AI multiplies the risk of false discovery.

RELATED FRAMEWORKS IN THIS COURSE

- **Fama-French Factors** · Slibrary 23
- **Risk-Profiling Matrix** · Slibrary 23
- **Prediction Machines Lens** · Slibrary 23
- **Monte Carlo Cone** · Slibrary 23

SOURCES (HARVARD REFERENCING STYLE)

López de Prado, M. (2018) Advances in Financial Machine Learning. Hoboken: Wiley.

Bailey, D.H. et al. (2014) 'Pseudo-mathematics and financial charlatanism', Notices of the AMS, 61(5).

Monte Carlo Cone

Distribution of possible outcomes

PROFESSIONAL DEFINITION

The Monte Carlo Cone visualises the distribution of possible future outcomes generated by simulating many random scenarios, replacing a single point forecast with a fan of probabilities.

WHO DEVELOPED IT

Based on the Monte Carlo method developed by Stanislaw Ulam and John von Neumann, widely applied in finance.

WHY IT WAS DEVELOPED

It was developed to represent uncertainty honestly, showing the range and likelihood of outcomes rather than a misleading single estimate.

FIGURE 1 · MONTE CARLO CONE

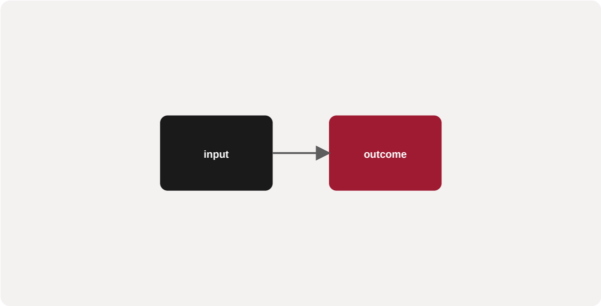

Figure 1. A schematic of the Monte Carlo Cone as applied in BARBM312 (illustrative).

HOW IT IS USED

Many simulated paths are generated from assumed distributions; the resulting cone shows the spread of plausible outcomes for planning.

WHEN IT IS USED

When modelling uncertain investment and financial outcomes.

BY WHOM IT IS USED

Analysts, risk managers and planners.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It communicates uncertainty as a distribution, countering false precision in forecasts — including AI-generated ones.

RELATED FRAMEWORKS IN THIS COURSE

Fama-French Factors · Slibrary 23
Risk-Profiling Matrix · Slibrary 23
Prediction Machines Lens · Slibrary 23
Backtesting Pitfalls Map · Slibrary 23

SOURCES (HARVARD REFERENCING STYLE)

Metropolis, N. and Ulam, S. (1949) 'The Monte Carlo method', Journal of the American Statistical Association, 44(247).
Glasserman, P. (2003) Monte Carlo Methods in Financial Engineering. New York: Springer.

Algo-Trading Pipeline

Signal → execution → monitoring

PROFESSIONAL DEFINITION

The Algorithmic Trading Pipeline describes the flow from signal generation through execution to monitoring, showing how automated strategies translate data into trades and manage their consequences.

WHO DEVELOPED IT

A construct from quantitative-finance and trading-systems practice.

WHY IT WAS DEVELOPED

It was developed to structure the design and governance of automated trading, where errors propagate at machine speed.

FIGURE 1 · ALGO-TRADING PIPELINE

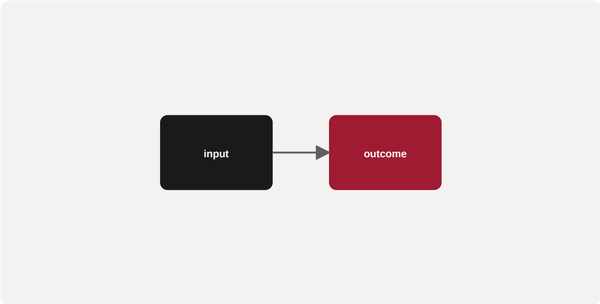

Figure 1. A schematic of the Algo-Trading Pipeline as applied in BARBM312 (illustrative).

HOW IT IS USED

Each stage — signal, execution, monitoring — is designed with controls; the pipeline clarifies where risk and oversight must sit.

WHEN IT IS USED

When designing or governing algorithmic trading systems.

BY WHOM IT IS USED

Quantitative traders, engineers and risk teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the end-to-end automated trading process explicit and governable, emphasising monitoring as integral, not optional.

RELATED FRAMEWORKS IN THIS COURSE

- **Latency Arms Race** · Slibrary 24
- **Market Microstructure Map** · Slibrary 24
- **Flash-Crash Risk Loop** · Slibrary 24
- **Execution Cost Frontier** · Slibrary 24

SOURCES (HARVARD REFERENCING STYLE)

Chan, E.P. (2013) Algorithmic Trading: Winning Strategies and Their Rationale. Hoboken: Wiley.
Aldridge, I. (2013) High-Frequency Trading. Hoboken: Wiley.

Latency Arms Race

Microseconds as competitive edge

PROFESSIONAL DEFINITION

The Latency Arms Race describes the competitive pursuit of ever-lower trading latency, where microseconds of speed advantage translate into profit, driving heavy investment in infrastructure.

WHO DEVELOPED IT

A phenomenon documented in market-microstructure and high-frequency-trading research.

WHY IT WAS DEVELOPED

It is described to explain the economics and risks of speed competition in modern markets.

FIGURE 1 · LATENCY ARMS RACE

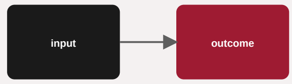

Figure 1. A schematic of the Latency Arms Race as applied in BARBM312 (illustrative).

HOW IT IS USED

The value of marginal speed is weighed against its cost and systemic risk; participants decide how far to compete on latency.

WHEN IT IS USED

When analysing high-frequency trading strategy and market structure.

BY WHOM IT IS USED

Trading-firm strategists, regulators and researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It captures how speed itself became a contested, costly source of edge, with systemic implications.

RELATED FRAMEWORKS IN THIS COURSE

Algo-Trading Pipeline · Slibrary 24
Market Microstructure Map · Slibrary 24
Flash-Crash Risk Loop · Slibrary 24
Execution Cost Frontier · Slibrary 24

SOURCES (HARVARD REFERENCING STYLE)

Budish, E., Cramton, P. and Shim, J. (2015) 'The high-frequency trading arms race', Quarterly Journal of Economics, 130(4).

MacKenzie, D. (2021) Trading at the Speed of Light. Princeton: PUP.

Market Microstructure Map

Order books, spreads and impact

PROFESSIONAL DEFINITION

The Market Microstructure Map describes how orders, order books, spreads and price impact interact to determine how trades are executed and prices formed at a fine-grained level.

WHO DEVELOPED IT

Grounded in the field of market microstructure, associated with scholars such as Maureen O'Hara and Larry Harris.

WHY IT WAS DEVELOPED

It was developed to understand price formation and trading costs at the level where automated and high-frequency strategies operate.

FIGURE 1 · MARKET MICROSTRUCTURE MAP

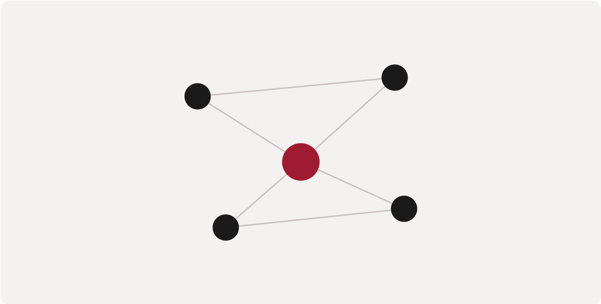

Figure 1. A schematic of the Market Microstructure Map as applied in BARBM312 (illustrative).

HOW IT IS USED

The components of the trading mechanism are mapped to analyse execution cost, liquidity and price impact.

WHEN IT IS USED

When designing execution strategy and analysing trading costs.

BY WHOM IT IS USED

Traders, execution and research teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It provides the fine-grained view of trading necessary to understand and manage execution in automated markets.

RELATED FRAMEWORKS IN THIS COURSE

- **Algo-Trading Pipeline** · Slibrary 24
- **Latency Arms Race** · Slibrary 24
- **Flash-Crash Risk Loop** · Slibrary 24
- **Execution Cost Frontier** · Slibrary 24

SOURCES (HARVARD REFERENCING STYLE)

O'Hara, M. (1995) Market Microstructure Theory. Cambridge: Blackwell.

Harris, L. (2003) Trading and Exchanges. Oxford: OUP.

Flash-Crash Risk Loop

How automation amplifies shocks

PROFESSIONAL DEFINITION

The Flash-Crash Risk Loop describes how automated trading can amplify shocks through feedback — rapid selling triggering further automated selling — producing sudden, severe market dislocations.

WHO DEVELOPED IT

Analysed in studies of events such as the 2010 Flash Crash and broader research on automation and market stability.

WHY IT WAS DEVELOPED

It was developed to explain how speed and automation can interact to destabilise markets, informing safeguards.

FIGURE 1 · FLASH-CRASH RISK LOOP

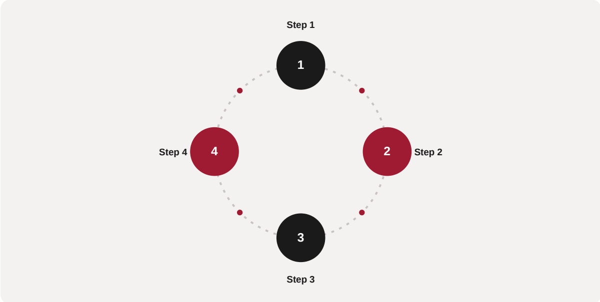

Figure 1. A schematic of the Flash-Crash Risk Loop as applied in BARBM312 (illustrative).

HOW IT IS USED

Feedback dynamics are mapped to identify amplification risks; circuit breakers and controls are designed to interrupt the loop.

WHEN IT IS USED

When assessing systemic risk in automated markets and designing safeguards.

BY WHOM IT IS USED

Regulators, risk managers and trading-system designers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames automation as a potential amplifier of shocks, motivating controls such as circuit breakers.

RELATED FRAMEWORKS IN THIS COURSE

- **Algo-Trading Pipeline** · Slibrary 24
- **Latency Arms Race** · Slibrary 24
- **Market Microstructure Map** · Slibrary 24
- **Execution Cost Frontier** · Slibrary 24

SOURCES (HARVARD REFERENCING STYLE)

Kirilenko, A. et al. (2017) 'The Flash Crash: High-frequency trading in an electronic market', Journal of Finance, 72(3).

CFTC-SEC (2010) Findings Regarding the Market Events of May 6, 2010.

Execution Cost Frontier

Speed vs. slippage trade-off

PROFESSIONAL DEFINITION

The Execution Cost Frontier expresses the trade-off between trading speed and market impact (slippage): trading faster moves the price against you, while trading slowly risks adverse moves.

WHO DEVELOPED IT

Grounded in optimal-execution research, notably the Almgren-Chriss framework.

WHY IT WAS DEVELOPED

It was developed to optimise trade execution by balancing the cost of speed against the risk of delay.

FIGURE 1 · EXECUTION COST FRONTIER

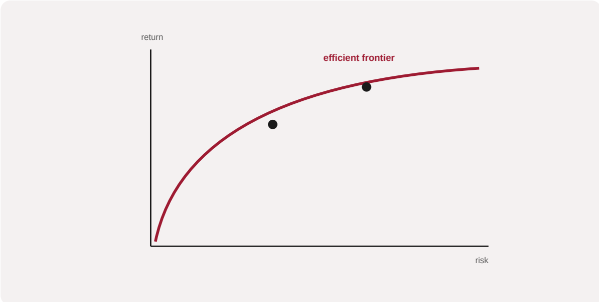

Figure 1. A schematic of the Execution Cost Frontier as applied in BARBM312 (illustrative).

HOW IT IS USED

Execution strategies are placed on the frontier; the schedule is chosen to minimise total expected cost given risk preferences.

WHEN IT IS USED

When executing large trades and designing execution algorithms.

BY WHOM IT IS USED

Traders, execution teams and quantitative researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It formalises the speed-versus-impact trade-off, underpinning algorithmic execution strategy.

RELATED FRAMEWORKS IN THIS COURSE

- **Algo-Trading Pipeline** · Slibrary 24
- **Latency Arms Race** · Slibrary 24
- **Market Microstructure Map** · Slibrary 24
- **Flash-Crash Risk Loop** · Slibrary 24

SOURCES (HARVARD REFERENCING STYLE)

Almgren, R. and Chriss, N. (2001) 'Optimal execution of portfolio transactions', Journal of Risk, 3(2).

Kissell, R. (2013) The Science of Algorithmic Trading and Portfolio Management. London: Academic Press.

AI Startup Valuation Stack

Team, data, traction, defensibility

PROFESSIONAL DEFINITION

The AI Startup Valuation Stack identifies what an AI-first company is actually valued on — team, proprietary data, traction and defensibility — beyond headline revenue multiples.

WHO DEVELOPED IT

A synthesis from venture-valuation practice applied to AI-first companies.

WHY IT WAS DEVELOPED

It was developed to look past revenue multiples to the strategic assets that justify and sustain an AI venture's valuation.

FIGURE 1 · AI STARTUP VALUATION STACK

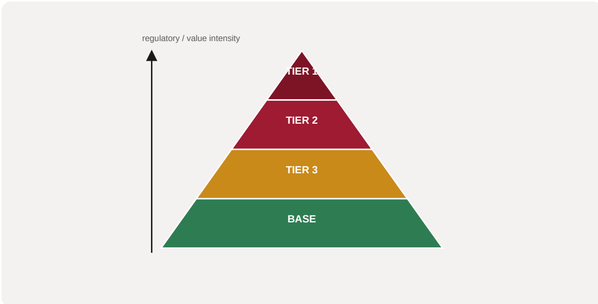

Figure 1. A schematic of the AI Startup Valuation Stack as applied in BARBM312 (illustrative).

HOW IT IS USED

The company is assessed layer by layer; weak layers (e.g. no defensibility) flag valuation risk despite strong revenue.

WHEN IT IS USED

During fundraising, investment due diligence and valuation.

BY WHOM IT IS USED

Founders, investors and corporate-development teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It grounds AI valuation in durable strategic assets rather than growth alone, exposing fragile high-valuation companies.

RELATED FRAMEWORKS IN THIS COURSE

- **Cost-Structure Lens** · Slibrary 25
- **Rule of 40** · Slibrary 25
- **Strategic Asset Map** · Slibrary 25
- **Investor Expectation Curve** · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

Damodaran, A. (2018) The Dark Side of Valuation. 3rd edn. Upper Saddle River: Pearson.

Gornall, W. and Strebulaev, I.A. (2020) 'Squaring venture capital valuations with reality', Journal of Financial Economics, 135(1).

Cost-Structure Lens

Gross margin reality of AI products

PROFESSIONAL DEFINITION

The Cost-Structure Lens examines the gross-margin reality of an AI product, recognising that inference, data and human-in-the-loop costs can erode the high margins assumed for software.

WHO DEVELOPED IT

A construct from SaaS and AI-economics analysis, prominent in the debate over generative-AI unit economics.

WHY IT WAS DEVELOPED

It was developed because AI products often carry variable costs (compute, review) that classic software margins ignore.

FIGURE 1 · COST-STRUCTURE LENS

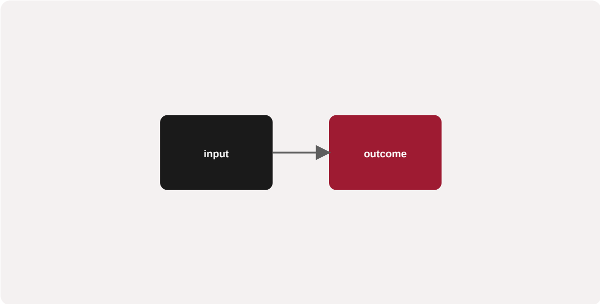

Figure 1. A schematic of the Cost-Structure Lens as applied in BARBM312 (illustrative).

HOW IT IS USED

The full cost of serving each unit is analysed; pricing and strategy adjust to the true, often lower, gross margin.

WHEN IT IS USED

When assessing the economics and scalability of AI products.

BY WHOM IT IS USED

Founders, finance leaders and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It punctures the assumption of software-like margins for AI, exposing the inference and labour costs beneath them.

RELATED FRAMEWORKS IN THIS COURSE

- **AI Startup Valuation Stack** · Slibrary 25
- **Rule of 40** · Slibrary 25
- **Strategic Asset Map** · Slibrary 25
- **Investor Expectation Curve** · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

a16z (2023) 'Navigating the high cost of AI compute'.

Bessemer Venture Partners (2023) 'State of the cloud'.

Rule of 40

Growth + profit threshold for SaaS/AI

PROFESSIONAL DEFINITION

The Rule of 40 holds that a healthy software or AI company's growth rate plus profit margin should sum to at least 40%, balancing growth against profitability.

WHO DEVELOPED IT

A heuristic popularised by venture and growth investors, attributed to Brad Feld and widely used in SaaS analysis.

WHY IT WAS DEVELOPED

It was developed as a quick test of whether a company's growth is being achieved at acceptable economic cost.

FIGURE 1 · RULE OF 40

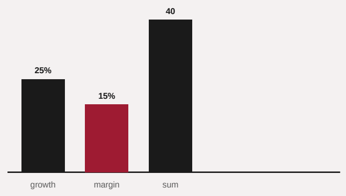

Figure 1. A schematic of the Rule of 40 as applied in BARBM312 (illustrative).

HOW IT IS USED

Growth rate and profit margin are summed; a result of 40 or more signals a healthy balance, prompting scrutiny if below.

WHEN IT IS USED

When assessing the health and fundability of growth-stage AI companies.

BY WHOM IT IS USED

Founders, investors and finance leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It provides a single, memorable test balancing growth and profitability, widely adopted in growth-stage investing.

RELATED FRAMEWORKS IN THIS COURSE

- **AI Startup Valuation Stack** · Slibrary 25
- **Cost-Structure Lens** · Slibrary 25
- **Strategic Asset Map** · Slibrary 25
- **Investor Expectation Curve** · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

Feld, B. (2015) 'The rule of 40% for a healthy SaaS company', Feld Thoughts.

Bain & Company (2021) 'The Rule of 40'.

Strategic Asset Map

What an acquirer actually pays for

PROFESSIONAL DEFINITION

The Strategic Asset Map identifies what an acquirer or investor actually pays for in an AI company — data, talent, technology, customers or market position — clarifying where durable value resides.

WHO DEVELOPED IT

A construct from corporate-development and M&A practice applied to AI companies.

WHY IT WAS DEVELOPED

It was developed to focus value creation on the assets that command a premium in acquisition or investment.

FIGURE 1 · STRATEGIC ASSET MAP

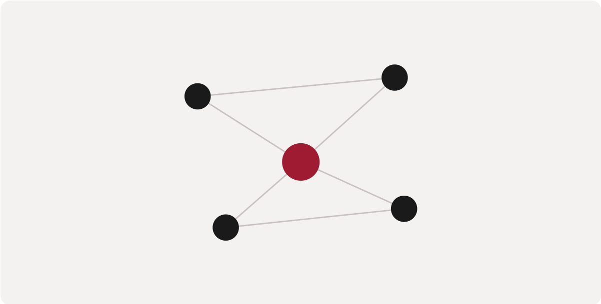

Figure 1. A schematic of the Strategic Asset Map as applied in BARBM312 (illustrative).

HOW IT IS USED

The company's assets are mapped and assessed for transferability and scarcity; strategy strengthens the most valuable.

WHEN IT IS USED

During strategy, fundraising and M&A preparation.

BY WHOM IT IS USED

Founders, corporate-development and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It clarifies which assets create acquisition or investment value, guiding where an AI company should build moats.

RELATED FRAMEWORKS IN THIS COURSE

- **AI Startup Valuation Stack** · Slibrary 25
- **Cost-Structure Lens** · Slibrary 25
- **Rule of 40** · Slibrary 25
- **Investor Expectation Curve** · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

Capron, L. and Mitchell, W. (2012) Build, Borrow, or Buy. Boston: HBR Press.
Damodaran, A. (2018) The Dark Side of Valuation. Upper Saddle River: Pearson.

Investor Expectation Curve

Milestones by funding stage

PROFESSIONAL DEFINITION

The Investor Expectation Curve maps the milestones investors expect at each funding stage, from seed through growth rounds, clarifying what a company must demonstrate to raise the next round.

WHO DEVELOPED IT

A construct from venture-financing practice describing stage-appropriate expectations.

WHY IT WAS DEVELOPED

It was developed to align founders' progress with the evidence investors require at each stage.

FIGURE 1 · INVESTOR EXPECTATION CURVE

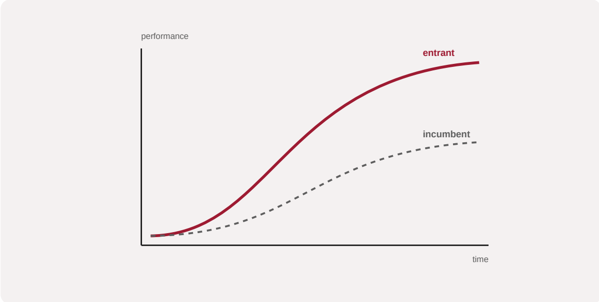

Figure 1. A schematic of the Investor Expectation Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

The company's stage is matched to expected milestones; the gap defines what must be achieved before the next raise.

WHEN IT IS USED

During fundraising planning across stages.

BY WHOM IT IS USED

Founders, investors and finance leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes stage-by-stage investor expectations explicit, helping AI ventures time and target their raises.

RELATED FRAMEWORKS IN THIS COURSE

AI Startup Valuation Stack · Slibrary 25
Cost-Structure Lens · Slibrary 25
Rule of 40 · Slibrary 25
Strategic Asset Map · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

Feld, B. and Mendelson, J. (2019) Venture Deals. 4th edn. Hoboken: Wiley.
Gompers, P. and Lerner, J. (2004) The Venture Capital Cycle. Cambridge: MIT Press.

Growth Loops

Self-reinforcing acquisition engines

PROFESSIONAL DEFINITION

Growth Loops are self-reinforcing systems where each cycle's output fuels the next cycle's input, producing compounding growth that linear funnels cannot capture.

WHO DEVELOPED IT

Articulated by growth practitioners, notably Brian Balfour and the Reforge community.

WHY IT WAS DEVELOPED

It was developed to capture how durable products grow through reinvested output rather than one-off acquisition.

FIGURE 1 · GROWTH LOOPS

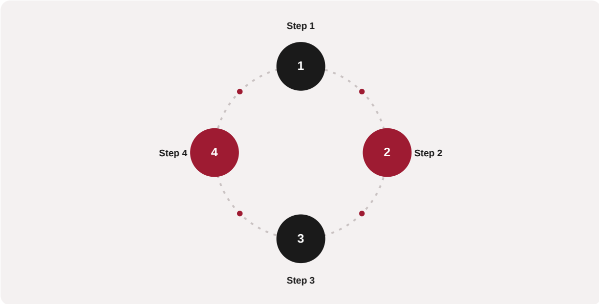

Figure 1. A schematic of the Growth Loops as applied in BARBM312 (illustrative).

HOW IT IS USED

The loop and its reinvestment mechanism are mapped; AI and product effort target strengthening the loop.

WHEN IT IS USED

When designing scalable growth engines for AI products.

BY WHOM IT IS USED

Growth, product and strategy leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes growth as compounding loops, focusing scaling effort on reinforcing mechanisms.

RELATED FRAMEWORKS IN THIS COURSE

- **Product-Led Growth** · Slibrary 26
- **Network Effects Map** · Slibrary 26
- **T2D3 Trajectory** · Slibrary 26
- **Flywheel Model** · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Balfour, B. (2017) 'Growth loops are the new funnels', Reforge.

Ellis, S. and Brown, M. (2017) Hacking Growth. New York: Currency.

Product-Led Growth

The product as the funnel

PROFESSIONAL DEFINITION

Product-Led Growth (PLG) is a strategy in which the product itself drives acquisition, conversion and expansion, with users experiencing value before, or instead of, a sales conversation.

WHO DEVELOPED IT

Popularised by the OpenView venture firm and product-led practitioners as a dominant software go-to-market model.

WHY IT WAS DEVELOPED

It was developed to lower acquisition cost and accelerate growth by letting the product do the selling.

FIGURE 1 · PRODUCT-LED GROWTH

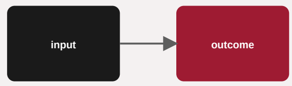

Figure 1. A schematic of the Product-Led Growth as applied in BARBM312 (illustrative).

HOW IT IS USED

The product is designed for self-serve value, virality and expansion; metrics track activation and product-qualified leads.

WHEN IT IS USED

When designing go-to-market for AI products with self-serve potential.

BY WHOM IT IS USED

Product, growth and go-to-market leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It shifts the growth engine from sales to the product, often lowering cost of acquisition for AI-enabled software.

RELATED FRAMEWORKS IN THIS COURSE

- **Growth Loops** · Slibrary 26
- **Network Effects Map** · Slibrary 26
- **T2D3 Trajectory** · Slibrary 26
- **Flywheel Model** · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Bush, W. (2019) Product-Led Growth. ProductLed.

OpenView (2021) 'The product-led growth playbook'.

Network Effects Map

Direct, indirect and data effects

PROFESSIONAL DEFINITION

A Network Effects Map distinguishes the types of network effect — direct, indirect (two-sided), and data — that can make a product more valuable as more people use it.

WHO DEVELOPED IT

Grounded in platform-economics theory, associated with scholars such as Katz and Shapiro and the platform work of Parker, Van Alstyne and Choudary.

WHY IT WAS DEVELOPED

It was developed to identify whether and how a product benefits from network effects, a key source of durable advantage.

FIGURE 1 · NETWORK EFFECTS MAP

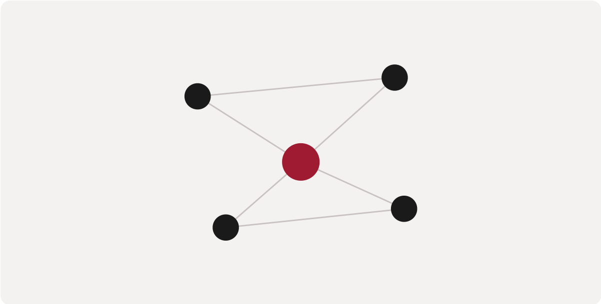

Figure 1. A schematic of the Network Effects Map as applied in BARBM312 (illustrative).

HOW IT IS USED

The product is examined for each type of effect; strategy strengthens the effects that compound advantage.

WHEN IT IS USED

When assessing platform and AI-product defensibility.

BY WHOM IT IS USED

Founders, product and strategy leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It distinguishes types of network effect rather than treating them as one, clarifying where compounding advantage actually arises.

RELATED FRAMEWORKS IN THIS COURSE

Growth Loops · Slibrary 26 | **Product-Led Growth** · Slibrary 26 | **T2D3 Trajectory** · Slibrary 26 | **Flywheel Model** · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Katz, M.L. and Shapiro, C. (1985) 'Network externalities, competition, and compatibility', American Economic Review, 75(3).
Parker, G., Van Alstyne, M. and Choudary, S.P. (2016) Platform Revolution. New York: Norton.

T2D3 Trajectory

Triple, triple, double, double, double

PROFESSIONAL DEFINITION

T2D3 describes a hyper-growth revenue trajectory — triple, triple, double, double, double — that some software companies follow on the path from early traction to scale.

WHO DEVELOPED IT

Articulated by venture investor Neeraj Agrawal of Battery Ventures as a benchmark for SaaS hyper-growth.

WHY IT WAS DEVELOPED

It was developed as a benchmark for the growth pace that defines breakout software companies.

FIGURE 1 · T2D3 TRAJECTORY

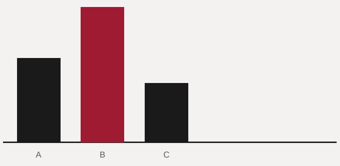

Figure 1. A schematic of the T2D3 Trajectory as applied in BARBM312 (illustrative).

HOW IT IS USED

A company's year-on-year growth is compared with the T2D3 path to gauge whether it is on a breakout trajectory.

WHEN IT IS USED

When benchmarking growth-stage AI and software companies.

BY WHOM IT IS USED

Founders and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It gives a memorable, ambitious growth benchmark, clarifying the pace expected of breakout ventures.

RELATED FRAMEWORKS IN THIS COURSE

- **Growth Loops** · Slibrary 26
- **Product-Led Growth** · Slibrary 26
- **Network Effects Map** · Slibrary 26
- **Flywheel Model** · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Agrawal, N. (2015) 'The SaaS adventure', TechCrunch.
Battery Ventures (2018) 'Scaling to $100M ARR'.

Flywheel Model

Momentum compounding over time

PROFESSIONAL DEFINITION

The Flywheel Model depicts business momentum as a heavy wheel that, once pushed consistently in one direction, accumulates and compounds momentum over time.

WHO DEVELOPED IT

Popularised by Jim Collins in Good to Great (2001) as a metaphor for how breakthroughs are built cumulatively.

WHY IT WAS DEVELOPED

It was developed to counter the myth of the single decisive move, showing that durable success compounds from consistent effort.

FIGURE 1 · FLYWHEEL MODEL

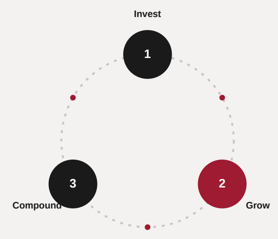

Figure 1. A schematic of the Flywheel Model as applied in BARBM312 (illustrative).

HOW IT IS USED

The reinforcing components of the business are identified and pushed consistently so momentum compounds rather than resets.

WHEN IT IS USED

When building and communicating a compounding growth strategy.

BY WHOM IT IS USED

Founders, executives and strategists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes success as cumulative momentum rather than a single breakthrough, guiding consistent strategic effort.

RELATED FRAMEWORKS IN THIS COURSE

- **Growth Loops** · Slibrary 26
- **Product-Led Growth** · Slibrary 26
- **Network Effects Map** · Slibrary 26
- **T2D3 Trajectory** · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Collins, J. (2001) Good to Great. New York: HarperBusiness.
Collins, J. (2019) Turning the Flywheel. New York: HarperBusiness.

Vertical vs. Platform 2×2

Industry depth vs. cross-sector reach

PROFESSIONAL DEFINITION

The Vertical-vs-Platform matrix contrasts vertical AI strategies that go deep in one industry with platform strategies that go broad across sectors, mapping industry depth against breadth of reach.

WHO DEVELOPED IT

A contemporary venture-strategy framing from the debate over how AI companies should position themselves.

WHY IT WAS DEVELOPED

It was developed to clarify a fundamental strategic choice for AI ventures: depth in a vertical or breadth as a platform.

FIGURE 1 · VERTICAL VS. PLATFORM 2×2

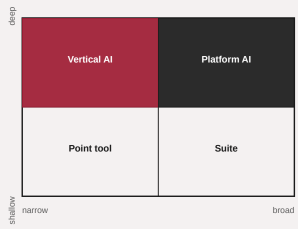

Figure 1. A schematic of the Vertical vs. Platform 2×2 as applied in BARBM312 (illustrative).

HOW IT IS USED

A venture is placed by its depth and breadth; the position implies distinct moats, go-to-market and scaling paths.

WHEN IT IS USED

When setting AI-venture strategy and positioning.

BY WHOM IT IS USED

Founders, investors and strategists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It names the depth-versus-breadth choice central to AI-venture strategy, with different defensibility on each path.

RELATED FRAMEWORKS IN THIS COURSE

- **Wedge-to-Platform Path** · Slibrary 27
- **Category Design Map** · Slibrary 27
- **TAM-SAM-SOM Funnel** · Slibrary 27
- **Platform vs. Pipe** · Slibrary 27

SOURCES (HARVARD REFERENCING STYLE)

a16z (2023) 'The economic case for vertical AI'.
Moore, G. (1991) Crossing the Chasm. New York: HarperBusiness.

Wedge-to-Platform Path

Narrow entry, widening expansion

PROFESSIONAL DEFINITION

The Wedge-to-Platform Path describes how a company enters with a narrow, sharp 'wedge' product and expands from that beachhead into a broader platform over time.

WHO DEVELOPED IT

A venture-strategy construct drawing on Moore's beachhead strategy and platform-expansion thinking.

WHY IT WAS DEVELOPED

It was developed to show how durable platforms are built — by winning a narrow wedge first, not by launching broad.

FIGURE 1 · WEDGE-TO-PLATFORM PATH

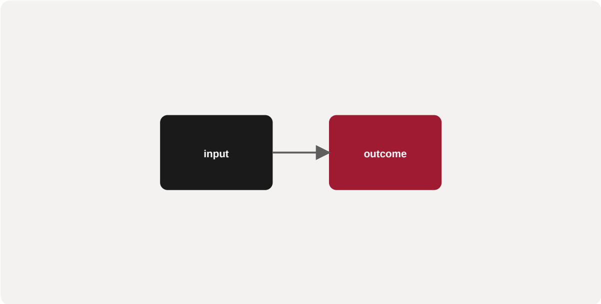

Figure 1. A schematic of the Wedge-to-Platform Path as applied in BARBM312 (illustrative).

HOW IT IS USED

The wedge is chosen for sharpness and expandability; expansion follows once the beachhead is secured.

WHEN IT IS USED

When planning AI-venture entry and expansion strategy.

BY WHOM IT IS USED

Founders, product and strategy leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It charts the realistic path from a focused entry point to a broad platform, countering premature breadth.

RELATED FRAMEWORKS IN THIS COURSE

- **Vertical vs. Platform 2×2** · Slibrary 27
- **Category Design Map** · Slibrary 27
- **TAM-SAM-SOM Funnel** · Slibrary 27
- **Platform vs. Pipe** · Slibrary 27

SOURCES (HARVARD REFERENCING STYLE)

Moore, G. (1991) Crossing the Chasm. New York: HarperBusiness.

Parker, G., Van Alstyne, M. and Choudary, S.P. (2016) Platform Revolution. New York: Norton.

Category Design Map

Create and own a new category

PROFESSIONAL DEFINITION

The Category Design Map guides a company in creating and owning a new market category, rather than competing within an existing one, by defining the problem and the category it solves.

WHO DEVELOPED IT

Articulated by Al Ramadan, Dave Peterson, Christopher Lochhead and Kevin Maney in Play Bigger (2016).

WHY IT WAS DEVELOPED

It was developed on the observation that category creators capture a disproportionate share of a category's value.

FIGURE 1 · CATEGORY DESIGN MAP

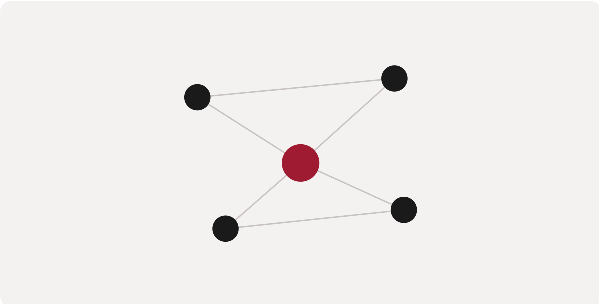

Figure 1. A schematic of the Category Design Map as applied in BARBM312 (illustrative).

HOW IT IS USED

The company defines a new category, frames the problem it addresses, and invests in making the category — and itself as its leader — known.

WHEN IT IS USED

When an AI venture can define and lead a new category.

BY WHOM IT IS USED

Founders, marketing and strategy leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes competition as category creation, where defining the game can be more valuable than winning an existing one.

RELATED FRAMEWORKS IN THIS COURSE

- **Vertical vs. Platform 2×2** · Slibrary 27
- **Wedge-to-Platform Path** · Slibrary 27
- **TAM-SAM-SOM Funnel** · Slibrary 27
- **Platform vs. Pipe** · Slibrary 27

SOURCES (HARVARD REFERENCING STYLE)

Ramadan, A. et al. (2016) Play Bigger. New York: Harper Business.

Lochhead, C. and Peterson, D. (2018) Niche Down. Self-published.

TAM-SAM-SOM Funnel

Sizing the addressable market

PROFESSIONAL DEFINITION

The TAM-SAM-SOM funnel sizes a market in three nested layers — Total Addressable Market, Serviceable Addressable Market and Serviceable Obtainable Market — to ground growth ambitions in realistic opportunity.

WHO DEVELOPED IT

A standard market-sizing construct in venture and strategy practice.

WHY IT WAS DEVELOPED

It was developed to discipline market-size claims, distinguishing the theoretical maximum from the realistically obtainable.

FIGURE 1 · TAM-SAM-SOM FUNNEL

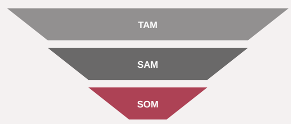

Figure 1. A schematic of the TAM-SAM-SOM Funnel as applied in BARBM312 (illustrative).

HOW IT IS USED

The total market is narrowed to what the firm can serve and then to what it can realistically obtain, grounding projections.

WHEN IT IS USED

During fundraising, planning and market-entry analysis.

BY WHOM IT IS USED

Founders, investors and strategists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It separates aspirational market size from obtainable share, tempering inflated AI-market claims.

RELATED FRAMEWORKS IN THIS COURSE

Vertical vs. Platform 2×2 · Slibrary 27 | **Wedge-to-Platform Path** · Slibrary 27 | **Category Design Map** · Slibrary 27 | **Platform vs. Pipe** · Slibrary 27

SOURCES (HARVARD REFERENCING STYLE)

Blank, S. (2013) The Startup Owner's Manual. Pescadero: K&S Ranch.

Mullins, J. (2017) The Customer-Funded Business. Hoboken: Wiley.

Platform vs. Pipe

Orchestrate value vs. produce it

PROFESSIONAL DEFINITION

The Platform-vs-Pipe distinction contrasts platforms that orchestrate value created by others with pipes that produce and sell value linearly, explaining why platforms often scale and defend better.

WHO DEVELOPED IT

Articulated by Sangeet Paul Choudary and colleagues in Platform Revolution (2016).

WHY IT WAS DEVELOPED

It was developed to explain the rise of platform businesses and why they outcompete traditional linear 'pipe' firms.

FIGURE 1 · PLATFORM VS. PIPE

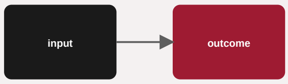

Figure 1. A schematic of the Platform vs. Pipe as applied in BARBM312 (illustrative).

HOW IT IS USED

A business model is classified as platform or pipe; platform strategy focuses on orchestrating and governing an ecosystem.

WHEN IT IS USED

When choosing and assessing AI-venture business models.

BY WHOM IT IS USED

Founders, strategists and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It crystallised why platforms beat pipes, reframing strategy around orchestration rather than linear production.

RELATED FRAMEWORKS IN THIS COURSE

- **Vertical vs. Platform 2×2** · Slibrary 27
- **Wedge-to-Platform Path** · Slibrary 27
- **Category Design Map** · Slibrary 27
- **TAM-SAM-SOM Funnel** · Slibrary 27

SOURCES (HARVARD REFERENCING STYLE)

Parker, G., Van Alstyne, M. and Choudary, S.P. (2016) Platform Revolution. New York: Norton.

Van Alstyne, M., Parker, G. and Choudary, S.P. (2016) 'Pipelines, platforms, and the new rules of strategy', Harvard Business Review, 94(4).

The Scaling Funnel

Seed → Series → scale survival rates

PROFESSIONAL DEFINITION

The Scaling Funnel depicts the steep attrition of startups across funding stages, from seed through to scale, illustrating the low survival rates and what distinguishes survivors.

WHO DEVELOPED IT

A construct drawn from venture-financing data and survival analysis of startups.

WHY IT WAS DEVELOPED

It was developed to set realistic expectations about scaling and to identify what separates companies that survive each stage.

FIGURE 1 · THE SCALING FUNNEL

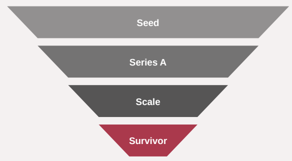

Figure 1. A schematic of the The Scaling Funnel as applied in BARBM312 (illustrative).

HOW IT IS USED

A company's stage and survival odds are assessed; effort focuses on the factors — defensibility, unit economics — that aid survival.

WHEN IT IS USED

When planning and assessing the scaling journey.

BY WHOM IT IS USED

Founders and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the brutal attrition of scaling explicit, focusing attention on the determinants of survival rather than valuation alone.

RELATED FRAMEWORKS IN THIS COURSE

- **Blitzscaling Lens** · Slibrary 28
- **Stages of Scale** · Slibrary 28
- **Unicorn Anatomy** · Slibrary 28
- **Beyond Unicorn — Durability** · Slibrary 28

SOURCES (HARVARD REFERENCING STYLE)

CB Insights (2021) 'The top 12 reasons startups fail'.

Gompers, P. and Lerner, J. (2004) The Venture Capital Cycle. Cambridge: MIT Press.

Blitzscaling Lens

Speed over efficiency to win the market

PROFESSIONAL DEFINITION

The Blitzscaling Lens prioritises speed over efficiency to capture a market before rivals when winner-takes-most dynamics apply, accepting inefficiency and risk in pursuit of scale.

WHO DEVELOPED IT

Developed by Reid Hoffman and Chris Yeh in Blitzscaling (2018), drawing on Silicon Valley scaling experience.

WHY IT WAS DEVELOPED

It was developed to explain when prioritising breakneck growth over efficiency is the right strategy — and when it is not.

FIGURE 1 · BLITZSCALING LENS

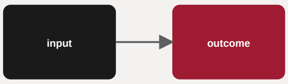

Figure 1. A schematic of the Blitzscaling Lens as applied in BARBM312 (illustrative).

HOW IT IS USED

Market dynamics are assessed for winner-takes-most conditions; if present, the company prioritises speed, accepting inefficiency.

WHEN IT IS USED

When market dynamics reward being first to scale.

BY WHOM IT IS USED

Founders, executives and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It articulates the deliberate choice of speed over efficiency under specific market conditions, with explicit acknowledgment of its risks.

RELATED FRAMEWORKS IN THIS COURSE

- **The Scaling Funnel** · Slibrary 28
- **Stages of Scale** · Slibrary 28
- **Unicorn Anatomy** · Slibrary 28
- **Beyond Unicorn — Durability** · Slibrary 28

SOURCES (HARVARD REFERENCING STYLE)

Hoffman, R. and Yeh, C. (2018) Blitzscaling. New York: Currency.

Sullivan, T. (2016) 'Blitzscaling', Harvard Business Review, 94(4).

Stages of Scale

Founders to professional org

PROFESSIONAL DEFINITION

The Stages of Scale describe how an organisation transforms as it grows — from a small founder-led team to a professionally managed company — with distinct challenges at each stage.

WHO DEVELOPED IT

Drawing on organisational-growth models such as Greiner's stages of growth.

WHY IT WAS DEVELOPED

It was developed to anticipate the organisational crises and transitions that accompany rapid growth.

FIGURE 1 · STAGES OF SCALE

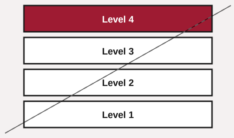

Figure 1. A schematic of the Stages of Scale as applied in BARBM312 (illustrative).

HOW IT IS USED

The company's stage is identified; leadership and structure are evolved to meet the next stage's demands.

WHEN IT IS USED

When scaling an organisation and anticipating growing pains.

BY WHOM IT IS USED

Founders, executives and HR leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames scaling as a series of organisational transformations, not merely getting bigger, preparing leaders for predictable crises.

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- **Unicorn Anatomy** · Slibrary 28
- **Beyond Unicorn — Durability** · Slibrary 28

SOURCES (HARVARD REFERENCING STYLE)

Greiner, L.E. (1972) 'Evolution and revolution as organizations grow', Harvard Business Review, 50(4).
Wasserman, N. (2012) The Founder's Dilemmas. Princeton: PUP.

Unicorn Anatomy

Common DNA of $1B+ AI firms

PROFESSIONAL DEFINITION

Unicorn Anatomy examines the common characteristics of companies that reach a billion-dollar valuation — large markets, strong defensibility, exceptional teams and compounding advantages — to identify the DNA of breakout success.

WHO DEVELOPED IT

A construct from venture analysis of high-valuation companies, building on Aileen Lee's coining of the term 'unicorn'.

WHY IT WAS DEVELOPED

It was developed to identify recurring traits of breakout companies, while cautioning that valuation alone is not durability.

FIGURE 1 · UNICORN ANATOMY

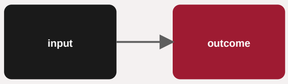

Figure 1. A schematic of the Unicorn Anatomy as applied in BARBM312 (illustrative).

HOW IT IS USED

A company is assessed against the recurring traits; gaps indicate where durability must be strengthened.

WHEN IT IS USED

When assessing breakout potential and durability of AI ventures.

BY WHOM IT IS USED

Founders, investors and strategists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It dissects what billion-dollar companies share, while warning that the valuation label is an output, not a strategy.

RELATED FRAMEWORKS IN THIS COURSE

- **The Scaling Funnel** · Slibrary 28
- **Blitzscaling Lens** · Slibrary 28
- **Stages of Scale** · Slibrary 28
- **Beyond Unicorn — Durability** · Slibrary 28

SOURCES (HARVARD REFERENCING STYLE)

Lee, A. (2013) 'Welcome to the unicorn club', TechCrunch.
Gornall, W. and Strebulaev, I.A. (2020) 'Squaring venture capital valuations with reality', Journal of Financial Economics, 135(1).

Beyond Unicorn — Durability

From valuation to lasting advantage

PROFESSIONAL DEFINITION

Beyond-Unicorn Durability shifts the focus from achieving a billion-dollar valuation to building lasting competitive advantage, arguing that durable value, not the valuation label, is the real goal.

WHO DEVELOPED IT

A synthesis from strategy and venture thinking emphasising durability over valuation milestones.

WHY IT WAS DEVELOPED

It was developed to counter the chasing of unicorn status, which can produce fragile companies, in favour of lasting advantage.

FIGURE 1 · BEYOND UNICORN — DURABILITY

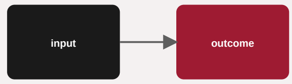

Figure 1. A schematic of the Beyond Unicorn — Durability as applied in BARBM312 (illustrative).

HOW IT IS USED

The company is assessed on durable advantage — moats, economics, adaptability — rather than valuation, and strategy reinforces durability.

WHEN IT IS USED

When setting long-term strategy beyond growth-stage milestones.

BY WHOM IT IS USED

Founders, executives and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes the goal from valuation to durability, the integrative lesson of the capstone unit.

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- **Stages of Scale** · Slibrary 28
- **Unicorn Anatomy** · Slibrary 28

SOURCES (HARVARD REFERENCING STYLE)

Helmer, H. (2016) 7 Powers. Deep Strategy.

Collins, J. and Porras, J.I. (1994) Built to Last. New York: HarperBusiness.